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LEVELLING  
INSTRUCTION BOOK

PART I - FIELD WORK

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His Majesty's Government of Nepal  
Survey Department  
GEODETIC SURVEY BRANCH

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**LEVELLING  
INSTRUCTION BOOK**

PART I - FIELD WORK

Revised and Prepared by:

Nara N. Shrestha (Vaidya)  
Chief Survey Officer  
Levelling Division

Approved by:

Narayan Krishna N. Pradhan  
Deputy Director General  
Geodetic Survey Branch

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## I. PREFACE

This field instruction is prepared mainly for the survey personnel to perform Geodetic Levelling. The first edition was published in 1975 and with the couple of years experience we have revised our earlier instruction and are now presenting you with this Second Edition.

Since our Levelling Division is immature (established in 1975) we cannot guarantee our field instruction to be perfect in all respects. Further revision will be necessary in time to come, in the mean time any suggestions from the readers to improve its quality will be very much appreciated.

**Levelling Division  
1988**

## 2. TERMINOLOGY

### Level Surface

is that surface which is at all points normal to the direction of gravity as indicated by a plumb line.

### Levelling

is an operation for the determination of the relative difference of height between any two points on the earth's surface.

### Level line

is a line which lies on one level surface and is therefore perpendicular to the direction of gravity at all points along the line.

### Horizontal Plane

passing through a point is the plane normal to the vertical at that point and therefore is tangential to the level surface at that point.

### Line (Ln)

is called the route followed by the method of levelling from point A to point B.

### Loop (Lp)

is called the closed system of a certain number of lines.

### Alignment (Al)

is called a sequence of coherent roads considered in one line. (e.g. The road Birganj - Pathlaiya - Hetauda - Naubise - Kathmandu is considered as one alignment).

**Datum Surface** is an adopted (assumed) level surface to which the elevation of points are referred. Mean Sea Level is most frequently used as Datum Surface.

**Mean Sea Level (MSL)** is the mean level of the sea at a selected place and is determined by taking the mean value of the height of the tide, measured at hourly intervals over a long period, with respect to a fixed mark.

**Height** of any point is its vertical distance over or below the reference datum surface.

**Bench Mark (BM)** is the common expression used for any kind of monumented reference mark of any material of which the height above the selected datum (Mean sea level) has been determined by the process of levelling.

**Datum Stations (DS)** are such BMs from which the height above Mean Sea Level for Nepal is derived. They are usually established as special-monumentations and are situated close to the boundaries of Nepal.

**Base Points (BP)** are such BMs which are established in geologically stable areas to serve always for the control and maintenance of the

levelling network. They are also established as special monumentations.

**Permanent Bench Mark (PBM)**

is called such a monumentation which outlasts at least the next 50 years without changing its relative position neither vertically nor horizontally. Such monumentations are established with a single BM or as a triple or quadruple group.

**Junction Point (JP)**

is called the intersection of a line or the point where two lines cross together. At such points triple or quadruple groups of BMs are established.

**Section**

is a line between any two permanent Bench Marks.

**Sub-Section**

is the distance within a line of such length that can be levelled with 6 to 8 instrument set-ups. i. e. sub-section is the sub-division of a section.

**Line of collimation (Line of sight)**

is the line joining the intersection of the cross hairs to the optical centre of the objective.

**Back sight**

is the first reading taken from any instrument position to a known height.

**Fore sight** is the last reading taken from any instrument position.

**Change point (Turning point)** is a staff position to which both a Foresight and a Backsight are taken.

**14-days section (14-ds)** consists of such numbers of sections that its length can be levelled within 14 days, by one team forward and at the same time, by the other team backward.

Observations of 14-ds must always begin and end on a triple or quadruple group of PBMs.

**Link (Lk)** is called such a line which is levelled for purposes other than for the establishment of a coherent levelling - network. Links depart and end always at PBMs.

e. g. Heights of Trigonometrical Stations are derived from a PBM by means of a link.

**Temporary Bench Mark (TBM)**

is called such a monumentation (usually with an iron rivet) which must outlast at least during the observation period. TBMs serve to check the accuracy of the sections as one compares immediately after finishing a 14-days section, the two differences of

height between two consecutive TBMs resulting from Forward and Backward Levelling. (Lev. Form 1.16)

**Destroyed BM**

is called such a BM, which can no more be found *after* its height was observed and computed. BMs which are discovered to be unstable or lost *at the time of the first* determination of their height, are not considered to be destroyed ones as they can be re-established and then treated as if nothing had happened.

**Forward Levelling** is the levelling carried out in the direction of the increment of the Bench Mark numbering (i. e. the forward direction.)

**Backward Levelling**

is the levelling carried out in the direction of the descending order of Bench Mark numbering. (i. e. in opposite direction).

### 3. INTRODUCTION

Levelling is the process of the determination of the relative altitude of points on the earth's surface. The altitude or height of any point is its distance above the datum surface (Mean Sea Level i.e. Geoid). Various heights are in use, which are referred to their respective datum such as Spheroidal and Geoidel Height.

The principles of Precise or Geodetic Levelling are the same as ordinary spirit levelling. The only differences in the methods of levelling are the following conditions imposed on Precise Levelling to eliminate all the possible sources of errors.

#### (a) Instrument refinement:

The use of high precision instruments and more accurate invar staves.

#### (b) Field procedure refinement:

(i) The application of more rigorous observing techniques under restricted climatic and environmental conditions.

(ii) Refinement in booking and reduction of readings.

Thus, this process results in precise Geoidel Height if extreme care is taken during the process of observation and computation.

#### Applications of High Precision Levelling:

The H. P. Levelling is normally used for:-

- (a) Basic levelling network of a country.
- (b) Geodetic Research, Tectonic, Crustal Movement, Mean sea Level studies etc.

- (c) Frame work of vertical control for major engineering works, dams, bridges and canals etc.

#### Accuracy

Various survey agencies have their own specifications. The standard of accuracy required for Precise Levelling will vary according to the purpose for which the levelling is required. For the Survey Department/H. M. G., Nepal, to provide a field check, the following tolerances are specified as the maximum acceptable misclosure between Forward and Backward Levelling on any section of levelling.

#### ORDER

#### TOLERANCE

|                      |                  |                              |
|----------------------|------------------|------------------------------|
| High Precision/First | $1.1\sqrt{K}$ mm | where K is<br>distance in Km |
| Second               | $2.0\sqrt{K}$ mm |                              |
| Third                | $5.0\sqrt{K}$ mm |                              |

#### Network

In laying out primary levelling network in a country, the whole area to be levelled should be covered by the network. That network consists of traverses (lines) in the shape of circles, rectangles and triangles. The length of their sides will vary depending upon the nature of the terrain.

#### High Precision/First Order

When selecting lines of levelling for the H.P. Levelling, the following points should be borne in mind.

- (a) Lines should follow routes of communication but due consideration must be given to avoid the vibration caused by excessive traffic.

- (b) If at all possible, a large change of height should be avoided.
- (c) If it is feasible, the gap crossing (i.e. rivers etc.) should be avoided.

In running the H.P./First Order Levelling, the following requirements are fulfilled.

- (a) A High Precision/Geodetic Levels with parallel plate micrometers are used.
- (b) A pair of invar staves with circular bubbles are used.
- (c) Maximum line of sight = 25 m and  
Backsight Distance = Foresight Distance
- (d) Staff reading below 0.5m and above 2.5m are not to be used.
- (e) Observation sequence should be followed strictly.
- (f) Forward and Backward Levelling are to be carried out by different equipment and observers.
- (g) All the possible sources of errors must be minimised and subsequent corrections are to be applied.

First Order Levelling network provides the basis for all other height control in a country.

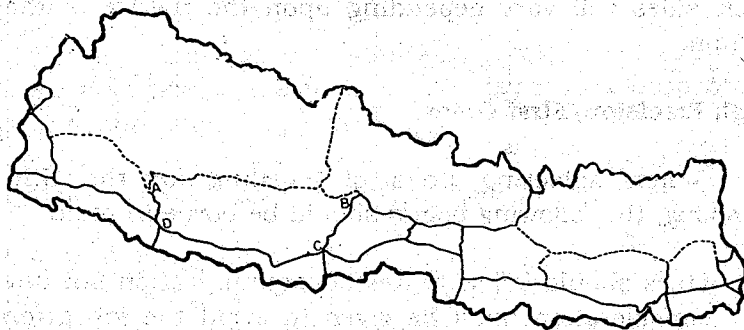


Fig. 1

## Second Order:

1. e. sub-division of First Order

- Less geodetic corrections than for the First Order Levelling.
- Shorter lines than First Order.
- Adjustment inside a First Order Loop.

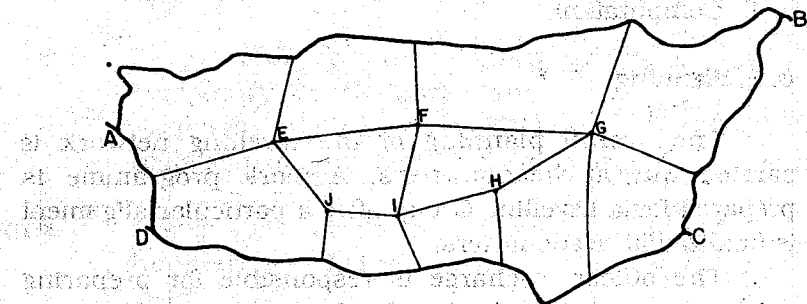


Fig. 2

## Third Order:

1. e. sub-division of Second Order

- No geodetic corrections (the earth's surface is considered as a plane).
- Adjustment inside the Second Order loop.

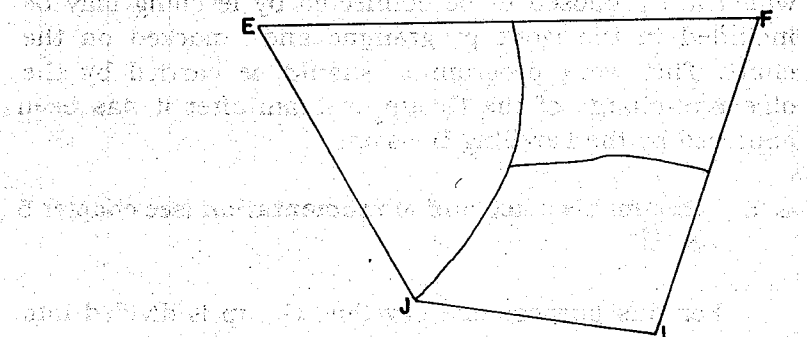


Fig. 3.

#### 4. WORKING PROCEDURE

The work of the Levelling Division can be divided into the following main headings.

4. 1. Planning
4. 2. Reconnaissance and Monumentation
4. 3. Observation
4. 4. Computation

##### 4. 1. Planning

The overall planning of the levelling network is carried out at headquarters. A work programme is prepared for a Levelling Group after a particular alignment is finalized at headquarters.

The officer in-charge is responsible for preparing detailed technical work plan for the execution of the task assigned to him. In addition to this, a chart in the scale of 1:50,000 or equivalent should be prepared, showing the proposed route of the levelling line together with all the existing Geodetic Levelling lines in the area with their alignment numbers and field season. The chart should also indicate the starting and closing Bench Marks and all the Permanent Bench Marks and Base Points proposed to be established in the new line. A list of Triangulation Stations which are proposed to be connected by levelling may be included in the work programme and marked on the chart. This work programme should be carried by the officer in-charge of the Group or Team after it has been approved by the Levelling Division.

##### 4. 2. Reconnaissance and Monumentation (see chapter 5 & 6)

For this purpose one Levelling Group is divided into four working teams as follows.

**Team 1:** Responsible for the monumentation of Permanent Bench Marks and their description cards.

**Team 2:** Responsible for the reconnaissance of the location of Temporary Bench Marks and their markings.

**Team 3:** Responsible for the drilling of holes for Temporary Bench Marks and plotting of the alignment and location of PBMs on a 1:50,000 scale map. (or any large scale map available)

**Team 4:** Responsible for the monumentation of TBMs numbering, sketching and measuring the bearing.

The quality of monumentation and description cards must be checked frequently by the Team Leader or the Group Leader.

The observers will be interchanged from one team to another team in rotation.

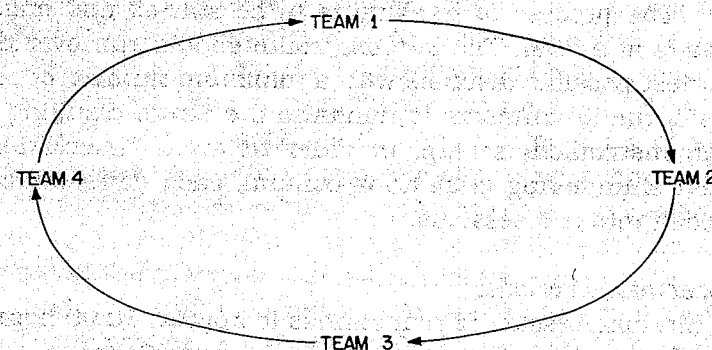


Fig. 4.

#### 4.3. Observation (for detail see chapter 9)

For this purpose the group is divided into two teams as follows.

**Team I:** ..... For Forward Levelling.

**Team II:** ..... For Backward Levelling.

**Note:-** In surveying, one measurement has no meaning unless there is a second measurement to support it. That is why double levelling is carried out (Forward and Backward).

Each team consists of:-

1. Assistant team leader (observer) ..... 1
2. Booker ..... 1
3. Staff supervisor ..... 1
4. Recce-supervisor ..... 1

##### 4.3.1. The work of a team:

The process of levelling is to be started and ended **always** at a PBM. The levelling route should run over the shortest possible distance with a minimum number of set-ups of the instruments to minimise the errors resulting at each instrument set-up. In order to avoid observations when shimmering is at its maximum, each day's work is divided into two sessions.

Observation Periods:

1. Morning session : From sunrise to approx. 10.00 hours.
2. Afternoon session : From 14.00 hours to Sunset.

The observation sub-team consists of:

Field personnel

- |                  |         |
|------------------|---------|
| Observer         | ..... 1 |
| Booker           | ..... 1 |
| Staff Supervisor | ..... 1 |

and the required number of chainmen for the mobilization of the survey equipments during the observation.

**Note:** Always the same staff man carries the same staff (so that the observer during a reading always knows where staff A or B is located). Take care to attach always the **Same thermometers** to the **same staff**.

Preparations before the observations:

**Immediately** after arriving at the place of work, the **observer** first **removes the instrument** from its Container and places it **in a shadow** which has freely circulating air for about **half an hour**, so that it can adapt to the observation conditions.

The **staff supervisor** goes quickly along the line to check whether the marks for the instrument and staff positions established by recce sub-team are still visible or not. At the same time he also checks the condition of BMs.

The **booker** supervises the quick unloading of all necessary materials. He checks the correct attachment of the thermometers. He is responsible for the **same staff** to be set up at the BM which was set up at that BM where levelling of the previous observation period ended.

The two staff men remove the material from the top rack of the vehicle. Then everybody moves with their own material to their respective positions as shown in fig. 5.

(The positions of staff A and B can be interchanged if the levelling of the previous observation ended with staff B.)

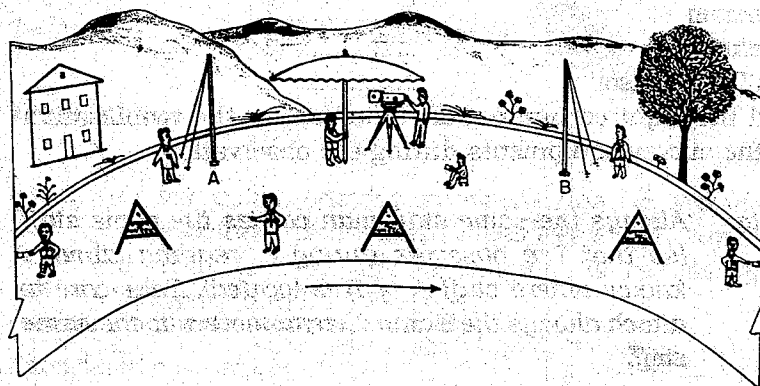


Fig. 5.

The observation is carried out by setting up the instrument and staves over the marks which were previously reconnoitred by the recce sub-team (see 5.4). The ground should be stable and more or less dry. As far as possible, **the distances** of the Backsight and the Foresight **must be equal** in order to minimize the influences of the collimation error, refraction, earth's curvature etc. (see chapter 10)

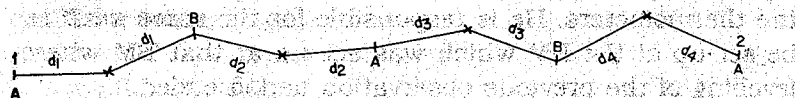


Fig. 6.

Referring to fig. 6 levelling from station 1 to station 2 is carried out by choosing several instrument positions.

The first instrument position is established at mid-point of the two staves A and B. For levelling of High Precision, staves with **double graduations** are only to be used. Thus the observation of the staves are taken as follows:

1. BACKSIGHT,  
left graduations, i.e. small numbers, (a) stadia hairs  
(b) centre hair
2. FORESIGHT,  
left graduations small numbers, (a) centre hair  
(b) stadia hairs
3. FORESIGHT,  
right graduation big numbers centre hair
4. BACKSIGHT,  
right graduation, big numbers centre hair

This rhythmic sequence is to be made with **equal intervals of time** between each reading. In particular the time lapse between 1 (b) and 2 (a) should be equal as near as possible to that between 3 and 4. At the next set-up the observation sequence is to be made as follows:

1. FORESIGHT, small numbers (a) stadia hairs  
(b) centre hair
2. BACKSIGHT, small numbers (a) centre hair  
(b) stadia hairs
3. BACKSIGHT, big numbers centre hair
4. FORESIGHT, big numbers centre hair

As previously the time lapse between the centre hair reading in small and big numbers must be kept equal.

Note:

1. The only difference in the two sequences given above is that once the backward staff is aimed first. And on

the next set-up the forward staff is aimed first, i. e. read always the same staff ('A') first.

2. Take the **observations as quickly as possible** one after another, but keeping in mind the time lapses mentioned above.
3. **Move as quickly as possible** from one instrument position to the next one and then start the observations as soon as possible. Otherwise the forward staff will have too much time lapse, during which its height may have changed without being noticed.
4. During changing from one set-up to the next, the foresight staff (change point) should be turned with **extreme care** so that it does not change its position.

The above mentioned sequences of observations and their speedy execution, eliminates most of the error introduced by the vertical movement of the staves.

Performing the observations in such a way, a minimum of 6 to 8 sub-sections can be levelled in one day (during the two observation periods).

#### 4.3.2. The work of the group:

Levelling of a line is to be carried out in two directions. One in the forward direction and the other in the opposite direction, for the investigation of gross error and for the increment of levelling accuracy.

Therefore the Backward Levelling must be done as soon as possible after the Forward Levelling or vice versa.

For this reason the whole line of levelling is divided into sections of such length, so that each section can be levelled within approximately 14 days.

#### 4.3.2.1. 14-days section (14-ds)

They always start and end at a **triple** or quadruple group of PBMs and **never** at a **single BM**.

As far as possible the camp is to be established in the middle of each 14-ds.

The number of sections and sub-sections in a 14-ds depend on the nature of the terrain. This number is to be decided upon before starting the levelling of that 14-ds. This is easily done in the following way:

In one day a minimum of 8 sub-sections can be levelled, i.e. in 14 days a minimum of 112 sub-sections can be levelled.

At the time of reconnaissance, a list of Bench Marks (Lev. Form 1.24) are prepared (see chapter 5). From the triple or quadruple group at the end of the last levelled 14-ds count in that list the 112 sub-sections in the direction of levelling. Now take the nearest triple group as the end of the new 14-ds.

The two teams level the 14-ds from opposite directions.

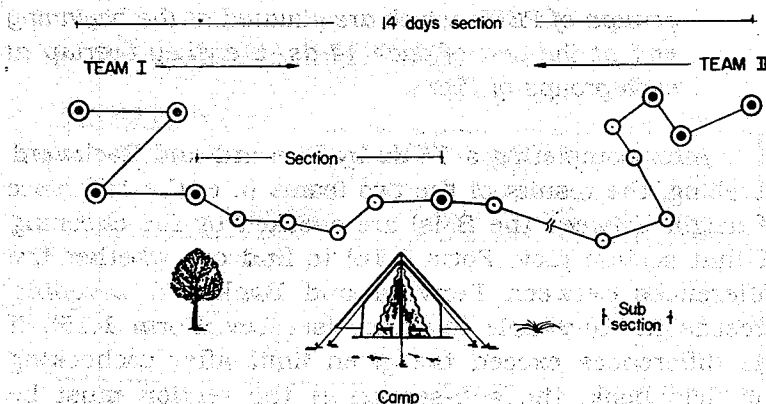


Fig. 7.

Note: The Forward and Backward Levellings are always carried out by a different observer and different equipments. The observations are carried out as shown in fig. 7.

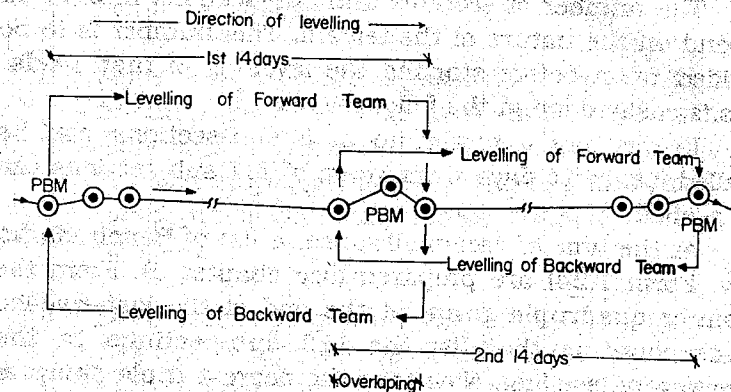


Fig 8.

Note: Each team levels twice over the **Whole triple groups** of PBMs which are situated at the beginning and at the end of each 14-ds, i. e. they overlap at such groups of PBMs.

After completing a 14-ds by Forward and Backward Levelling, the results of the two teams (i. e. the difference of height between the BMs) are checked by the checking of that section (Lev. Form 1.16) to find out whether the differences between Forward and Backward Levelling exceeds the allowable tolerance (see Lev. Form 1.15). If the differences exceed the given limit after rechecking the field book, the sub-section or the section must be repeated. Once everything is satisfactory and within its limit, the next 14-ds is observed.

#### 4.3.2.2. Check Levelling

If a line of levelling starts and closes on the Permanent Bench Marks, which had been previously established and levelled with a definite accuracy, the stability of such Bench Marks should be verified by check levelling. The check levelling is carried out at the group of 3 to 4 existing Permanent Bench Marks and the result should agree within the permissible limit as given in Appendix G.

The Bench Marks for check levelling should be very carefully selected so that these are adequately spaced to safe guard against the uniform local movement (sinking & upheaval).

This check levelling should also be carried out whenever the levelling line crosses the old one. During the check levelling, the description cards of the old Bench Marks should be updated if required.

The abstract of the check levelling is prepared in duplicate and a copy should be sent to head quarter for approval.

#### ABSTRACT OF CHECK LEVELLING

| ALIGNMENT NO.               |      |           |      |            |                               |                      |            |     |         |
|-----------------------------|------|-----------|------|------------|-------------------------------|----------------------|------------|-----|---------|
| BENCH MARK OF OLD LEVELLING |      |           |      | DIST. (Km) | OBSERVED DIFFERENCE OF HEIGHT |                      |            | TOL | REMARKS |
| NO.                         | TYPE | CONDITION | DATE |            | OLD LEVELLING (Cm)            | CHECK LEVELLING (Cm) | DIFFERENCE |     |         |
|                             |      |           |      |            |                               |                      |            |     |         |
|                             |      |           |      |            |                               |                      |            |     |         |

#### 4.4. Computations

This instruction book deals mainly with field work, so the computation aspect is not described here. For field plotting and checking see chapter 13.

#### 5. RECONNAISSANCE

Reconnaissance is the basic and most important step in all surveying work. The importance of this step is sometimes overlooked by the observer, but the utility of a thorough reconnaissance cannot be overemphasised, the time spent being amply repaid in the greater ease with which the levelling can be executed.

Generally reconnaissance is done in three stages:

- (a) choosing the best suitable alignment on the basis of the available topographical and geological maps of the area. This stage is very important and it is advisable to consult the geologist for this purpose. Normally this stage of reconnaissance is carried out at headquarters.
- (b) The determination of the **exact location of each PBM and TBM** in the field is of vital importance. Especially for DS and BP it is advisable to have a geologist on the site.
- (c) The determination of the exact locations of the instrument and the staves.

This third step is to be carried out during observations, but the first two stages are to be carried out **before** (and not during) the **monumentation** work of the concerned line.

##### 5.1. Alignments

Suppose the levelling between any two given points A and B could be done by different alignments but only one is needed, in such cases the suitable alignment can be selected keeping in view the following conditions. The

marks are allocated in order of priority, thus the alignment which scores higher marks is the best suitable alignment.

| Conditions                                                                      | Marks for |     |     |
|---------------------------------------------------------------------------------|-----------|-----|-----|
|                                                                                 | 1st       | 2nd | 3rd |
| order                                                                           |           |     |     |
| - The alignment proceeds through geologically stable areas                      | 9         | 9   | 3   |
| - Instruments and staves to be set on stable ground                             | 8         | 8   | 9   |
| - Traffic does not affect the stability of staves and instruments by vibration. | 7         | 7   | 7   |
| - Good rock for monumentation is available all along the alignment.             | 6         | 6   | 4   |
| - A minimum of BMs proposed to be monumented by pillars                         | 5         | 4   | 8   |
| - The alignment can be levelled by a minimum of instrument set-ups.*            | 4         | 3   | 5   |
| - The road construction is stable                                               | 3         | 2   | 2   |
| - The alignment proceeds through an area where BMs are needed.                  | 2         | 5   | 6   |
| - The levelling team can move along the alignment by vehicle.                   | 1         | 1   | 1   |

\* The number of necessary instrument set-ups for the levelling of an alignment depends on the length L (in km) and the gradient g (in %) of the alignment and on the total of absolute differences of heights  $\sum |\Delta h|$

$$\sum |\Delta h| = |\Delta h_1| + |\Delta h_2| + |\Delta h_3| + \dots + |\Delta h_n|$$

$$g = \frac{\Delta h \text{ (in metres)}}{10 \text{ XL (in km)}} \% \dots\dots\dots (5.1.a)$$

If  $g \leq 4\%$   $n_1 = 20 \times L \dots\dots\dots (5.1. b.)$

$\therefore$  max. line of sight = 25 m. Thus 1 km needs 20 set-ups.

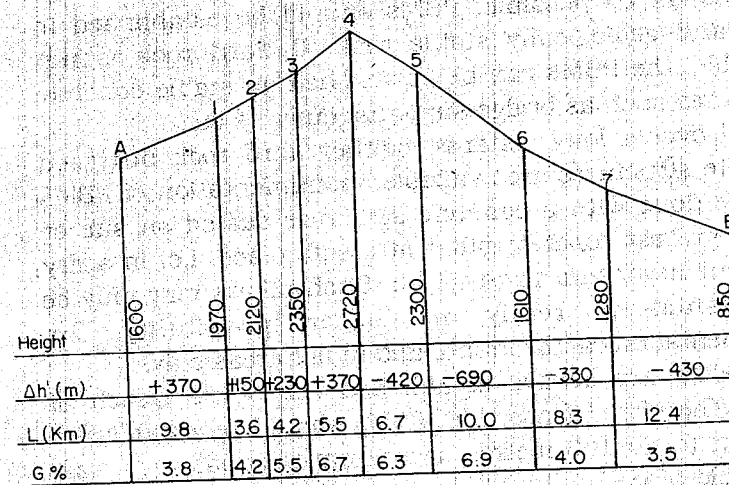
If  $g \geq 4\%$   $n_2 = \Delta h / 2 \dots\dots\dots (5.1.c)$

$\therefore$  In one set-up max. of 2 m  $\Delta h$  can be levelled.

So the number of necessary set-ups to level an alignment becomes

$$n = n_1 + n_2 \dots\dots\dots (5.1.d)$$

**Example:** (Profile of the alignment not to scale.) The height and distances are to be taken from a suitable map.



From A to 1 and from 6 to B  $g \leq 4\%$ .

Therefore  $n_1 = (9.8+8.3+12.4) \times 20 = 610$

From 1 to 6 g > 4%

Therefore  $\sum \Delta h$

$$= |+150| + |+230| + |+370| + |-420| + |-690|$$

$$= 150 + 230 + 370 + 420 + 690$$

$$= 1860$$

$$n_2 = 1860 / 2 = 930$$

$$n = n_1 + n_2 = 1540.$$

Thus a levelling traverse from A to B requires at least 1540 set-ups. From this, one can calculate the time required for the completion of the observation.

### 5.2. Permanent Bench Marks

The approximate distance between two PBMs is said to be 2 km but it can be longer or shorter (1.5 km to 2.5 km) depending on the existence of suitable places for monumentations.

As far as possible, PBMs should be established in hard and geologically stable rock. If hard rock is not available, the PBMs can be established on stable concrete structures such as bridge supports, culverts etc.

If over a long distance neither hard rock nor good concrete structures are available, monumentation of PBMs may be done with a concrete structure casted on site or with a precast concrete pillar at a safe place, i.e. in a dry field (but never in a paddy-field). Such pillars may only be established if really no other possibility for monumentation can be found, and in flat areas only.

PBMs are always established out of the reach of traffic. Therefore, if possible, PBMs should be established at about 50 to 100 metres away from the road on a safe accessible place.

PBMs may not be established near big trees, as growing roots may disturb their stability. Always keep in

mind, that it must be possible to set up the staff on the BM, i.e. the area 2 metres around and at least 3.20 metres above the BM must be clear.

If PBMs are established as a group, the BMs are to be established close together, so that the difference of height between two PBMs (or between the TBM and PBM) can be determined with two to four instrument set-ups only. During reconnaissance a sketch has to be drawn which shows the location of reconnoitred BMs along the alignment and the proposed location of the PBMs are marked with enamel paint.

### 5.3. Temporary Bench Marks

The reconnaissance of their locations is carried out after the reconnaissance of PBMs.

TBMs are established always where PBMs are more than one length of sight from the road. If there is a group of PBMs the TBM is established in the centre of that group very close to the road.

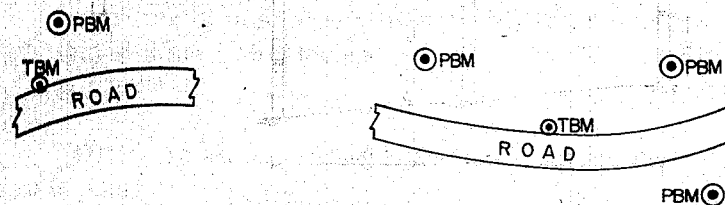


Fig. 9.

TBMs are established so close together along the alignment, that the difference of height between two consecutive TBMs (or between a PBM, which is situated very close to the road and a TBM respectively) can be levelled with 6 to 8 instrument set-ups, i.e. within approximately one hour.

It is obvious, that in flat areas the intervals between TBMs are longer than in the slope areas. The length of such intervals is determined by the following procedure.

Two sticks, one for the instrument height (1.50m) and another for the staff (3.20m), are prepared. The staff stick is marked at two places, i.e. 0.50 and 2.50m above ground level. Two persons are required for this job, one as observer and the other as staff-man.

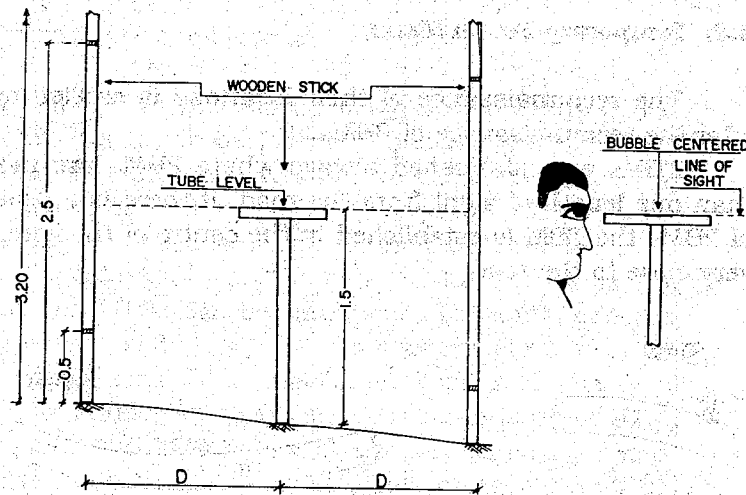


Fig. 10.

1. The staff-man with his stick stands at the place of a PBM or a previously reconnoitred TBM.
2. The observer with his stick and tube-level paces off the distance (maximum 25m) from the staff position

and locates a suitable position for the instrument, so that his line of sight, i.e. the **horizontal line of the tube-level** over the 1.50m stick is between the two marks of the staff-stick. If it is OK, he signals to the staff-man "O.K.", if not, he changes his position accordingly.

3. The staff-man after receiving the signal, **paces the direct** distance from his staff position to the instrument (i.e. D in the figure above) and **paces exactly the same distance** from the instrument towards the new staff position.
4. After the staff is set in the new position, the observer checks whether his line of sight lies between the two marks of the staff or not. If not, the positions of the staff and of the instrument are changed accordingly.  
When it is O.K., the observer counts the number of instrument set-ups and changes his instrument position to the new position.
5. After executing this procedure six times (i.e. six instrument positions are set out), one reaches the place where the location of the next TBM is to be looked for.

TBMs are always established very close to the road on a safe place.

They are established in rock or good concrete structures whichever is available on the ground.

Check always that it is possible to set up the staves at the reconnoitred places. The reconnoitred situations of TBMs are to be marked with paint.

During reconnaissance a sketch has to be drawn which shows the approximate location of reconnoitred TBMs along the alignment.

#### 5.4. Reconnaissance of the positions of staves and instrument

This work is of great importance for the execution of the High Precision Levelling and therefore it must be carried out with great care. This reconnaissance must be carried out at least one day prior to the observations. The aim of this work is to expedite the observations so that a minimum time is taken in connecting one Bench Mark to another. Thus the observations can be run smoothly.

For this reason the reconnaissance sub-team has to perform the following tasks.

1. Uncover and check the conditions of all BMs.
2. Remonumentation of destroyed BMs if any, well in advance before the observations.
3. Reconnaissance of the minimum of necessary staff and instrument positions between two consecutive BMs making sure that there are always an even number of set-ups.
4. Foresight and Backsight for each set-up must be of equal length.
5. The positions for the staves and the instrument must be chosen on stable ground, but if possible not on asphalt.

Best positions are on

- fine gravel or sand.
- concrete surface.
- big, embedded stones.

6. If there is no other way than to proceed over grass-land, the grass-layer is dug out and the instrument/staff is placed over it.
7. All reconnoitred positions are marked in such a way that the marking must last until the time of observation. It is done either by pick or by paint.

- Material required
- 30m nylon rope.
  - 1.5m long wooden stick (instrument).
  - 3.20 m long wooden stick (staff) with one mark 0.50m and another mark 2.50m from the bottom of the stick.
  - Tube level.
  - Marking paint and brush.
  - Pick.

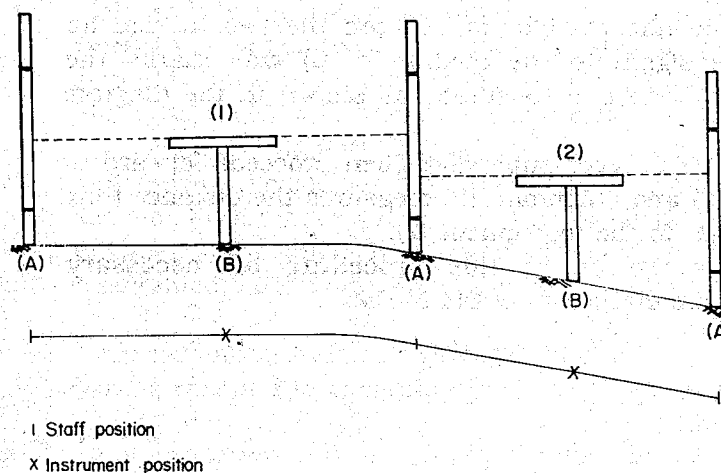


Fig. 11.

#### Operation:

One chainman (A) with the staff and one end of the rope erects his "staff" over the TBM or PBM, the reconnaissance supervisor with instrument stick and tube level locates the suitable position for the instrument and checks the staff (line of sight must be between the two marks of the staff). When he gives the signal, the other chainman (B) measures the distance of the staff-

instrument by marking the other end of the rope, the chainman (A) with the staff and one end of the rod proceeds forward till his rope is tight and chooses a suitable (i.e. stable) place for the staff.

*Note: The selection of stable staff positions as well as the equality of Backsight and Foresight are of utmost importance.*

The recce-supervisor advises the chainman in selecting the stable staff positions. When the position is chosen, he checks the staff reading from the instrument position (1).

If the line of sight is between the two marks, he gives the signal to the chainman (B) who marks the instrument and staff positions as shown in the diagram above.

The recce-supervisor then proceeds forward to position (2) and chainman (B) measures the distance from the staff (A) to the instrument (2).

By repetition of this procedure all necessary positions are marked from BM to BM.

## 6. MONUMENTATION

Monumentation is the process, by which the levelling station markers called "Bench Marks" are established in a reconnoitred place (see. reconnaissance). The Bench Marks thus established should remain intact for ever, thus extreme care must be taken while carrying out this monumentation. For this reason, the Bench Marks are established in a safe and protected area and in the case of a Base Point & Datum Station, a protection wall or railing should be constructed along with a marble/brass plate indicating its name and year of monumentation. During monumentation one should explain to spectators in a friendly and polite way about the importance of Bench Marks so that it will help for their protection.

All monumentation should be carried out just before and during the monsoon season, if possible at least one season in advance. However the following time gap between monumentation and observation must be maintained for the stability of the Bench Marks.

Monumentation in rock: at least two weeks.

Monumentation in concrete structures: at least two months.

Monumentation with concrete pillars: at least six months.

Monumentation of Datum Stations or Base Points in a concrete foundation: at least one year.

### 6.1. Kinds of marks

There are various kinds of marks used for the Bench Marks.

1. Brass mark
2. Steel mark

### 3. Iron Rivet.

Normally marks 1 & 2 are used for Permanent Bench Marks but mark 2 i.e. Steel Mark is preferred. Mark 3 is used as a Temporary Bench Mark.

## 6.2. Types of Marks

Previously the Levelling Division started with only two types of Permanent Bench Marks namely a brass mark on the rock/concrete structure and on the 70cm precast concrete pillars. But from the experience gained so far, considering the stability of Bench Marks over the years, the various types of marks have been classified so as to be applicable for the various natures of the terrain.

Thus during monumentation one or a group of these types are used as permanent marks.

- Type A Permanent Bench Mark on bed rock
- Type B Permanent Bench Marks on heavy Concrete foundation
- Type C Permanent Bench Marks on Concrete/Pipe foundation
- Type D Permanent Bench Marks on exposed Concrete foundation
- Type E Permanent Bench Marks on rock
- Type F Permanent Bench Marks on Concrete structure
- Type G Permanent Bench Marks on precast Concrete pillars.
- Type H Bench Marks established horizontally or Temporary Bench Marks.

See appendix A for details of these types.

Monumentation of these different types of Bench Marks can be classified into 3 categories.

- (a) Bench Marks on rock surfaces/concrete structures/stable foundations. (i.e. Type A, E, F & H).
- (b) Bench Marks on concrete foundations cast on site. (i.e. Type B, C & D)
- (c) Bench Marks on precast concrete pillars. (i.e. Type G)

### Layout of monumentation

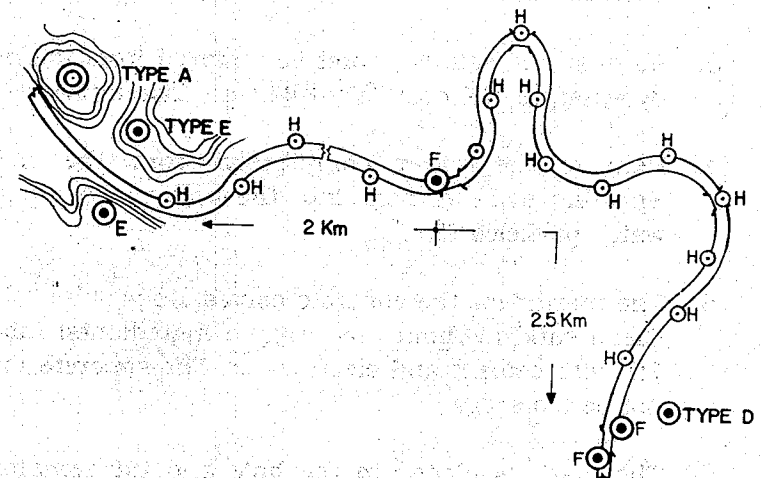


Fig. 12.

PBM to PBM is termed as a section and its average distance is about 2 km.

PBM to TBM or TBM to TBM is termed as a sub-section and its average distance is about 300-400m.

### 6.3. Bench Marks on rock surfaces/concrete structures/stable foundations

1. Prepare the surface of the rock/concrete by cutting a cubical hole of (15 x 15 x 10)cm.
2. In the centre of that area, drill a vertical hole. This hole should be as narrow as possible, i.e. of the same diameter as that of the kind of mark being used. It must be at a depth that only the head of the mark (1 cm of the steel/brass-mark or rivet) is exposed out of the hole, i.e. the base of the marks are supported by rock (or concrete in concrete structures) but not by filled in cement.
3. All dust and soil etc. must be removed from the hole by water i.e. the hole is washed out thoroughly.
4. Remove the water completely from the hole. Sprinkle some cement into the hole to dry up the water particles.
5. The mortar for the concrete consists of:- Clean sand (without any organic ingredients) mixed 1:1 with cement and clean water. The concrete must not be too soggy.
6. The mark is placed in the hole and the remaining cavity is fully filled in with concrete. Make sure that the mark rests on the base of the hole by slightly tapping the mark down. The compaction of the concrete is carried out with a thin stick or rod. Be careful not to destroy the smooth spherical surface of the mark.

7. Make sure, that the staff is supported by the Bench Mark and not by surrounding rock/concrete.
8. Cover the BM with damp soil or something similar to prevent it from drying up quickly.

### Example of good monumentation

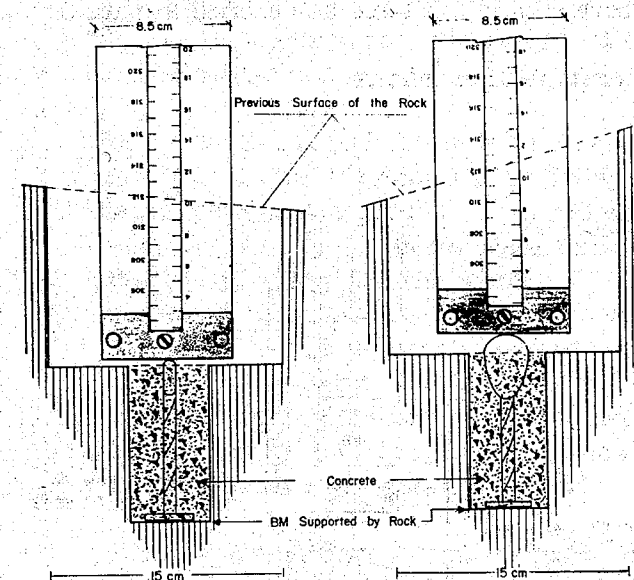


Fig. 13

9. After the monumentation of the mark, the important stage is its protection. The surface of the Bench Mark is covered by a Bench Mark Cover or by some other arrangement so that the surface remains intact. The hole is then filled in with sand or soil and

the upper surface of the rock/concrete structure is plastered to its original level.

10. A good description card will help to relocate the position of the mark and thus by breaking the plaster surface, the Bench Mark can be exposed for the observation.
11. One should make sure that the Bench Mark exposed during observation or found to be exposed, must be covered and plastered as described in para 9.

A bad example is as follows

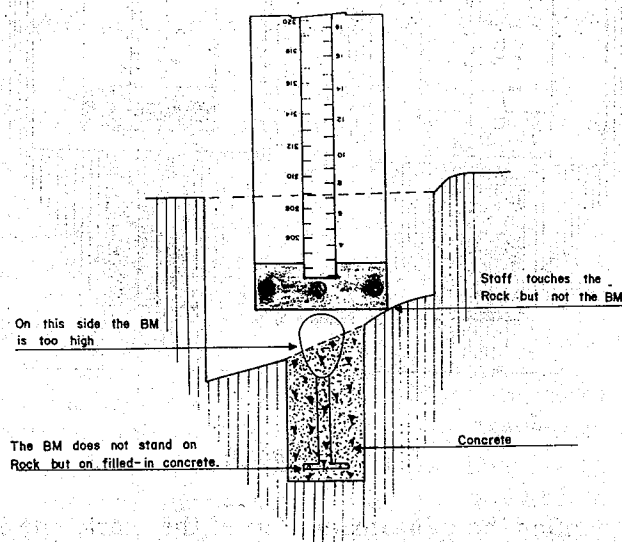


Fig. 14

#### 6.4. Bench Mark on a concrete foundation cast on site

For the monementation of these types of Bench Mark, the excavation of the earthwork and the concrete foundations are carried out as per the specification of those types of marks specified in appendix A. Para 9 to 11 of 6.3 are also applicable to this.

#### 6.5. Bench Mark on precast concrete pillars

1. Dig a hole not larger than 80 x 80 cm. square. it must be deep enough so that the surface of the pillar is 30 cm below ground.
2. Level the bottom of the hole and ram it with some gravel or pieces of brick in such a way that the pillar is supported by a compact rammed surface. i.e. the pillar must not stand in the loose soil.
3. Place the pillar in the centre of the base and surround it completely with stones. Press these stones by hammering each against the another and against the pillar, so that the pillar does not move laterally. Cover the stones by an approximately 10cm thick layer of soil. Press the soil down thoroughly. Ensure that the pillar stands vertically.
4. Place in again 10cm soil and press it down. Thus the soil is compacted in the hole by slowly pressing down several times.
5. Repeat step 4 until approximately the upper 15cm of the pillar is visible.
6. Surround the pillar again with stones and proceed as in para 3.

7. After finishing the dimension, description of that pillar, the BM is capped by a cover and the pillar is covered with a layer of approximately 10cm soil, without any stones. Press it down gently, but not too much. Then put a flat stone in the centre of the hole (i.e. above the mark) to protect the mark. Top up the hole fully with soil up to ground level and press it down firmly.

#### Example

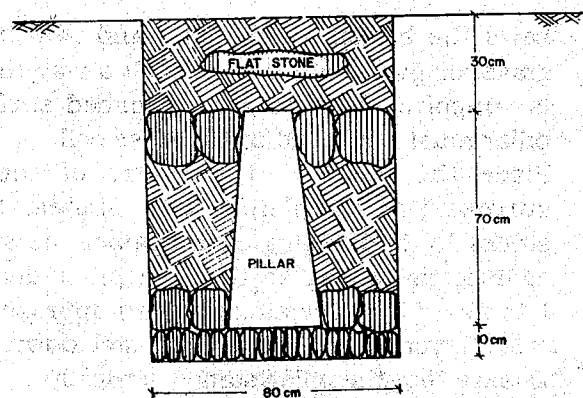


Fig. 15

#### 6.6. Temporary Bench Mark (Iron Rivet) in pipe

This type of monumentation for a TBM is only carried out if there is absolutely no rock or concrete structure in the surrounding area.

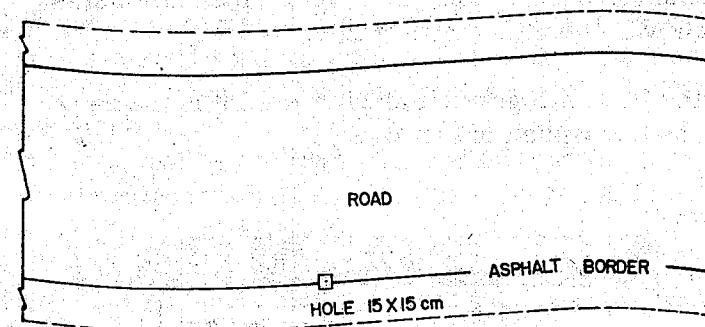


Fig. 16

1. Dig a hole at about 15cm square at the edge of the asphalt road.
2. After removing the asphalt layer, a long chisel is driven into the ground in the centre of the hole.
3. Judging the nature of the ground, choose either a 30 cm or .60cm long pipe (diameter approximately 2.5cm).
4. The chisel is removed from the hole and the hole is partly filled with small gravel. Then the pipe is driven into the hole by hammering down, using a piece of wood held over the pipe to prevent damage till it is approximately 10cm below the surface of the asphalt.
5. The pipe is then filled with gravel and it is compacted thoroughly leaving space for the iron rivet only.

6. Fit the iron rivet into the pipe and take the dimension description, i.e., some measurements which will indentify the rivet later on.
7. Fill the hole with gravel and sand until it is the same level as the asphalt of the road.

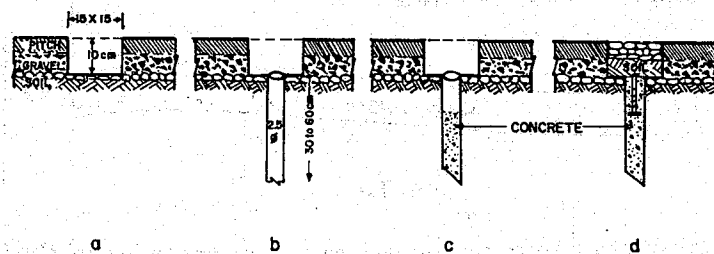


Fig. 17

## 7. NUMBERING OF BENCH MARKS

Correct numbering of each Bench Mark is as essential as a correct reading of the staves: Expecially, one number is allocated to one single BM only, i.e. each BM has a separete number.

Therefore, during the time of numbering in the field, a list of allocated numbers is to be sent monthly to headquarters for registration.

### 7.1 Fundamental Bench Marks, Datum Stations and Base Points

Such Bench Marks are characterized by the name of the place where they are situated (i.e. name of the respective town, village, temple, etc.). Such stations are established by a single BM only, to avoid confusion. However, to be able to control the stability of such BMs (and eventually to restore them) at least two other types of BMs are established near them. But these additional BMs are either treated as Base Points or as first order PBMs within that line.

### 7.2. First Order Levelling (High Precision Levelling)

While establishing a levelling network in a country, levelling is carried out from one known reference point to another unknown point following a line. Thus the numbering of Bench Marks is adopted in terms of alignment or line number.

#### 7.2.1. Alignment Number

The First Order alignment/line number is designated by 3 digits and the starting digit is number 1 indicating

the order of work. This enables one to number 100 different alignments i.e. 100 to 199. After assigning the alignment number of the first order lines a list is prepared and issued.

| e.g. | Alignment                     | Number |
|------|-------------------------------|--------|
|      | Birganj - Kathmandu           | 100    |
|      | Naubise - Mugling - Pokhara   | 101    |
|      | Pokhara - Butwal - Bhairahawa | 102    |
|      | Butwal - Hetauda              | 103    |
|      | Pathalaitya - Kankadbhitta    | 104    |

### 7.2.2 Numbering of PBMs

Three digits are assigned (i.e. from 001 to 999) for the numbering of Permanent Bench Marks. The numbering is carried out continuously for one alignment in the direction the levelling proceeds (Forward Levelling). This makes it possible to number maximum of 999 Bench Marks in one alignment.

Thus the final number of a First Order PBMs consists of the three digits for the alignment and three digits for the BMs separated by a hyphen.

|           |                                                                |
|-----------|----------------------------------------------------------------|
| e.g.      |                                                                |
| 103 - 050 | 1 - representing order of work                                 |
|           | 103 - alignment/line number with 3 digits                      |
|           | 050 - Bench Mark number in the direction of Forward Levelling. |

101 - 005 denotes the Permanent Bench Mark as no 5 of the First Order Line between Naubise and Mungling (Al. No. 101).

Since this system of numbering alignments restricts up to 100 different alignments only, one has to be careful

in numbering the alignments in such a way that the numbered alignment should be as long as possible.

Example:

Instead of numbering such a figure like this

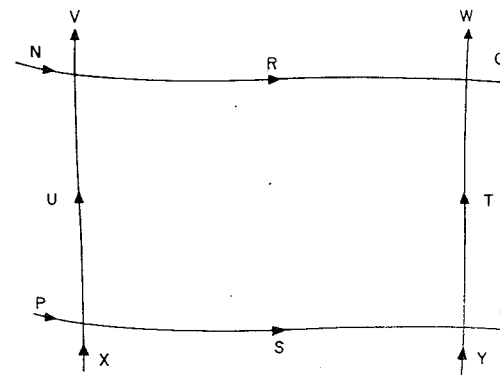


Fig. 18.

Number it like this

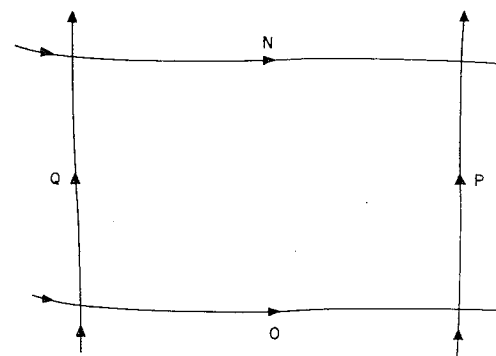


Fig. 19.

First Order PBMs are established alternately as single points and as groups of three to four, the distance between a group and a single being approximately 2km.

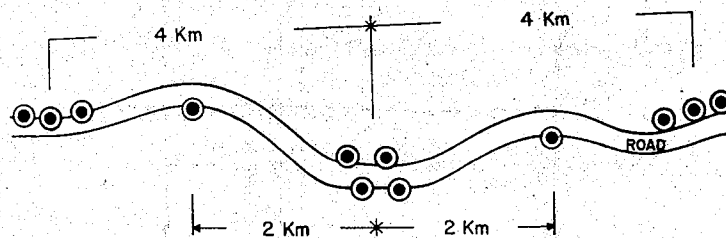


Fig. 20

From the sketch above it can be seen that over a distance of 4 km an average 4.5 PBMs are required for a distance of 1 km. an average (4.5/4) are required For a distance of L km the total number T of PBMs required is

$$T = (4.5/4) \times L$$

$$= (9/8)L \dots\dots\dots (7.2.2)$$

Thus, for a distance of 200 km, the total number of First Order PBMs required are:-

$$T = 200 \times 9/8 = 225$$

#### 7.2.2.1. Numbering of destroyed or additional PBMs

If some PBMs are found destroyed prior to the observation, such PBMs can be reestablished close to the previous PBM positions. These PBMs will carry the same previous numbers but the old description cards should be superseded by the new ones.

If some PBMs are found destroyed or missing after the observation, the number of such PBMs are cancelled from the list and a remark "Destroyed" is noted on its description card.

It may happen that the destroyed PBM positions need to be replaced by a new PBM or some additional PBMs need to be established between the existing old PBMs. In such cases the numbering of the new PBMs will be carried out as follows.

Example:-

Alignment No. 103

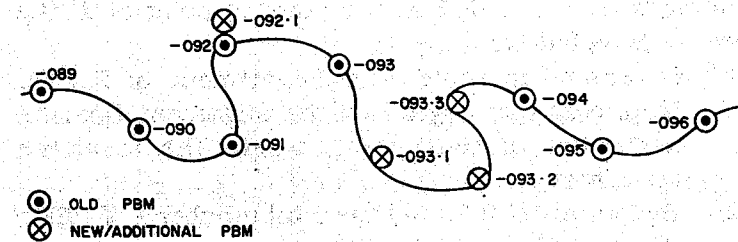


Fig. 21

Let us assume that one PBM needs to be established instead of 103-092, which was destroyed after observation. The new PBM number will carry the same old number and one extra digit after the decimal point.

i.e. 103 - 092.1

It may be required to have some additional PBMs between the existing old PBMs say between 103 - 093 and 103-094, for the densification of Bench Marks. Such new PBM numbers will carry the preceding PBM number and one extra digit after the decimal point. This extra digit will have the number 1 to 9 depending on how many additional PBMs are established.

i.e. 103 - 093.1  
 103 - 093.2  
 103 - 093.3

This system enables one to establish 9 additional PBMs between the old PBMs.

i.e. in 2 km (length between the PBMs)

There is a possibility of establishing an additional 9 PBMs at every 200 m interval.

### 7.2.2.2 Intermediate Numbering

It sometimes happens that levelling of an alignment does not start at the very beginning of that alignment. To have later on a continuous numbering for each alignment beginning with ..... 001 at the starting point of it, the procedure is as follows:

1. Determine in some way (Department of Roads, Map, etc.) the length  $L$  of the alignment from its beginning to that point, where the levelling actually starts.
2. By formula (7.2.2) find the total number  $T$  of PBMs required for this part of the alignment.
3. The starting PBM number will be  $N = (T+1)$

Thus the numbering is carried out continuously from  $N$  onwards.

In doing so there may be few spare numbers left over, which can be noted and recorded.

The lines connecting the junction points  $C, E, D, B$  are observed. To complete the loop  $CEDBC$ , it was decided to observe the line  $BC$  of the alignment 119 before the line  $AB$  of that alignment.  $B$  is already numbered as 118 - 237.

$AB = L = 160 \text{ km}$

Total number of PBMs ( $T$ ) required for the line  $AB$

$$T = (9/8) \times 160 = 180.$$

Therefore the starting PBM number for the line  $BC = 119 - 181$

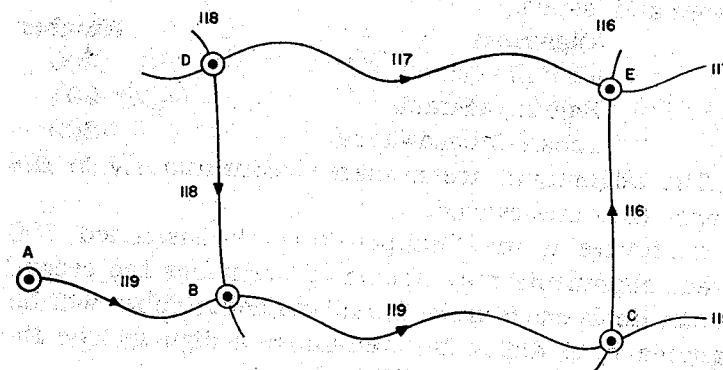


Fig. 22

### 7.2.2.3 Maximum length of an alignment

The maximum numbers of PBMs in one alignment is

$$T_{\max} = 999$$

$$T = (9/8) \times L \text{ ..... (from 7.2.2)}$$

$$L_{\max} = (8/9) T_{\max}$$

$$= (8/9) \times 999 \text{ km} = 888 \text{ km}$$

Thus while planning an alignment it can be no longer than 880 km.

## 7.3 Second Order Levelling

As in the First Order, the numbering of Bench Marks is adopted in terms of alignment/line number.

### 7.3.1 Alignment Number

The Second Order alignment/line number, is initially designated by 3 digits and the starting digit is number 2

indicating the order of work. This enables one to number 100 different alignments, i.e. 200 to 299. After assigning the alignment number of the Second Order lines, a list is prepared and issued.

| e.g. | Alignment           | Number |
|------|---------------------|--------|
|      | Charali - Illam     | 200    |
|      | Rajbiraj - Siraha   | 201    |
|      | Lahan - Bhagawanpur | 202    |

The alignments are numbered continuously in the sequence they are levelled.

In future it may happen that the assigned 100 different alignments may not be sufficient for the second order. **In such cases only** the alignment number will be designated by 4 digits, but the starting digit will be the same (i.e. 2 indicating the order of work).

This will enable one to number an extra 1000 alignments

|      |                          |          |               |
|------|--------------------------|----------|---------------|
| e.g. | Last alignment number in | 3 digits | 299           |
|      | first                    | "        | 4 digits 2000 |
|      | last                     | "        | 4 digits 2999 |

### 7.3.2 Numbering of PBMs

The Permanent Bench Marks are numbered with three digits ( from 001 to 999) continuously for each separate alignment in the direction the levelling proceeds (Forward Levelling).

Thus the final number of Second Order PBMs consists of three digits for alignment and for PBMs as well.

|      |           |   |                                                            |
|------|-----------|---|------------------------------------------------------------|
| e.g. | 201 - 098 | 2 | - Representing order of work                               |
|      | 201       |   | Alignment/line number with 3 digits.                       |
|      | 098       |   | - Bench Mark number in the direction of Forward Levelling. |

201 - 098 - denotes the PBM no 98 of the Second Order alignment/line between Rajbiraj and Siraha (Al. No. 201).

Second Order Bench Marks are established alternately as a group of three (Triple) and as a single, the distance between a triple and a single is approximately 2 km.

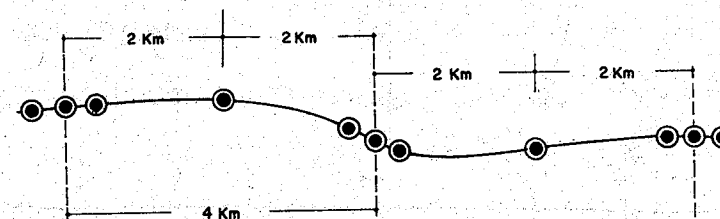


Fig. 23

From the sketch above it is clearly seen that the total number of Bench Marks required (T) is as in the first order.

e.g.  $T = (9/8) L$  (see 7.2.2)

#### 7.3.2.1 Numbering of destroyed or additional PBMs

The process is the same as in the First Order. (see 7.2.2.1)

Example:- Numbering of additional PBMs between the old PBMs.

218 - 055 and 218 - 056 are as follows.

218 - 055.1

218 - 055.2

218 - 055.3

### 7.3.2.2 Intermediate numbering

The process is the same as in the First Order. (see 7.2.2.2.)

Example:-

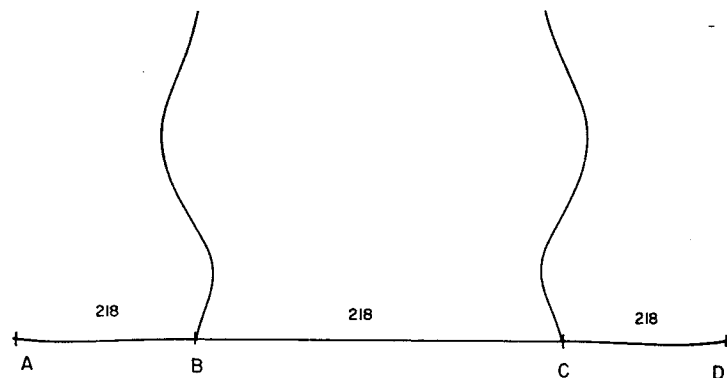


Fig. 24

Alignment No. of ABCD = 218

Length AB =  $L = 168$  km

Total number of PBMs required =  $T = (9/8) L = 189$

Therefore, the starting PBM number for the section BC of the alignment =  $N = T + 1$

Thus the numbering of the PBMs in the section BC is carried out continuously from 190 onwards.

### 7.4 Third Order Levelling

In this case as well the numbering of Bench Marks is adopted in terms of alignment/line number.

### 7.4.1 Alignment

The Third Order alignment/line number is initially designated by 3 digits and the starting digit is number 3, indicating the order of work. This enables one to number **100 different** alignments i.e. 300 to 399.

As in the case of the Second Order, the assigned 100 different alignments may not be sufficient. **In such cases only** the alignment number will be designated by 4 digits but the starting digit will be the same (3 indicating the order of work). This will enable one to number an extra 1000 alignments.

|      |                          |            |      |
|------|--------------------------|------------|------|
| i.e. | last alignment number in | 3 digits = | 399  |
|      | first       "       "    | 4 digits = | 3000 |
|      | last       "       "     | 4 digits = | 3999 |

### 7.4.2 Numbering of PBMs and additional PBMs

The system and procedure is exactly the same as for the second order PBMs. (see 7.3.2 & 7.3.2.1)

Example:- 303 - 054 denotes the PBM No. 54 of the Third order alignment No. 303.

### 7.5 Temporary Bench Marks

The system and procedure to number them is exactly the same for First, Second and Third order levelling.

Temporary Bench Marks are established between two PBMs always after 6 to 8 instrument set-ups. The TBM numbers are given as fractional numbers, i.e.  $125/1$ ,  $125/2$ . Where the numerator denotes the Bench Mark numbers from which the line started and the denominator

denotes the serial number of the Temporary Bench Marks in that line.



Fig. 25

It may often happen that a section does not start or end at a PBM (see the diagram), but at a TBM which is situated somewhere in a central position on the road between the triple group of the PBMs.

It is recommended to number such a central TBM with the highest PBM number in that group of PBMs. (see 121 - 222/1)

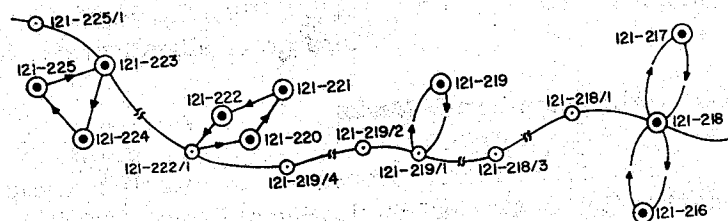


Fig. 26

If PBMs lie in the line of levelling, the following TBMs will be numbered with the highest PBM number in that group. (see 121 - 225/1).

## 7.6. Numbering of Staff Positions

If the staff is set up on a PBM, TBM or Trigonometrical Station (i.e. on any mounumented point) it gets the number of that particular BM or TS.

If it is set up on the ground plate it will be numbered with the number of the previous PBM or TBM number and in addition with small letters continuously.

Example:-

|          |               |              |                |
|----------|---------------|--------------|----------------|
| Staff on | PBMs          | 121 - 218    | in short - 218 |
| "        | Ground plates | 121 - 218/a  | -218/a         |
| "        | "             | 121 - 218/b  | -218/b         |
| "        | "             | 121 - 218/c  | -218/c         |
|          |               | etc.         |                |
| "        | TBMs          | 121 - 218/1  | -218/1         |
| "        | Ground plates | 121 - 218/1a | -218/1a        |
| "        | "             | 121 - 218/1b | /1b            |
|          |               | etc.         |                |
| "        | TBMs          | 121 - 218/5  | -218/5         |
| "        | Ground plates | 121 - 218/5a | /5a            |
|          |               | etc.         |                |

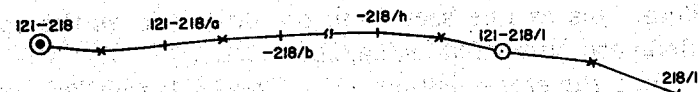


Fig. 27

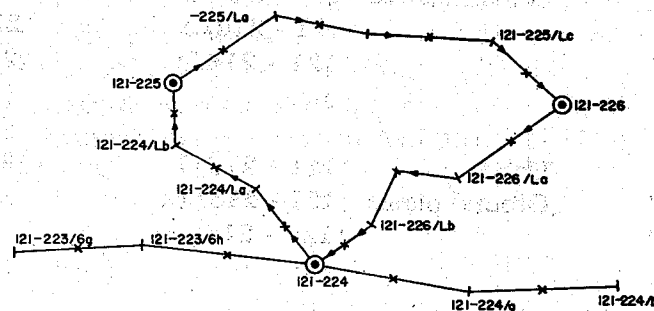
## 7.7 Numbering of Staff Positions and TBMs in Links

For the computations in the Head Office it is essential to know precisely, which data serves to the

computation of the differences of heights between two consecutive BMs of that line of levelling and which ones are data of links.

Such link observations do not serve the main line of levelling, but always start and end at PBMs of a line. The positions of the staves in such cases will be numbered as follows:

**Example:-**



**Fig. 28**

Here the small loop 121 - 224 → 121 - 225 → 121 - 226 → 121 - 224 does not belong to the line of levelling. This can be seen from the fieldbook, as the staff positions are numbered as/La/Lb, .....

Thus the staff-position 121 - 224/La is in a link, but the staff position 121 - 224/a is in a line.

Similar is that for a TBM, if it is needed in long links:

Examples:- 121 - 227/L1 is a TBM in a link

121 - 227/L1a is a staff position in a link

121 - 227/1 is a TBM in a line

121 - 227/1a is a staff position in a line.

**Note:-** In the example of fig. 28 the PBM 121 - 224 is common for the line and the small loop, hence it's number will be in the fieldbook of the line and in the fieldbook of the link.

## 8. DESCRIPTION CARDS

The preparation of description cards are the vital part in the process of monumentation. This consists of all the necessary information and measurement of the Bench Mark locality, thus serving as an identity for that particular Bench Mark. By the help of this, the Bench Mark can be recovered any time in future.

Almost all the Bench Marks are monumented under ground level or well inside the rock surface to protect them from destruction. That is why the description cards must be neatly completed during monumentation.

### 8.1 Completion of Description Card

The following information must be filled in Lev. Form 1.20.

- (a) District
- (b) Panchayat
- (c) Ward No.
- (d) Grid Sheet No.
- (e) One Inch Sheet No.
- (f) Type of Bench Mark
- (h) Kind of Land
- (i) Land owner

#### 8.1.1 Written Description

This description must be very short (in Nepali) but in detail. This must give the exact location, any prominent feature and the distance from the nearest village or town. The location of a Bench Mark **above** or **below** ground level must also be given here.

### 8.1.2 General Sketch

A general sketch (oriented to the north) covering a large area must be drawn to give the rough location of a Bench Mark. The prominent features such as buildings, trees, temples etc. must also be sketched. The shape of the terrain must be shown by the help of contour lines. If there are not any man made or physical features around the Bench Mark, the general sketch must be extended up to the nearest prominent feature.

For example:-

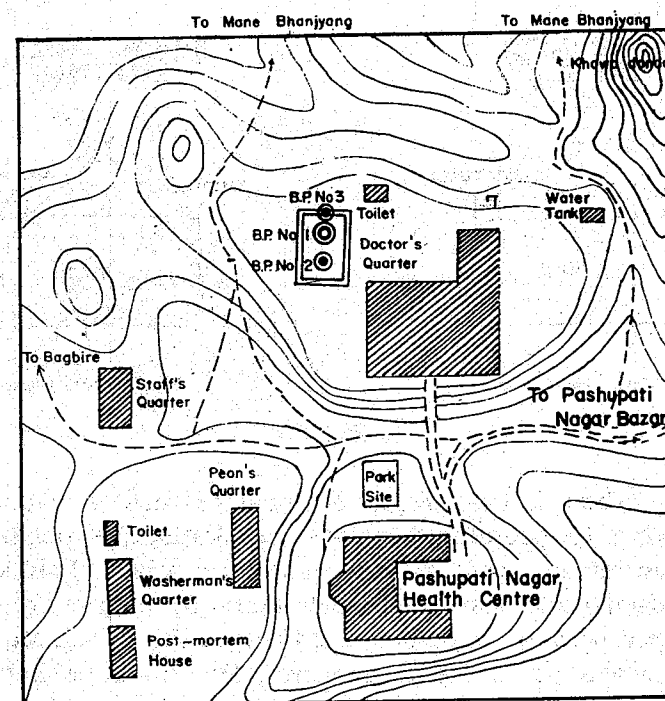


Fig. 29

### 8.1.3 Dimentional Description

This must be done in large scale with the proper orientation and all the possible measurement must be carried out. These measurements are carried out for the recovery of the station in the future, that is why as for as possible, only physical features must be chosen for measurements. A minimum of 3 best cutting measurements will be sufficient rather than to have several poor cutting measurement.

For example:-

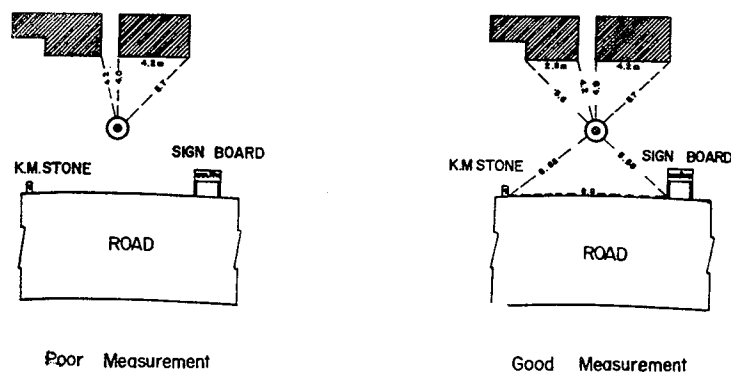


Fig. 30

When not a single physical feature is available, bearing to prominent points could also be used. The measurement could also be done from a large boulder or exposed rock near the Bench Mark by engraving a "V" shaped witness mark pointing towards the Bench Mark. It is always advisable to measure the distance between physical features (measured) or witness marks.

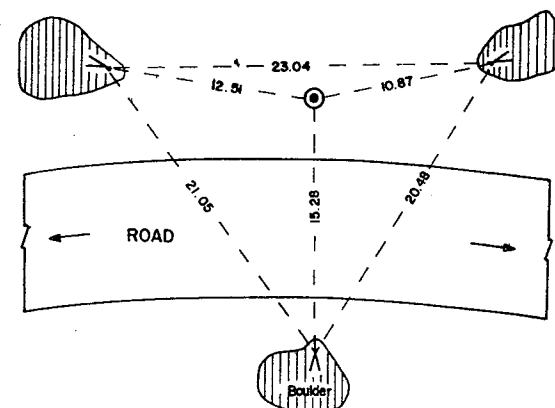


Fig. 31

All the features must be depicted as in the conventional signs.

(See Appendix C)

## 9. OBSERVATION

The observation procedure here is described for High Precision/First and Second Order Levelling. Before observation, the location of the BMs and the markings of staff and instrument positions (previously done by the recce supervisor) are to be reconnoitred by the staff supervisor. This recce is to be done immediately before the observations (see chapter 5). The staff supervisor is responsible, that **all BMs are levelled** and that not a single one is left out. Furthermore, that the staves are set up at the reconnoitred places.

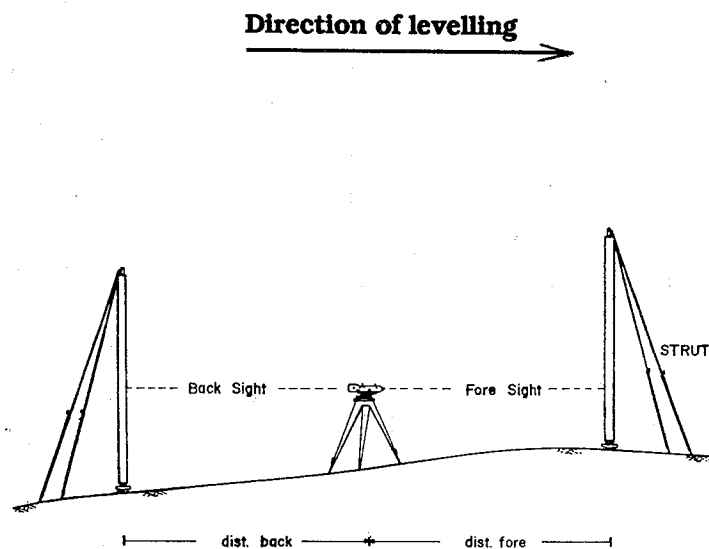


Fig. 32

The observation has to proceed very **quickly without any break** from one fixed point to another fixed point(PBM/TBM).

### 9.1 Operation

- (1) The staff supervisor establishes the same staff on the BM, which was set up there at the end of the previous session. The staff is set vertically by centering exactly the circular bubble attached to the staff. Now the staff is rotated one full turn and bubble is centred again. The rotation of the staff must be smooth. If not, check whether the staff touches somewhere else rather than the BM or the ground plate knob, clean the bottom of the staff and repeat the whole procedure. There after, the staff supervisor signals to the observer "Ready" and goes to the forward staff.
- (2) At the same time, the observer sets up the instrument insuring stability over the mark, **centres the tilting bubble** accurately and starts the readings to the backward staff as soon as the observer thinks that after these reading, the forward staff will be ready.

**Before the first reading to the staff**, the observer measures the height of the instrument, which is booked immediately into the field book.

- (3) The staff supervisor erects the forward staff.

*Note:- While moving from one set-up to the next one, the **staff** which does not change its position, **may no more be rotated a full turn: But only the amount**, which is necessary to direct it exactly towards the new instrument position. This motion is executed **very carefully. Centre the bubble thereafter again** if required.*

- (4) The observer takes all necessary readings.
- (5) The booker computes all the necessary checks. If every thing is o.k. the booker gives the signal to move to the next set-up.

## 9.2. Setting up the staves

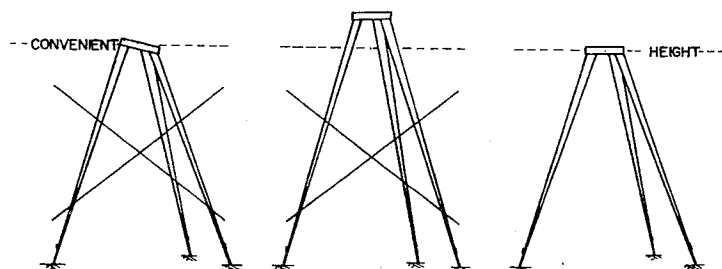
- (1) If the staff is to be erected **over a BM**, the **ring** on the bottom of the staff **is removed**.
- (2) **The ground plates** must be pressed with a hard stroke into the ground, so that they remain in the same position within a hundredth of a millimetre as long as the staff is set over them.  
They must be approximately horizontal. Their spikes are to be exchanged and repointed if they are no longer well pointed.
- (3) The bottom of the staff is first placed **near, but not on** the ground plate (or BM).
- (4) The staff is now erected approximately vertical.
- (5) The struts are placed so that they,
  - form an approximately right angle with the bottom of the staff.
  - support the staff **against the wind**
  - are not reaching too far **in the road**
  - cannot slip from their position.
- (6) The bottom of the staff is now placed **carefully** on the ground plate or on the BM, cleaning it from sand and dust.

- (7) The staff's circular bubble is now centred accurately.
- (8) **Never set a ground plate on asphalt.**
- (9) If there is no choice other than to set up the staff on asphalt, the staff supervisor first hammers a roadnail into the asphalt and the staff is set on this one.
- (10) The staff supervisor removes the road-nails after use.

## 9.3 Setting up the instrument

- (1) Before fixing the instrument into the tripod, the tripod head has to be cleaned.
- (2) The fixing screw must not be screwed in too tightly, otherwise the footscrews will be too tight.
- (3) The instrument is set up over the mark given by the recce supervisor. Remember - if any - the corrections to equalize the Back and Foresight distances.
- (4) The legs of the tripod are set so that
  - the head of the tripod is approximately horizontal.
  - they are not based on asphalt, but on large stones, gravel, sand or concrete.
  - they are free of any tension.
  - they are not placed where the observer must stand for the observations.

- the height of the instrument is set in such a way that the observer's eye is on the same level as the eyepiece.



**Fig. 33**

- (5) The tripod is set up in such a way that the legs are pressed firmly into the ground. Be careful not to introduce any tension in the legs of the tripod in doing so.
- (6) Turn the telescope towards staff A.
- (7) Centre the circular bubble accurately, watching it in the rotatable mirror.
- (8) Start the observations.

*Note: The instrument cover is always fitted over the instrument when it is not protected by the umbrella, or when carrying the instrument from one set-up to another.*

#### **9.4 Focussing the reticule of cross-hairs**

A parallax-free field of view is of utmost importance for accurate observations. parallax exists if:

- (i) the reticule cross-hairs are not focussed correctly,

- (ii) the field of view is not focussed correctly,
- (iii) the observer's eye has got tired.

*Note: To prevent the eyes from strain, correct focussing of the reticule cross-hairs is very important. The correct procedure is as follow:*

- (1) Set up the instrument on a tripod.
- (2) Sight an object at about **10 km away** (or infinity) and focus the field of view with the focussing screw.
- (3) Hold a sheet of white paper at about 1m distance in front of the telescope.
- (4) Sighting the paper, the focussing screw must remain in its position of step 2.
- (5) The telescope eyepiece is now turned very slowly until the reticule cross-hairs appear sharp and absolutely black.
- (6) Turn the telescope eyepiece very slowly anticlockwise until the reticule cross-hairs start to go out of focus.
- (7) A very small clockwise rotation of the telescope eyepiece will again focus the hairs correctly.
- (8) Read and book the number of the **dioptric scale**.
- (9) Rotate the telescope eyepiece half a turn and repeat steps 5 to 8 again twice.

- (10) The smallest and the greatest of the three readings of step 8 **may not differ by more than 1 unit of the dioptric scale**. If the difference is larger, the whole procedure is repeated.
- (11) Calculate the mean of the three readings of step 8.
- (12) Book the number of the instrument and the personal dioptric scale number (i.e. the mean of step 11) into the observer's note-book.

- Note:-
1. The personal dioptric scale number is always set and checked before the first reading with an instrument.
  2. The number **must not be changed** during observations.
  3. If during observation the field of view cannot be established free of parallax, check the setting of the dioptric scale. If that is correct, the observer's eye got tired.
  4. The number may be a different for another instrument and for another person.

### 9.5. Reading

The instrument is always protected from the sun, either by an umbrella (during reading) or by the instrument-cover (while moving from one set-up to the next).

The housing of the instrument must never be touched, but only the star shaped baseplate, footscrew, reflector knob, horizontal drivescrew, horizontal clamp, focussing knob, tilting screw and micrometer drive knob. Also take care not to touch the tripod in any way, even with your own clothing.

During observation nobody stands or walks round the instrument apart from the observer himself and the chainman with the umbrella. The booker sits near the instrument, but does not move around during the reading.

Each reading is a clearly spoken command to the booker, but not a discussion with him.

The observer after levelling the instrument accurately and setting the personal dioptric scale number starts the observations in the following sequence.

READING SEQUENCE

| STAFF | BACK SIGHT |             |        |  | FORE SIGHT  |      |        |  |
|-------|------------|-------------|--------|--|-------------|------|--------|--|
|       | STADIA     | CENTRE HAIR |        |  | CENTRE HAIR |      | STADIA |  |
|       |            | READING     | MEAN   |  | READING     | MEAN |        |  |
| +     | A          | R   1       | R   3  |  | R   5       |      | R   7  |  |
|       |            |             | R   4  |  | R   6       |      |        |  |
|       | B          | R   2       | R   11 |  | R   9       |      | R   8  |  |
|       |            |             | R   12 |  | R   10      |      |        |  |
| ++    | B          | R   7       | R   5  |  | R   3       |      | R   1  |  |
|       |            |             | R   6  |  | R   4       |      |        |  |
|       |            | R   8       | R   9  |  | R   11      |      | R   2  |  |
|       | A          |             | R   10 |  | R   12      |      |        |  |

Fig. 34

- + Reading sequence for the 1st, 3rd, 5th ..... set-up
- ++ Reading sequence for the 2nd, 4th, 6th ..... set-up

Note: At each set-up, staff A (same staff) is always read first, to minimize the effect of refraction.

Fig. 35 The reticule cross-hairs

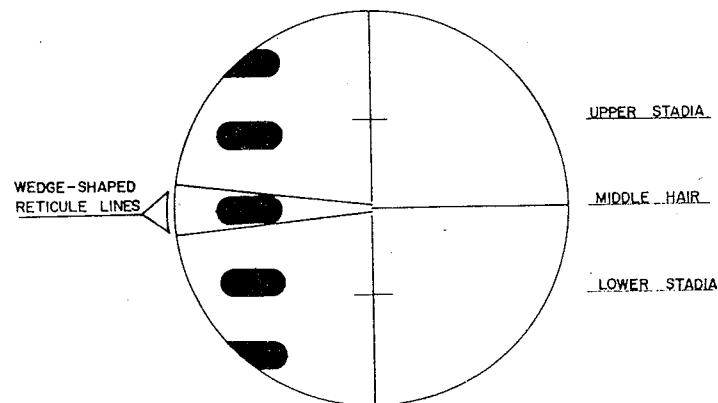


Fig. 35

As illustrated in fig. 34 the sequence of observation is as follows:

- (1) Lower stadia hair of staff A
  - (2) Upper stadia hair of staff A
  - (3) centre hair of staff A
  - (4) Turn micrometer, set centre hair again, check bubble, read micrometer again
  - (5) centre hair of staff B
  - (6) 2nd micrometer reading
  - (7) Lower stadia hair of staff B
  - (8) Upper stadia hair of staff B
  - (9) Centre hair of staff B
  - (10) 2nd micrometer reading
  - (11) Centre hair of staff A
  - (12) 2nd micrometer reading
  - (13) Rest of the information i.e. - height of instrument  
- time  
- temperature  
- air-pressure etc.
- ..... small numbers
- ..... big numbers

**Note:** The time is read at each set-up immediately after the last reading to a staff, i.e. after R 12.

Remember: 4.15 p.m. is to be booked as 16.15.

### 9.5.1. Reading the distance, i.e. the stadia hairs

- (1) Focussing (if focussing has to be changed significantly from staff A to staff B, the observer commands the booker "beware of distance" as Foresight and Backsight distances will not be of equal length).
- (2) Centre the split - bubble.
- (3) Set the lower stadia hair by micrometer to the nearest centimeter of the **small numbers**.
- (4) Check that **no parallax** exists between the staff image and the reticule cross-hairs.
- (5) Check 'split-bubble, if it is no longer centred, repeat steps 2 to 5.
- (6) Calling the booker "distance" take the reading in two groups  
(i) staff graduation and (ii) micrometer reading.  
  
e.g. 151.74  
Two figures only are read.
- (7) Set the upper stadia hair by micrometer to the nearest centimetre of the staff.
- (8) Check the split-bubble.

- (9) Take the reading of the upper stadia hair reading in the same way as mentioned in step 6.

### 9.5.2 Reading the Centre hair

The term "Centre hair" in this context always means the same as "wedge-shaped reticule lines" (see fig. 35) the procedure for the reading of the centre hair is the same for small numbers as it is for big numbers :

- (1) Centre the split-bubble.
- (2) Set the wedge-shaped reticule lines by micrometer knob to the appropriate centimetre graduation line of the staff. The wedge-shaped lines must just touch the graduation line, but they may neither lie inside nor outside of it.

Fig. 36 Correct setting of the wedge-shaped lines:

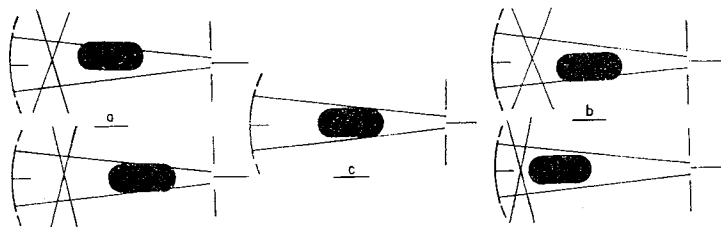


Fig. 36

- (3) Check that no parallel exists.
- (4) Check Split - bubble, if no more centred, repeat steps 1 to 4.

- (5) Check Centre hair, if no more centred, repeat steps 2 to 5.

- (6) Take the reading of the Centre hair as follows in **two groups-**

- (i) cm of the staff graduation, e.g. one four eight
- (ii) micrometer reading, e.g. six five six

148

. 656

148 . 656 cm.

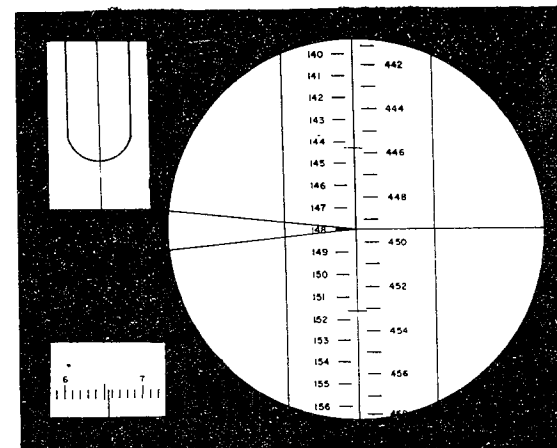


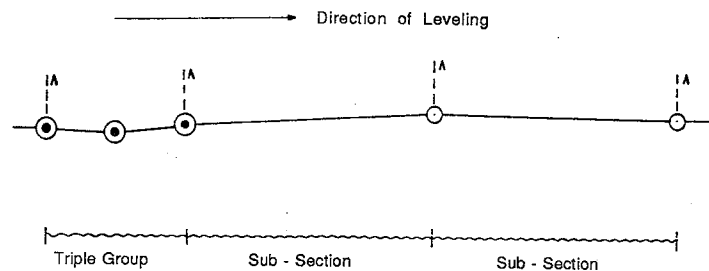
Fig. 37

- (7) Turn the micrometer half a turn.
- (8) Repeat steps 4 and 5.
- (9) Take the second micrometer reading.

**Note:-** On the micrometer reading the last place is estimated to only even umbers e.g. instead. 657 read 656 or 658 respectively.

## 9.6 Booking

For each new sub-section and for the levelling of a triple group of PBMs a **new field-book page** must be used.



A Begin here a new page of the field book

| H.M.G. SURVEY DEPARTMENT              |         |            |                       |       |                |                                           |         |           |                                 |       |                      |                     |         |  |
|---------------------------------------|---------|------------|-----------------------|-------|----------------|-------------------------------------------|---------|-----------|---------------------------------|-------|----------------------|---------------------|---------|--|
| LEVELLING OF HIGH PRECISION           |         |            |                       |       |                |                                           |         |           |                                 |       |                      |                     |         |  |
| PROJECT <u>Kapurbastu</u>             |         |            | DATE <u>19/5/1982</u> |       |                | STATION <u>333064</u>                     |         |           | OBSERVER <u>Dinesh P. Karna</u> |       |                      | PAGE NO. <u>127</u> |         |  |
| FROM <u>223.228</u> TO <u>223.239</u> |         |            | WEATHER <u>Clear</u>  |       |                | STARTED <u>4.591</u> STOPPED <u>4.581</u> |         |           | BOOKER <u>Laxman Mishra</u>     |       |                      |                     |         |  |
| POINT NO                              | STAFF   | BACK SIGHT |                       |       | FORE SIGHT     |                                           |         | DISTANCES |                                 | TEMP  | DIFFERENCE OF HEIGHT | BEARING             | REMARKS |  |
|                                       |         | STANDA     | CENTRE READING        | MEAN  | CENTRE READING | MEAN                                      | STANDA  | BACK      | FORE                            | ° F   | IN                   |                     |         |  |
| 1                                     | 223.228 | A          | 2.150                 | 2.150 | 150.000        | 150.000                                   | 150.000 | 171.50    | -0                              | -     | -                    | + 2.65              |         |  |
| 2                                     | 223.228 | A          | 1.940                 | 1.940 | 500.705        | 500.705                                   | 150.000 | 160.15    | +1                              | -     | -                    | + 2.65              |         |  |
| 3                                     | 223.228 | A          | 2.530                 | 2.530 | 150.000        | 150.000                                   | 150.000 | 25.34     | -1                              | 25.34 | 25.32                | -6.53               | + 2.65  |  |
| 4                                     | 223.228 | A          | 1.600                 | 1.600 | 150.000        | 150.000                                   | 150.000 | 151.21    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 5                                     | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 6                                     | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 7                                     | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 8                                     | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 9                                     | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 10                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 11                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 12                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 13                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 14                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 15                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 16                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 17                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 18                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 19                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 20                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 21                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 22                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 23                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 24                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 25                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 26                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 27                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 28                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 29                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 30                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 31                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 32                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 33                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 34                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 35                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 36                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 37                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 38                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 39                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 40                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 41                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 42                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 43                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 44                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 45                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 46                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 47                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 48                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 49                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 50                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 51                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 52                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 53                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 54                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 55                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 56                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 57                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 58                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 59                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 60                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 61                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 62                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 63                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 64                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 65                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 66                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 67                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 68                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 69                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 70                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 71                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 72                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 73                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 74                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 75                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 76                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 77                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 78                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 79                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 80                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 81                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 82                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 83                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 84                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 85                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   | 150.000 | 150.00    | +1                              | 25.60 | 25.20                | -                   | + 2.65  |  |
| 86                                    | 223.228 | A          | 1.800                 | 1.800 | 150.000        | 150.000                                   |         |           |                                 |       |                      |                     |         |  |

Fig. 38

While booking, the booker must repeat the reading loudly so that the observer can check it to avoid

misbooking. During the observation the booker must **compute and check** the readings as soon as possible and inform the observer for any blunders (gross errors) before shifting the instrument.

### 9.6.1 Reduction and checks

(a) WHEN STAFF A IS IN BACK SIGHT

| LEVEL STAFF, R.I.S. IN BACK SIGHT |            |             |      |             |      |        |            |      |   |   |           |    |                 |
|-----------------------------------|------------|-------------|------|-------------|------|--------|------------|------|---|---|-----------|----|-----------------|
| STAFF                             | BACK SIGHT |             |      |             |      |        | FORE SIGHT |      |   | V | DISTANCES |    | DIFF. OF HEIGHT |
|                                   | STADIA     | CENTRE HAIR |      | CENTRE HAIR |      | STADIA | BACK       | FORE |   |   |           |    |                 |
|                                   |            | READING     | MEAN | READING     | MEAN |        |            |      |   |   |           |    |                 |
| A                                 | R 1        | R 3         |      | R 5         |      | R 7    |            |      |   |   |           |    |                 |
|                                   | R 2        | R 4         |      | R 6         |      | R 8    |            |      |   |   |           |    |                 |
| B                                 |            | R 11        |      | R 9         |      |        |            |      |   |   |           |    |                 |
|                                   |            | R 12        |      | R 10        |      |        |            |      |   |   |           |    |                 |
| A                                 |            |             | 2    |             | 6    |        | 16         |      | 4 | 1 | 9         | 7  | 19              |
|                                   |            | 3           |      |             |      |        |            |      |   |   |           |    | 21              |
|                                   |            |             | 14   |             | 11   | 8      | 13         |      |   |   |           |    | 20              |
| B                                 | 1          |             | 15   |             | 12   | 7      | 17         |      | 5 |   | 18        | 10 | 22              |

(b) WHEN STAFF A IS IN FORE SIGHT

| STAFF | BACK SIGHT |             |      |        |             |      | FORE SIGHT |      |      |  |  |  | V | DISTANCES |  | DIFF. OF EIGHT |
|-------|------------|-------------|------|--------|-------------|------|------------|------|------|--|--|--|---|-----------|--|----------------|
|       | STADIA     | CENTRE HAIR |      | STADIA | CENTRE HAIR |      | STADIA     | BACK | FORE |  |  |  |   |           |  |                |
|       |            | READING     | MEAN |        | READING     | MEAN |            |      |      |  |  |  |   |           |  |                |
| B     | R 7        | R 5         |      | R 3    |             | R 1  |            |      |      |  |  |  |   |           |  |                |
|       | R 8        | R 6         |      | R 4    |             | R 2  |            |      |      |  |  |  |   |           |  |                |
| A     |            | R 9         |      | R 11   |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            | R 10        |      | R 12   |             |      |            |      |      |  |  |  |   |           |  |                |
| B     |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
| A     |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            |             |      |        |             |      |            |      |      |  |  |  |   |           |  |                |
|       |            | </          |      |        |             |      |            |      |      |  |  |  |   |           |  |                |

Fig. 39

### Explanations to fig. 39

In the following, the numbers in brackets ( ) represent the steps of computations as shown in fig. 39. Numbers without brackets represent numericals.

- (1) R1 - R2
- (2) Mean of R3 and R4

**Check:** The difference  $m$  between the two micrometer readings is  $< \pm 0.015$  cm (15 units of micrometer).

- (3) **Check:**  $(1) \div 2 + R2 = (2) \pm 0.05$  cm
- (5) Sum of all distances in that column
- (6) Mean of R5 and R6

**Check:**  $m < \pm 0.015$  cm

- (7) R7 - R8
- (8) **Check:**  $(7) \div 2 + R8 = (6) \pm 0.05$  cm
- (10) Sum of all distances in that column
- (11) Mean of R9 and R10

**Check:**  $m < \pm 0.015$  cm

- (12) (11) - (6)
- (13)  $v = 301.550 - (12)$ ; Tolerance:  $\pm 0.020$  cm  
This value is to be booked in units of 1/100 mm (e.g.  $0.014 = 14$ ). Beware of the signs (plus or minus).

- (14) Mean of R11 and R12

**Check:**  $m < \pm 0.015$  cm

- (15) (14) - (2)
- (16)  $v = 301.550 - (15)$ ; Tolerance  $\pm 0.020$  cm (see (13))
- (17) Upper minus lower value in that column =  $\Delta v$   
i.e.  $v_1 - v_2 = \Delta v$ ;  $v_1$  of the backward staff  
 $v_2$  of the forward staff.

**Check:**  $\Delta v \leq \pm 0.020$  cm  
and  $|v_1|/|v_2| \leq \pm 0.010$  cm.

**Note:-** The values  $m$ ,  $v_1$ ,  $v_2$  and  $\Delta v$  may not be systematic in their sign, i.e. the sign should be alternately positive and negative. The values mentioned must also be within the given limits. If they are not, possible reasons for bigger differences are:

#### (a) If their sign, is not systematic

- The personal dioptric scale number is set wrong.
- bad focussing of the field of view.
- The observers eye has got tired.
- shimmering is too intensive; stop the work after reaching the next BM.

#### (b) If their sign is systematic

- if the signs of  $v_1$ ,  $v_2$  and  $\Delta v$  are positive at each set-up, it means that the instruments have sunk or that the staves have risen; therefore: the instrument must be set up at a more stable place.
- if the signs are always **negative**, the **instrument** has **risen** or the staves have sunk; therefore: the ground-plates must be set more carefully and at more stable places.

- (18) **Check:** (5) = (10)

If not, inform the observer at the next set-up to equalize the sums.

- (19) (2) - (6) if staff A is backward  
(6) - (2) if staff A is forward
- (20) (14) - (11) if staff A is backward  
(11) - (14) if staff A is forward

(21) **Check:** (20) + (17) = (19)

(22) Mean of (19) and (20)

If there are any gross errors in the readings, or if the readings are out of tolerances, the booker informs the observer **after finishing all readings on that set-up** and not somewhere in between readings.

If the booker finds all readings to be correct after performing **all** of the above mentioned checks up to step (18) and if **all** data of that set-up are **recorded**, the booker gives the command "Finish move."

In the column **Remarks** of the field-book all the unusual happenings outside the normal procedure should be recorded, especially:

- condition of BMs
- steel ball, road nail or any other points used for turning point
- Sketch showing how a triple group and Trig-pillar was levelled.
- sudden change of weather
- problems in observations ( e.g. shimmering)
- Beginning and end of a 14-days section etc.

### 9.7 Sum Foresight-Distances = sum Backsight-Distances

The subsequent positions of the instrument and the staves are already marked by the reconnaissance party, making the Backsight equal to the Foresight. However, differences may be discovered during observations by measuring the distances accurately with the instrument.

In such a case, the booker commands the observer to slightly alter the marked instrument position on the next set-up so as to equalize the sum of Backsight-Distances and the sum of Foresight-Distances.

### Example:

| Distance |    |      |    |
|----------|----|------|----|
| Back     |    | Fore |    |
| 78       | 72 | 78   | 61 |
| 18       | 16 | 19   | 29 |
| 96       | 88 | 97   | 90 |

As the sum of Foresight distances is 1m longer than the sum of Backsight distances the booker commands the observer.

"Backsight 0.50m longer:"

**Note:-** Only **half of the difference** must be corrected to equalize the sums.  
As to avoid confusion, the correction necessary is to be given always in terms of the backsight.

The positions of the **last instrument set-up** between two BMs (PBM or TBM) is to be determined in such a way, that the sum of Backsight distances and the sum of Foresight distances between the two BMs become equal to within  $\pm 0.020\text{m}$ .

### 9.8 Moving to the next set-up

The instrument is protected against the sun with the instrument cover, while moving to the next set-up.

After reading the temperature at staff A, the staff supervisor sends the chainman with that staff to the next position and he himself goes to staff B. He revolves this

staff so that it faces exactly towards the new instrument position. He checks the staff's circular bubble. He now goes to the position of staff A and establishes it over the mark

- Note:-** 1. Take extreme care of that staff which keeps its position for the next set-up; i.e. staff in the Turning point.
2. If the line of levelling leads over vegetable mounds, this is to be dug out to set the ground plates and the tripod on gravel or sand, which are the best bases for stable set-ups.

In this set-up the same sequence of observation is repeated as described earlier.

**Note:-** for the relevant sequence of observation and booking when staff A is on Backsight or on Foresight see fig. 39.

Starting from a fixed point (PBM/TBM) the sequence of observation is repeated at every set-up until it reaches another fixed point (PBM/TBM). This is the process of levelling in the field.

## 9.9 Specification

|                                                          | H.P./1st order                              | 2nd order                                                                           | 3rd order                                                                           |
|----------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Level Instrument                                         | Wild N <sub>3</sub> or equivalent           | wild N <sub>3</sub> or NA <sub>2</sub> with parallel plate micrometer or equivalent | Wild N <sub>2</sub> or NA <sub>2</sub> with parallel plate micrometer or equivalent |
| Staff                                                    | 3m invar staves with double graduation      | 3m invar staves with double graduation                                              | Invar staves or equivalent                                                          |
| Length of sight (max)                                    | 25m                                         | 30m                                                                                 | 35m                                                                                 |
| Double levelling                                         | Yes                                         | Yes                                                                                 | Yes                                                                                 |
| Forward and Backward Levelling is carried out by         | Different instrument and different observer | Different instrument and different observer                                         | Different/same instrument and different/same observer                               |
| Observation time                                         | Morning and Afternoon                       | Morning and Afternoon                                                               | Wholoday                                                                            |
| Tolerance for Back and Fore Levelling (distance K in km) | $\pm 1.1 \sqrt{K}$ mm                       | $\pm 2.0 \sqrt{K}$ mm                                                               | $\pm 5.0 \sqrt{K}$ mm                                                               |
| Number of set-ups between BMs                            | even                                        | even                                                                                | even                                                                                |

## 10. ERRORS AND THEIR ELIMINATION

There are three different kinds of errors:

- (a) Gross errors
- (b) Systematic errors
- (c) Random errors.

### 10.1 Gross errors

These errors are caused by carelessness. They really are errors in the proper sense of the word and can be avoided by careful work. Above all, they are **discoverable** by repeating the measurements or by check-computations.

#### Examples:

- (a) Reading to the left scale of staff A: 87.746 cm  
Reading to the right scale of staff A: 391.302 cm  
The difference is 303.556 cm  
The difference should be 301.550 cm

Here we can see that the gross error is introduced in one of the above two readings. To determine the gross error the readings are to be repeated.

- (b) A 223.65 m. long sub-section was levelled forward and backward. The difference of height between the two BMs was determined by team I as 81.72364 m. and by team II as 83.65738 m.  
difference = 1.93374 m.  
tolerance = 0.00019 m.

It is obvious, that either one or both of the differences of heights are grossly wrong.

**Notes:-** (1) Gross errors can occur, where ever observations are taken.

- (2) They are disclosed by **two independent observations** of the same values (staves with double graduations, Forward and Backward Levelling, etc.) and by careful execution of **computing checks**. Each computation is to be carried out by two different persons.

- (3) The procedure of the levelling work as described in this instruction book allows one to discover each and every gross error present in the field.

- (4) Therefore: **Gross errors should not exist in any datas sent from field to the headquarter.** This is one of the responsibilities of the Group Leader. He cannot delegate this responsibility to anybody else.

### 10.2 Systematic errors

A systematic error is one that occurs always with the **same sign**, and often with a similar magnitude in a number of consecutive or otherwise related observations. Systematic errors are caused by:

- (1) Poor measurement (e.g. staff not on stable ground)
- (2) The influence of the atmosphere (refraction)
- (3) Badly adjusted instruments.

This error is one of the most dangerous things in the process of levelling. Repetition of an observation does little or nothing to reduce the ill effect of **systematic errors**, which therefore **are a most undesirable feature** in any

observations. The influence of the systematic errors may be reduced to a certain extent by:

- (1) Establishing and **following** a correct measuring sequence.
- (2) Taking into account mathematically the previously determined extent of their influence .
- (3) Careful adjustment of the instruments.

#### 10.2.1. Staff graduation error

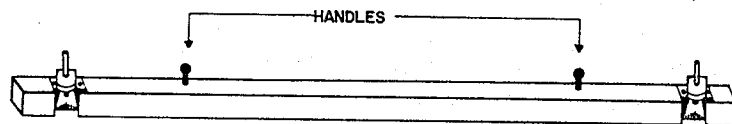
This is the error of the graduation of the staff, caused by elongation or contraction of the invar tape due to change of its temperature and tension.

This variation in length of the staff **is to be checked.**

- (1) At the beginning and end of each field season.
- (2) **Immediately after the staff falls down.**

The calibration of the staves are carried out at the beginning and the end of a field season by the Double Image Comparator at head quarters.

In the field, whenever necessary the calibration is carried out by an Invar (nickel-steel) Testmeter (R. & A. Rost). This is a rod of 1110 mm length, 20 mm width and 10 mm thick. Near each end of it a fine line is engraved.



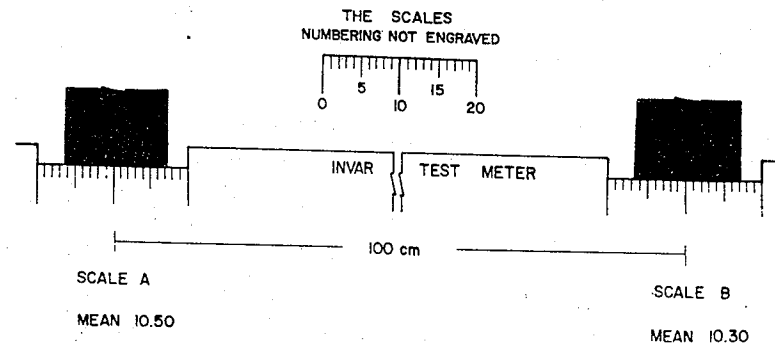
**Fig. 40**

On either side of both lines, which are  $(1.0 + c)m$  apart, (see **certificates** of each Testmeter for the exact correction  $c$ ) 10 graduation lines at an interval of 0.1 mm are engraved, i.e. there is a two mm long scale at each end of the Testmeter. **A microscope is fixed above each scale**

**which allows direct reading of 1/10 mm and an estimation up to 1/100 mm.**

The calibration of the staves, i.e. the investigation of the deviation of their graduations from the international metre is to be performed as follows:

- (1) This test is to be carried out **in shadow.**
- (2) The staff (staves) which are to be tested are placed in shadow **at least half an hour before the test.**
- (3) The staff is to be laid horizontally, supported **at both ends and in the middle.**
- (4) The Testmeter fits exactly into the hollow portion of the staff, so that it can be placed directly on to the invar tape.
- (5) The engraved scale (B) on the right side of the Testmeter is placed over that line on the invar-tape of the staff, **which is numbered as 10** on the staff. The left hand graduation thus falls on the line of 110. (see fig. below)
- (6) Viewing through the microscopes, the readings are taken as shown in fig.



**Fig. 41**

- (7) The readings are taken on the left **and** on the right border of the line. **Without touching** the Testmeter, the readings are taken on both microscopes, i.e.

four readings at each position of the Testmeter. The booking is done in Lev. Form 1.10.

- (8) The Testmeter is now displaced by 10 cm, i.e. to the number 20 and 120 respectively. **Caution. While displacing the Testmeter, lift it slightly off the invar-tape in order not to rub off (with the time) its graduation.**
- (9) In this way **19 positions of the Testmeter are observed** the last being at the numbers 190 and 290 respectively.
- (10) In the similar way the graduation of the "big" numbers on the same staff are observed, beginning at the numbers 310 and 410 respectively and ending on 490 and 590 respectively.

*Note:- Readings and bookings are to be performed in units of 1/10 mm, the distance between two consecutive lines of the graduation is 1/10 mm.*

*The temperature is to be recorded in the Lev. Form 1.10 immediately before each displacement of the Testmeter.*

### 10.2.2. Zero error

This is an error if the zero point of the graduation of a staff is not exactly at the base of the staff.

This error is to be checked:- (1) Before beginning of each 14-days section.  
(2) after any accident with the staff.

In levelling, the difference of height between two consecutive BMs are measured with the help of staves. The zero error is eliminated if it is started and end up by the same staff, but it has to be applied if a section is ended by a different staff.

Zero error=(Reading on staff A) - (Reading on staff B). For this purpose the staves are set up at the same **BM** and the readings are taken for both staves from the **same position of the instrument**.

The investigation of the zero error is to be performed as follows:

- (1) Establish the instrument at about 10 m from a PBM.
- (2) Staff A\* at the PBM.
- (3) Reading  $a_1$  on both scales of staff A and check **301.550**.
- (4) Staff B\* at the same PBM.
- (5) Reading  $b_1$  on both scales and check.
- (6) Again staff A at the PBM.
- (7) Reading  $a_2$  on both scales and check.
- (8) If reading  $a_2 = \text{reading } a_1 \pm 0.003 \text{ cm}$  (to ensure that the instrument has not moved)  
Then zero error = reading  $a_1$  minus reading  $b_1$   
 $= a_1 - b_1$ .  
If  $a_2$  is not  $a_1 \pm 0.003 \text{ cm}$ , then repeat steps 1 to 8.
- (9) The booking is performed in Lev. Form **1.12**.

\* The letters are written on the staves by the factory.

### 10.2.3. Collimation error

This is the error caused by the line of collimation (line of sight) not being horizontal when the split bubble is centred, i.e. if the split bubble is out of adjustment.

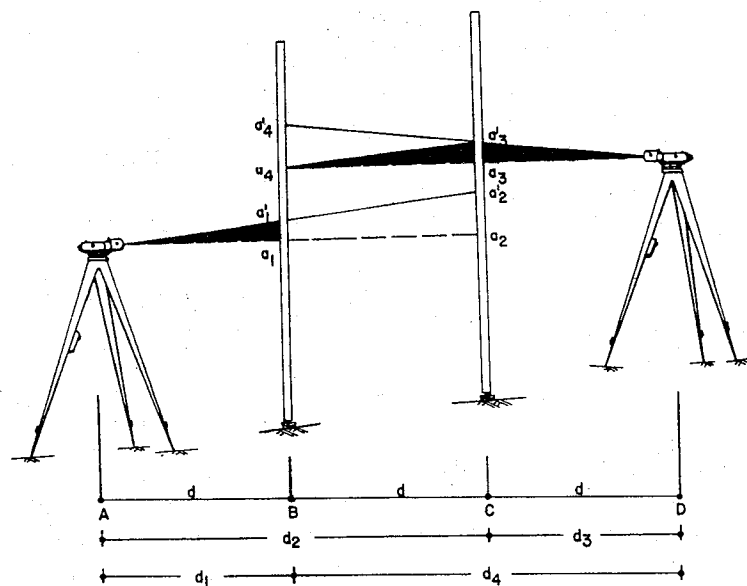
The test of the horizontality of the line of collimation (line of sight) is carried out:

- (a) at the beginning of each 14-days section.
- (b) immediately after each accidental hard shock of the instrument.

- (c) Never in the line of levelling, but always after or before carrying out the actual levelling work.

**The test is carried out as follows:**

- (1) In more or less **flat terrain** a distance between 45 and 60 m long is selected and **divided accurately** with the aid of a tape into three **equal** sections of the length  $d$ . The points A, B, C and D are marked on the ground.



**Fig. 42**

- (2) The invar staves are set up on an iron ground plate at the intermediate points B and C.
- (3) The instrument is first set up at A (centred accurately above the mark) levelled up and the staff

readings  $a'_1$  and  $a'_2$  as well as the stadia readings for the distances are observed and booked in Lev. Form 1.13.

- (4) Set up the instrument at D and record the staff readings  $a'_3$  and  $a'_4$ .

- (5) **Check:** (distances from fieldbook)

$$2d_1 = d_2 \pm 10 \text{ cm and } 2d_3 = d_4 \pm 10 \text{ cm}$$

$$\text{and } d_1 = d_3 \pm 10 \text{ cm and } d_2 = d_4 \pm 10 \text{ cm}$$

If not, correct the distances accordingly and repeat all observations.

- (6) In fig. 42  $a_1, a_2, a_3, a_4$  are the readings on the staves if the line of sight is absolutely horizontal.

$a'_1, a'_2, a'_3, a'_4$  are the observed readings on the staves (here the line of sight is inclined).

If an imaginary line parallel to  $(a'_1 - a'_2)$  is projected through  $a'_3$ , it will cut the staff at B in the correct position  $a_4$ , thus giving the required value for a true horizontal line of sight from D.

All shaded triangles are **congruent** and  $a'_1, a'_2, a'_3$  and  $a'_4$  is a parallelogram. Therefore

$$a_4 - a'_1 = a'_3 - a'_2$$

$$a_4 = a'_1 - a'_2 + a'_3$$

It is seen, that the correct value  $a_4$  can be computed from the "wrong" observed values.

- (7) The computation of  $a_4$  is carried out in Lev. Form 1.13,  $(a_4 - a'_4)$  is the collimation error for the

distance  $d_4$ ,  $d_4$  being the distance from instrument position D to that staff whose centre hair reading is  $a'_4$ .

collimation correction  $c' = a_4 - a'_4$  (see fig. 42).

- (8) Divide  $c'$  by  $d_4$ ;  $c'$  in 1/100 mm and  $d_4$  in m.  
 $c'/d_4 = c$ ,  $c$  being the collimation correction in 1/100 mm per meter. If  $c$  is greater (absolute value) than 0.010cm, the instrument must be adjusted.

- Notes:** (a) Any **adjustment** of the levelling instrument may be carried out by specially trained people only, (Mechanic at headquarter, Group Leader and Team Leader).  
 (b) The collimation error is cancelled out, if the Fore and Backsight distances are made equal.  
 (c) Therefore: If you disclose a collimation error which ought to be adjusted and if you are **not trained** to carry out the adjustment, **do not try to adjust** anything on the level.  
 (d) **Do not** carry out adjustments **more often** than absolutely necessary.  
 (e) Before adjustment, the test of the collimation error is to be carried out once again, to confirm the value of  $c$ .

#### 10.2.3.1 Adjustment by means of the coverglass

If  $c$  dash (see Lev. Form 1.13) is not greater than say up to **two or three mm**, the adjustment can be performed by turning the cover glass of the instrument's objective. (See Fig. 43).

- (1) The locking screw of the cover glass (1) is loosened by half a turn. When the instrument is at D.

- (2) The computed millimetre value of  $a_4$  is set on the micrometer reading.  
 (3) Staff at position B is sighted (See Fig. 42).  
 (4) Split-bubble is carefully centered.  
 (5) The cover glass is turned, until the required staff centimeter graduation (of  $a_4$ ) is straddled symmetrically and accurately by the wedge-shaped reticule lines

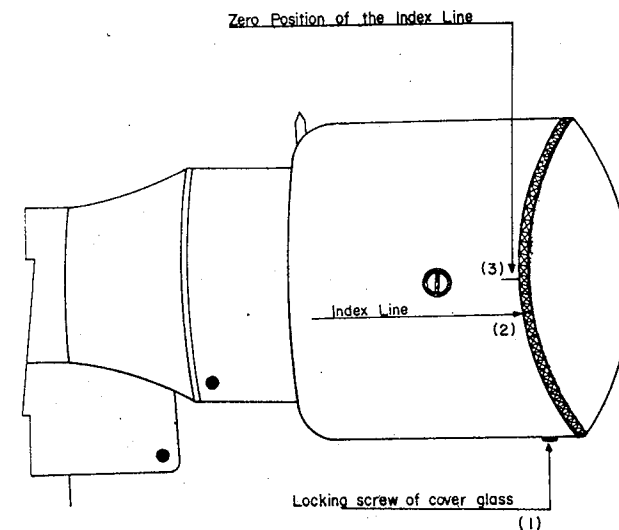


Fig. 43

- (6) Check steps 2, 4 and 5. If not O.K., repeat them all.  
 (7) The locking screw is then tightened. **Caution:** Tightening is to be performed gently, but not too much.  
 (8) The position of the index line (2) on the cover glass shows how much the cover glass has been turned from its zero position (3). If the line (2) is **more than 2cm** away from the index, it must be **turned**

back to zero and the level must be adjusted as described in 10.2.3.2.

- (9) The test of collimation error is to be carried out once again, to see whether the adjustment was done correctly or not.

**Note:-** For the execution of an adjustment of the collimation error, the **staff at B** and the **instrument at D** must keep **exactly the same position** as they had been during collimation error checking.

### 10.2.3.2 Adjustment by means of tubular level

If the value of **c dash (c')** is greater than **3mm** the adjustment is carried out as follows:

- (1) The staff at B and the instrument at D are still exactly in the same position as described in 10.2.3.
- (2) The micrometer reading is set to the computed millimetre value of  $a_4$ .
- (3) The staff at position B is sighted (See Fig. 42).
- (4) The **tilting screw** is turned until the required staff centimeter graduation (of  $a_4$ ) is straddled symmetrically and accurately by the wedge-shaped reticule lines.
- (5) In doing this, the ends of the split-bubble are no more in coincidence. The perfect coincidence of the split-bubble is now restored by means of the **vertical adjustment** of the level vial holder (see fig. 44).
- (6) The outer horizontal screw must be loosened slightly, so that the bubble ends can follow the adjustment movements of the vertical screws.
- (7) If the **left** bubble end is **low**, the lower adjustment screw is **loosened** slightly and the upper one is tightened by **equal** amounts as seen from the

objective, **both** are turned to the left (i.e. both always in the same sequence).

- (8) The adjustment is continued in this way, **step-by-step**, until the bubble ends are in coincidence. Turn the adjustment screws **always a minute amount only**.
- (9) The outer horizontal screw, originally loosened, is again tightened.
- (10) Check that the micrometer reading is still correct ( $a_4$ ) when the bubble-ends are in coincidence.
- (11) Finally, the horizontality of the line of sight of the **adjusted instrument** is re-checked, to see whether the adjustment was performed correctly or not.

**Note:** (a) This adjustment is **not easy**:

- (b) All the turned screws should be tightened carefully and moderately, otherwise the adjustment will not last. They must not be tightened **too much**.
- (c) The last movement of any screw is always in its tightening sense.

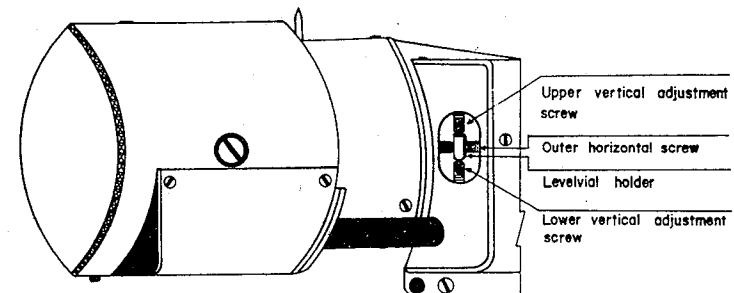


Fig. 44

The collimation error of the instrument thus determined can be applied in the observed difference of height or can be cancelled by keeping the Backsight and the Foresight distance equal.

#### 10.2.4 Non-verticality of the instrument's vertical axis

The vertical axis of the instrument is set vertically by centering the instrument's circular bubble. If the circular bubble is not in correct adjustment, the vertical axis of the instrument can no more be set vertically. This will hamper speedy work and it may cause small errors during observations.

Therefore, the adjustment of the instrument's circular bubble is to be checked. The check is performed before the beginning of each 14-days section.

##### Check and adjustment

- (1) Set up the tripod, with instrument attached, on stable ground and protect it from the sunshine by an umbrella.
- (2) Using the three footscrews the circular bubble is centred **exactly** in the middle of the black setting circle. A **parallax-free** viewing of the circular bubble is to be made in the rotatable mirror.
- (3) Rotate the telescope by 200°.
- (4) Watch the circular bubble in the rotatable mirror. If it has changed its position by **more than 1 mm**, it has to be adjusted otherwise nothing has to be done.
- (5) **Adjustment:** Clamp the telescope in its position. Do not touch the circular bubble with fingers.
- (6) **Exactly half** of the bubble's displacement from the centre is to be corrected by the footscrews.

**Example:** Note that the bubble must move along the straight line while turning the footscrews.

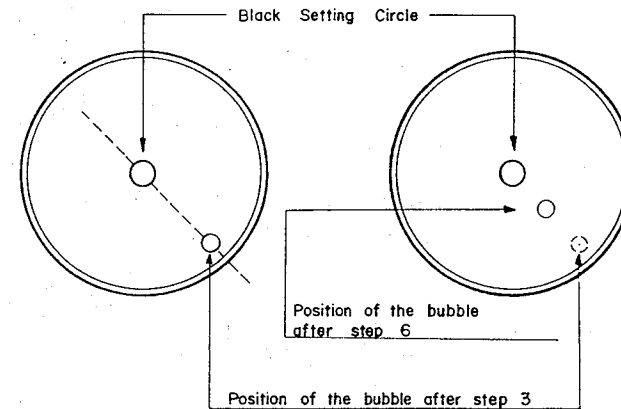
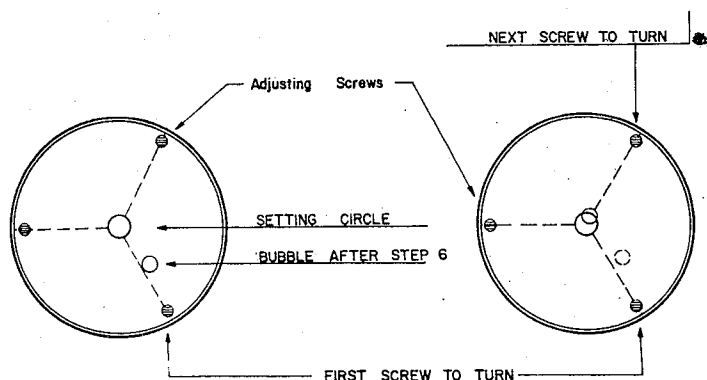


Fig. 45

- (7) The rest of the displacement, i.e. the **other half** of it, is to be corrected by the adjusting screws (at the bottom of the circular bubble), using the adjusting pin as follows.
- (8) **The first** adjustment screw **to be turned**, is the one that is the **nearest** to being in line with the middle of the bubble and the centre of the black setting circle. It is turned slowly, until the bubble reaches the centre of the circle or until the centre of the bubble reaches the nearest possible line from the centre of the circle to any of the other two adjusting screws.
- (9) If the bubble cannot be brought into the centre by step 8, then the next adjusting screw to be

turned is the one, that lies in the same line as the centre of the circle and the centre of the bubble.



**Fig. 46**

**Note:** The screws must not be turned more than is necessary to complete the adjustment.

### Check of the adjustment

Rotate the telescope always a small amount only and check after each such small movement the position of the circular bubble. It must now remain in the centre, whatever direction the telescope is pointing.

### 10.2.5 Non-verticality of the staff

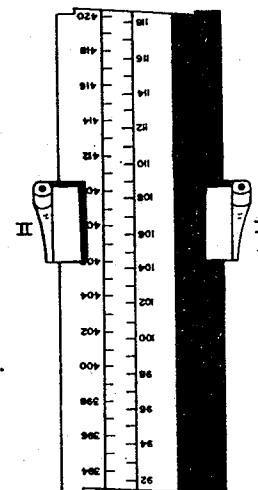
The staff is set vertically by centering its circular bubble. If the bubble is not in correct adjustment, the staff cannot be set vertically. This means that each reading was taken too high on the staff.

Therefore, the adjustment of the staff's circular bubble is to be checked. The check is to be carried out

before the beginning of each 14-days section.

### (a) Check and adjustment by the help of round box bubble

- (1) Set up the staff, well stable on a ground plate.
- (2) Centre the staff's circular bubble by turning the handles of the staff's struts.
- (3) Rotate the staff a full turn and direct it so that its bubble is placed in shadow.

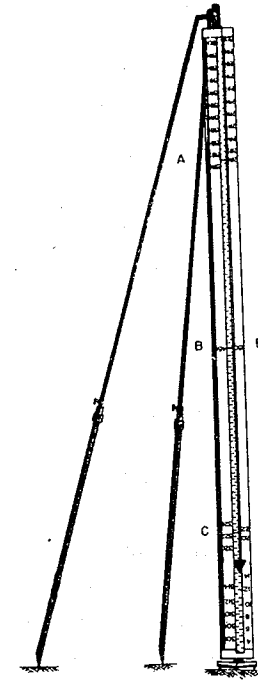


**Fig. 47**

- (4) Take a round box bubble and hold it firmly against a corner of the staff, approximately at the height of the staff's circular bubble. This being position I of the round box bubble. Be careful, **not to bend the staff** in doing so. Level the round box bubble by turning the handles of the struts.

- (5) **The same round box bubble** is now held firmly of that corner of the staff, which lies diametrically opposite to the former one. This being position II.
- (6) If the round box bubble is still levelled in its position II, go to step 8.
- (7) If the round box bubble is no longer levelled in position II, the half of the deflection is corrected by turning the handles of the struts. (Compare with 10.2.4. step 6)
- (8) **The circular bubble of the staff** shows now its true displacement of the bubble. The full deflection of the bubble from the black setting circle is to be adjusted by the help of struts and the adjusting screws on the bottom of the circular bubble. The adjustment is to be carried out only, if the displacement of the bubble is **more than 1 mm.**  
**Adjustment.**
- (9) See 10.2.4. steps 6 to 9.  
Do not touch the circular bubble with fingers.  
**Check of the adjustment**
- (10) Hold the round box bubble in position I as mentioned in step 4. **Remember** now the position of its bubble.
- (11) Hold **the same** round box bubble in position II. If now the deflection is the same, but in exactly opposite direction as in step 10, the adjustment was carried out correctly, otherwise the whole procedure must be repeated.

(b) **By the help of plumb bob**



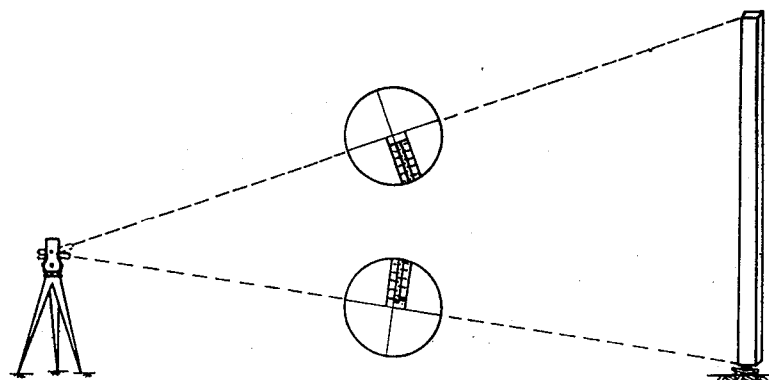
**Fig. 48**

This test is only possible when there is no wind or breeze in the field. In this test a plumb bob with a sufficiently long string is hung from the top of the staff, set vertically by the help of the circular bubble in such a way that the plumb bob hangs freely in the air (as shown in the figure) making the string and face of the staff parallel to each other. Check visually  $AA' = BB' = CC'$ .

If it is same i.e. plumb bob is parallel to the staff, the staff is vertical and the staff bubble is in

adjustment, otherwise the bubble is out of adjustment. This method is used only for checking the bubble but not for adjustment.

(c) **By the help of Theodolite**



**Fig. 49**

- (1) Set the theodolite at about 15-20m, from the staff held vertically by the help of the circular bubble.
- (2) Bisect the base of the staff in such a way that the vertical cross hair coincides with the edge of the staff. Now check the verticality of the staff by slowly raising the telescope till it reaches the top of the staff.
- (3) At the top, if the vertical cross hair remains in the same position as it was in the bottom, the staff bubble is in adjustment in the direction perpendicular to the line of sight of the telescope.
- (4) If there is any displacement 'd' the bubble is out of adjustment.

**Adjustment of the staff bubble**

The displacement 'd' seen is eliminated by inclining the staff in the appropriate direction (perpendicular to the direction of the line of sight). By doing this if the displacement of the circular bubble is more than 1mm, adjust the bubble as described earlier. (see 10.2.5 step 7,8 and 9)

- (5) After adjustment, check this verticality again till the bubble is in adjustment.
- (6) Now turn the staff by 100° and repeat the steps 2 to 5.
- (7) When the staff is vertical in both directions then only will the bubble be in perfect adjustment.

**10.3 Random errors**

They have different algebraic signs (+ or -). In a large number of observations there will be the same number of errors with a positive sign as there are errors with a negative sign. Also the magnitude of a random error is different for most observations. In contrast to gross errors and systematic errors, the random errors **cannot be avoided**. They are a result of the imperfections of the human senses and of the instruments. Even after adjustments the instrument will not be free of all errors.

**10.3.1 Cross error**

This error is defined as the difference between the direction of the tubular level and the direction of the line of sight of the telescope, i.e. the amount of horizontal non-parallelity between the two.

The test (and if necessary the adjustment) is to be carried out before the beginning of each 14-days section. **Before** the investigation of the cross error, the collimation error (10.2.3) and the instrument's circular bubble (10.2.4) have to be checked and if necessary adjusted.

### Test

- (1) Set up a staff at a **PBM**.
- (2) Set up the instrument over the rigid type tripod 20m from the staff. The footscrews of the instrument are oriented so that any two footscrews are in a line perpendicular to the telescope when the telescope is directed towards the staff.
- (3) Centre the split-bubble, using the tilting screw. **Watch the bubble for a while**, to check whether it remains centred or not. If it does not, check instrument stability and repeat steps 2 and 3.
- (4) Book the following readings into Lev. Form 1.12. Stadia hairs and small numbers only (micrometer twice).
- (5) Turn the tilting screw a half turn. Set the split-bubble again in coincidence and read again, then book the small numbers once again (micrometer twice). The booker computes the mean  $m_1$  of all readings of steps 4 and 5.  
If  $v$  is greater than  $\pm 0.005\text{cm}$ , step 4 and 5 are to be repeated.
- (6) **The footscrew on the left of the eyepiece is now given a half turn.**
- (7) The micrometer reading is set to the computed millimetre value of  $m_1$ .
- (8) Looking through the telescope, the **right footscrew** is now turned very slowly until the

centimetre graduation of  $m_1$  is once again straddled symmetrically by the wedge-shaped reticule lines. Check again the micrometer setting.

- (9) Centre the split-bubble using the tilting screw.
- (10) Execute steps 4 and 5. The booker calculates the mean  $m_2$  of these four readings.
- (11)

$$\text{Compute } e = \frac{m_1 - m_2 \left( \text{in } \frac{1}{100} \text{ mm} \right)}{\text{distance (in m)}}$$

- (12) If  $e$  is smaller than  $\pm 1.5$  the test is finished and no adjustment has to be carried out as the cross error in this case is considered as negligible.

### Adjustment

- (13) The micrometer reading is set to the computed millimetre value of  $m_1$ .
- (14) Using the tilting screw, the centimetre graduation of  $m_1$  is straddled symmetrically by the wedge-shaped reticule lines.
- (15) The ends of the split-bubble are now to be brought in coincidence by turning the horizontal adjusting screws of the tubular level. The procedure is as follows (See Fig. 44).
- (16) One adjustment screw is **loosened first** slightly and the other is tightened by an **equal** amount (both being turned in the same direction).
- (17) The adjustment is continued in this way, step-by-step, until the bubble ends are in coincidence. The screws should be **tightened carefully and moderately**, otherwise the adjustment will not last.

- (18) Once the bubble ends are in coincidence, the whole test is to be repeated again (centre the circular bubble first) to see whether the adjustment is carried out successfully i.e. to see whether  $e$  is now smaller than  $\pm 1.5$ . If it is not, the adjustment is to be repeated.
- (19) When  $e$  is smaller than  $\pm 1.5$  the collimation error (10.2.3) is to be rechecked and, if necessary, be adjusted again.

## 11. STAFF AND THEIR DUTIES

### 11.1. Allocation of staff

#### Levelling Group

- 1. Group Leader
- 1 Team Leader
- 1 Computer
- 1 Kharidar
- 1 Driver
- 1 Orderly

#### Team I

- 1 Assistant Team Leader (Observer)
- 1 Booker
- 1 Staff Supervisor
- 1 Recce Supervisor

#### Team II

- 1 Assistant Team Leader (Observer)
- 1 Booker
- 1 Staff Supervisor
- 1 Recce Supervisor

The field work is carried out in two stages.

1. Reconnaissance and monumentation
2. Observation

For the first stage the Group Leader allocates the field personnel as per his own discretion. For the second stage the allocation is as follows:

|                      |   |                                     |
|----------------------|---|-------------------------------------|
| Observation Sub-team | 1 | Assistant Team Leader (as observer) |
|                      | 1 | Booker                              |
|                      | 1 | Staff Supervisor                    |
|                      |   | Chainmen (as helper)                |
| Marking Sub-team     | 1 | Recce Supervisor                    |
|                      |   | Chainmen (as helper)                |

Since the observations are carried out by two teams (forward and backward direction) the above mentioned setting will be applicable for both teams.

### 11.2 Allocation of work and responsibilities

Group Leader : His direct superior is the head of the Levelling Division. He is responsible for:

1. The reconnaissance of Base Points and Permanent Bench Marks.
2. The organisation of the reconnaissance of all other Bench Marks and for all monumentations.
3. The quality of all data produced by his group.
4. The dispatching of all the field data (checked & completed) to headquarters.

He supervises and advises his Team Leader for the smooth running of daily work. He visits the field camp frequently in a month and submits the field report quarterly. He always emphasises on quality rather than on quantity.

Team Leader : His direct superior is the Group Leader. He is responsible for all the work carried out by his two teams and for their co-ordination. He issues the proper instructions to the respective observers to maintain the qualitative work. He signs daily the diaries of his observers. He rechecks the field book and other forms checked by the computer and sends them to the Group Leader. (He checks in

particular that there is no over-writing in the observed reading in any field books.) He is responsible for the smooth running of the field work.

Computer : His direct superior is Team Leader. He is entirely responsible for the technical side. He daily receives the field book from the booker after checking that it is neat and complete. He thoroughly checks the field book (see 9.6.1 reduction and checks) and entries the final  $\Delta H$  and distance at the end of every sub-section. These results are then transferred to Lev. Form 1.16 for section checking. While checking the field books, any correction in the reduction column can be changed but the actual (original) observation may never be altered without any sound reason.

He is responsible for the checking whether the sub-section is inside the given tolerance or not. The field books of those sub-sections which are outside tolerance are to be thoroughly rechecked and then only they are to be listed in repetition list. After performing the required repetition, finally he checks the tolerance for all the sections. When all the measurements are satisfactory he then declares the end of the observation.

He should be critical in checking the field book and if any doubt arises he should consult his superior and should be strict for its quality control.

Observer :  
(Assistant  
Team Leader)

His direct superior is the Team Leader. He is responsible for all the work carried out by his team. The level instrument is his personal responsibility. He is responsible for the adjustment and safety of the instrument and for accurate observations. The instrument should be thoroughly checked, cleaned and protected when he is in-charge. While handing over to the next man, the instrument must be clean from dust and moisture.

The time of observation has been fixed according to the season by the consent of Group Leader and Team Leader and he is responsible for its execution. He allocates the work to be done in his team. If the observations are carried out without the stability of the instrument or staff and unequality of Back and Fore distances, such observations are to be cancelled and a new set is to be taken.

He maintains a daily diary containing all special events of technical interest that happened during the work.

Booker

: He is responsible for all the observations (observed by the observer) to be booked neatly and correctly into the provided field book and to carry out the necessary computations and checks in the field book. Overwriting in the field book is strictly prohibited. If some reading after checking it twice, has to be corrected in the field, a completely new reading is taken cancelling the previous reading. The Back and Foresight distances are to be balanced at each set-up and at the last set-up of a sub-section it must be equalized. (See 10. Errors and their elimination). Each cancelled observation must have some sound reason testified in the remarks column. After completing the field book and testifying it with his and the observer's signature, he hands over to the computer daily without fail.

Staff  
Supervisor

: He is responsible for both the staves and for their quick setting up at the reconnoitred places. If the reconnoitred place is not found to be stable he changes its position and later warns the recce supervisor about it.

He maintains the adjustment of the circular bubble of the staves. He checks the verticality of the

staves at each set-up before the first reading to any staff.

Proper handling of the staff is very important. If a staff has fallen down, the graduation and the adjustment of the circular bubble must be checked immediately thereafter.

Temperatures at each staff are recorded and conveyed to the booker for recording.

Recce  
Supervisor

: His job is to carry out the reconnaissance of the instrument and staff positions (see 5.4).

Proper reconnaissance (choosing the stable point for staff and instrument positions and equal Back and Foresight distance) is the vital part in achieving the qualitative work.

During reconnaissance he prepares a recce list which consists of: condition of Bench Mark, number of set-ups in the sub-section and any other remarks. After the team crosses, both recce supervisors, interchange their list. Any damage of TBMs/PBMs are immediately informed to the staff supervisor.

## 12. MATERIAL AND MAINTENANCE

The materials required for levelling work have been categorised into two groups-

- (a) for reconnaissance and monumentation
- (b) for the observation.

The officer in-charge is responsible for checking all the materials required for his group before the work starts. Each item and everything are to be handled with special care. The Team Leader should check the materials before dismantling the camp. If there is any problem with the equipment or any damage or loss he should immediately report to his superior.

### **Maintenance:**

Generally it is the duty of a surveyor to take utmost care of the instrument, which he is using. All the measurements depend on the instrument used, if the instrument is kept in good condition and is used with utmost care, one can achieve a very good result. One should never neglect the Survey instruments, these should be carefully used and maintenance should be carried out regularly.

This aspect of "maintenance or care of the instrument" is very vital for a surveyor, but most of them tend to neglect it. One is called a good and honest surveyor who loves one's instrument and keeps it always in a neat clean and working condition. It is not a hard job to clean and maintain the instrument in good condition, but one should feel that it is the part and parcel of his duty.

### **Instruments:**

The following instruments at present are used for H.P/First Order and other Orders of Levelling.

- (a) Wild N<sub>3</sub> Precision level - for High Precision/  
First Order and  
Second order  
Levelling.
- (b) Wild NA<sub>2</sub> Automatic Level - This instrument may  
with parallel plate  
micrometer be used for Second  
and Third Order  
Levelling.
- (c) Wild N<sub>2</sub> with parallel  
plate micrometer - For Third Order  
Levelling.

Here the term maintenance means that the working instruments should be well looked after and therefore should be kept in working condition. If there are any repairs or adjustments to be made on the instruments, these should never be carried out in the field but should be sent to the mechanical workshop in the headquarters with a full report of the repairs or adjustments that are required.

For the field personnel when they receive the instruments, first of all they should go through the instruction manuals thoroughly and maintain the instruments accordingly. Daily cleaning and checking of all the instruments are of utmost importance.

Beside these, the observer should be aware of the following facts regarding the level instrument and Invar staves.

- (a) No liquid (i.e. water, kerosene, petrol etc.) should be used for cleaning.
- (b) Never touch any glass part (objective and eyepiece) with fingers.
- (c) While transportation the instrument should be carried on a soft pad or suspended.
- (d) If it is mounted on the tripod, always have somebody there.
- (e) Always store in a dust free, cool and dry place.
- (f) Always use the instrument cover (made of light cloth or plastic) while carrying the instrument during the observation.
- (g) Do not expose directly to sun, rain or frost, always use an umbrella to protect it.
- (h) Before starting the observation, the instrument should be taken out of the box at least half an hour earlier to adapt to the atmospheric temperature.
- (i) Invar staves and their numbering are to be cleaned regularly, but never scratch the invar band.
- (j) Regular greasing of struts should be carried out regularly.

### 13. FIELD PLOTTING AND COMPUTATION

#### Plotting

During reconnaissance and monumentation the alignment of the levelling traverse if it is not plotted in the map, will be plotted in a 1" = 1 mile map or the nearest available large scale map by careful map reading. The positions of all the Bench Marks established along the levelling line will be plotted after careful identification of all the local details in the map. The map may be made up to date by filling in the village name or town name and any other prominent feature as required. The position of the Bench Mark should be marked in ink by a circle and dot and numbered clearly and properly. When there are group of Bench Marks located so near that these cannot be shown separately in the map, in such a case a single dot and circle will signify all of them but the numberings are given just beside the dot. As shown below:

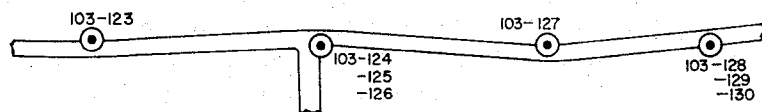


Fig. 50

#### Field Computation

The levelling procedures are adopted in such a way that the results are self checking and the errors are minimized. There is no chance of introducing any gross error in the levelling line, if the checking is performed seriously during the booking. The checks, which are to be performed during observation by the booker have already been described in chapter 9.

The error check forms (Lev. Form 1.10, 1.12 & 1.13) are completed and checked by the respective observer and booker and these are finally rechecked by the computer.

The field books duly completed and signed by the observer party are handed over to the computer. The level books thus received will have to be reduced at the conclusion of each days work and will not be left until the end of the section is reached.

Following are the list of checks and corrections to be applied:

- Thorough checking of the field book and mean difference of height.
- Application of distance correction (additive constant).
- Application of collimation and zero error.

After these checks and corrections the results are transferred to section checking form (Lev. Form 1.16). In this form, it is checked whether the levelling work carried out is within the allowable limit or not. If there are any sub-sections or sections, which are out of limits, these are sent for repetition until these are in the allowable limit. Once the check is performed the team moves to another section.

Keeping in mind that the field data sent to headquarters must be complete and error free, the completion of field data are rechecked by the Team Leader and the Group Leader. After the completion of the alignment, a technical report and statistical data (Lev. Form 1.29, 1.30) should be prepared and a check should be made to ensure that the work has been accomplished according to the assigned work programme.

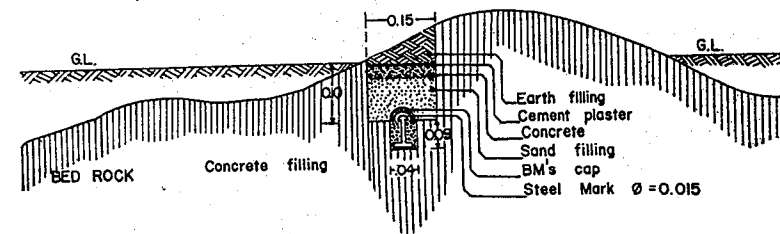
On satisfactory completion of a stretch or alignment, all level books should be sent by the quickest means to the officer in-charge for final checking and then forwarded to headquarters.

The list of technical data to be sent to headquarters  
are listed in appendix L.

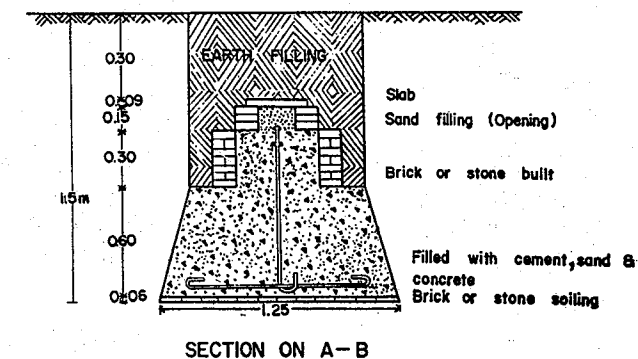
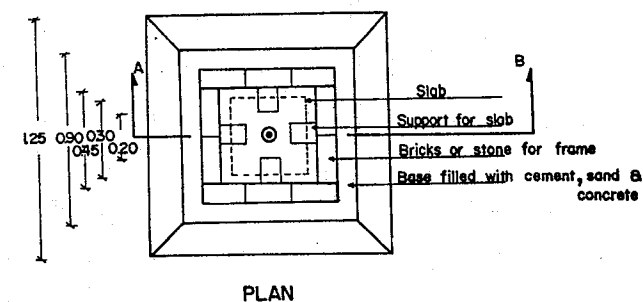
## APPENDIXES

# APPENDIX A/1

## TYPE "A" BASE POINT ON BED ROCK

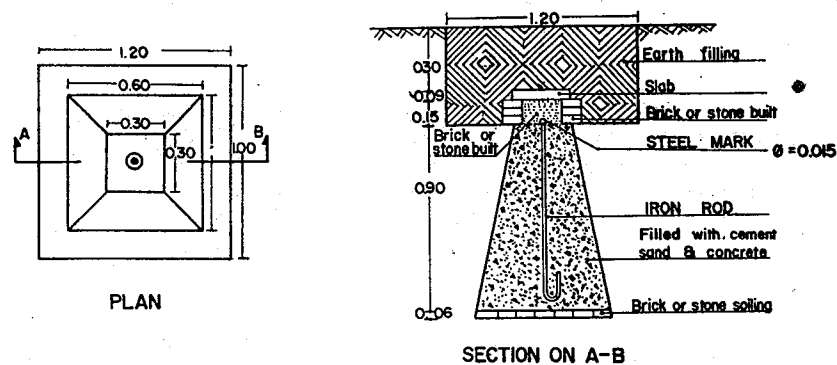


## TYPE "B" BASE POINT ON CONCRETE FOUNDATION

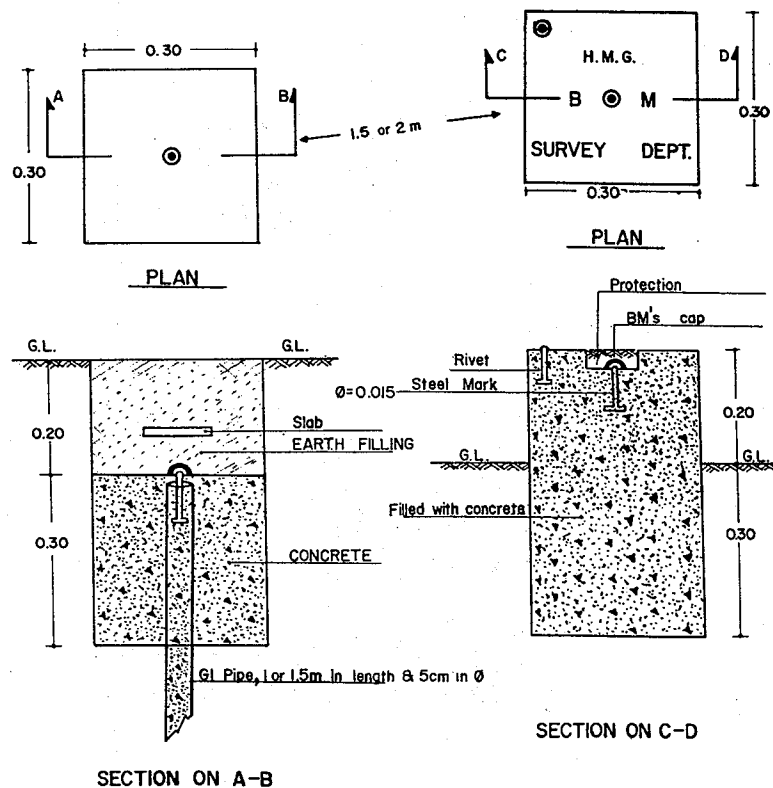


APPENDIX A/2

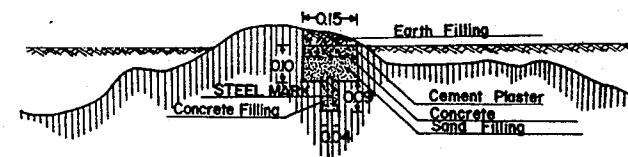
TYPE "C" BENCH MARK ON CONCRETE FOUNDATION



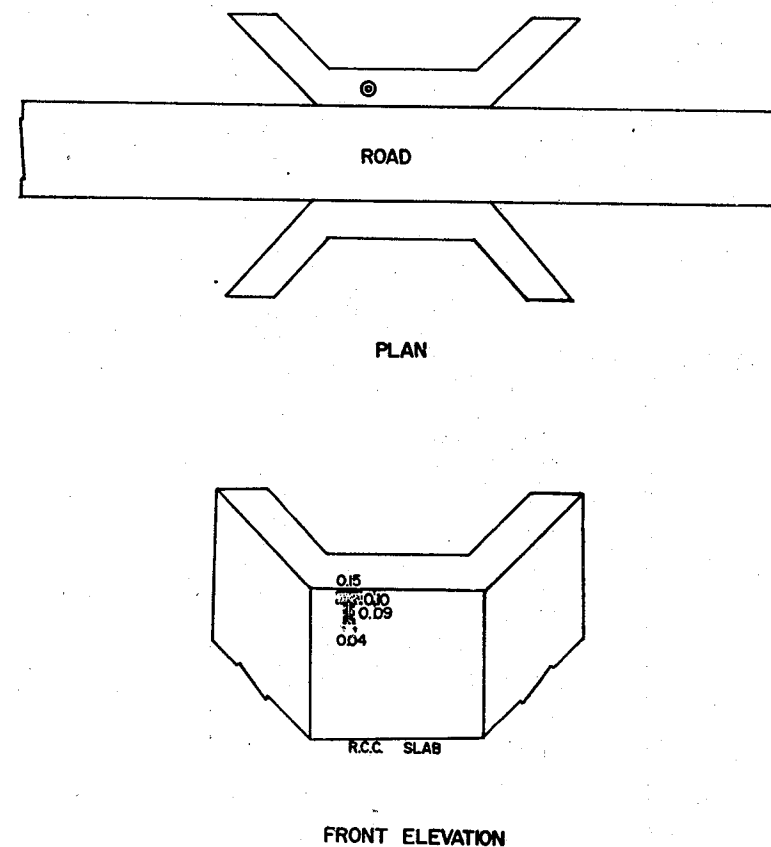
TYPE "D" BENCH MARK ON PIPE FOUNDATION



TYPE "E" PERMANENT BENCH MARK ON ROCK

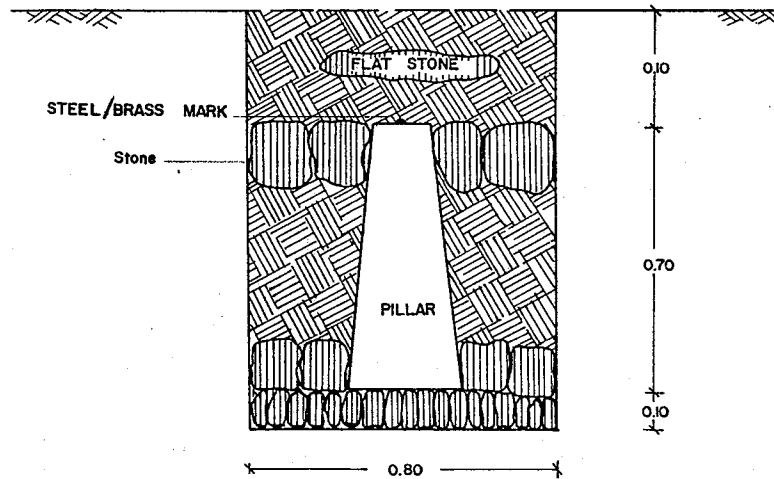
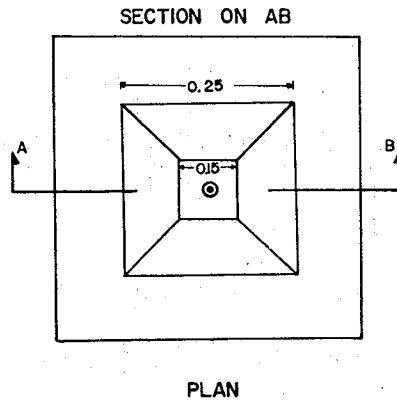


TYPE "F" PERMANENT BENCH MARK ON BRIDGE/CULVERT



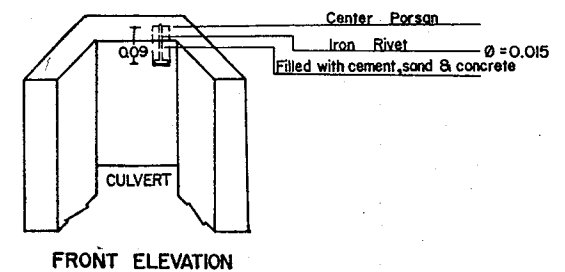
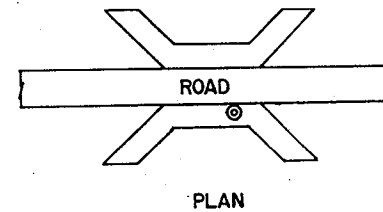
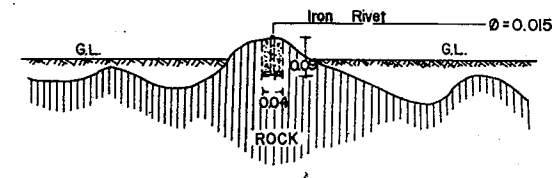
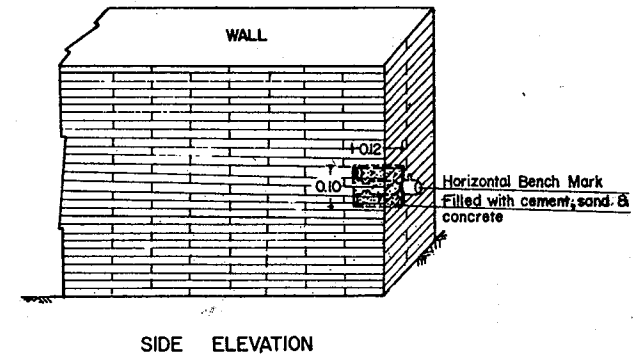
# APPENDIX A/4

## TYPE "G" PERMANENT BENCH MARK ON CONCRETE PILLAR



# APPENDIX A/5

## TYPE "H" HORIZONTAL BENCH MARK OR RIVET



## H.M.G. SURVEY DEPARTMENT DESCRIPTION OF BENCH MARKS

DISTRICT BHAKTAPURPAGE NO. 14PANCHAYAT TUKUCHAGRID SHEET 102WARD NO. 7ONE INCH SHEET 72 E/6

## WRITTEN DESCRIPTION

105-027 - यो उचाई स्मारक विन्दु नगरकोटको रेडियो एण्ड टाइम सर्भिस हाउसको अगाडी जमिनको सतह भन्दा करिब 0.24 m तल स्थापना गरिएको छ।

105-028 - यो विन्दु नगरकोटको अवजरमेटोरी हाउसको चौरमा दक्षिण पश्चिम पट्टि जमिनको सतह भन्दा करिब 0.17 m तल स्थापना गरिएको छ।

105-029 - यो विन्दु नगरकोटको अवजरमेटोरी हाउसको उत्तर पश्चिम कुना पट्टि जमिनको सतह भन्दा करिब 0.5 m तल स्थापना गरिएको छ।

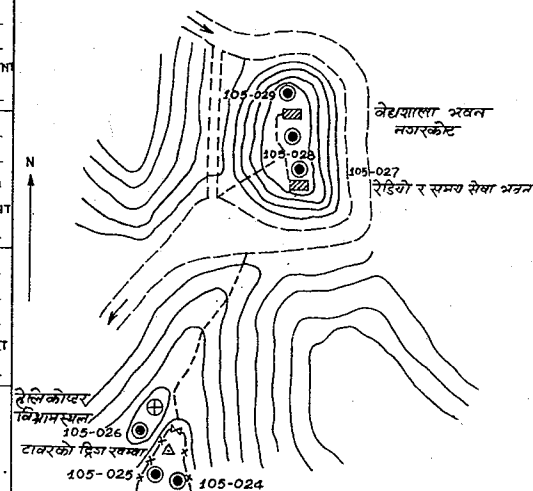
PBM NO. 105-027  
TYPE: BRASS MARK / PILLAR ✓  
KIND OF LAND UNCULTIVATED  
LANDOWNER SURVEY DEPARTMENT  
SIGNATURE \_\_\_\_\_

PBM NO. 105-028  
TYPE: BRASS MARK / PILLAR ✓  
KIND OF LAND UNCULTIVATED  
LANDOWNER SURVEY DEPARTMENT  
SIGNATURE \_\_\_\_\_

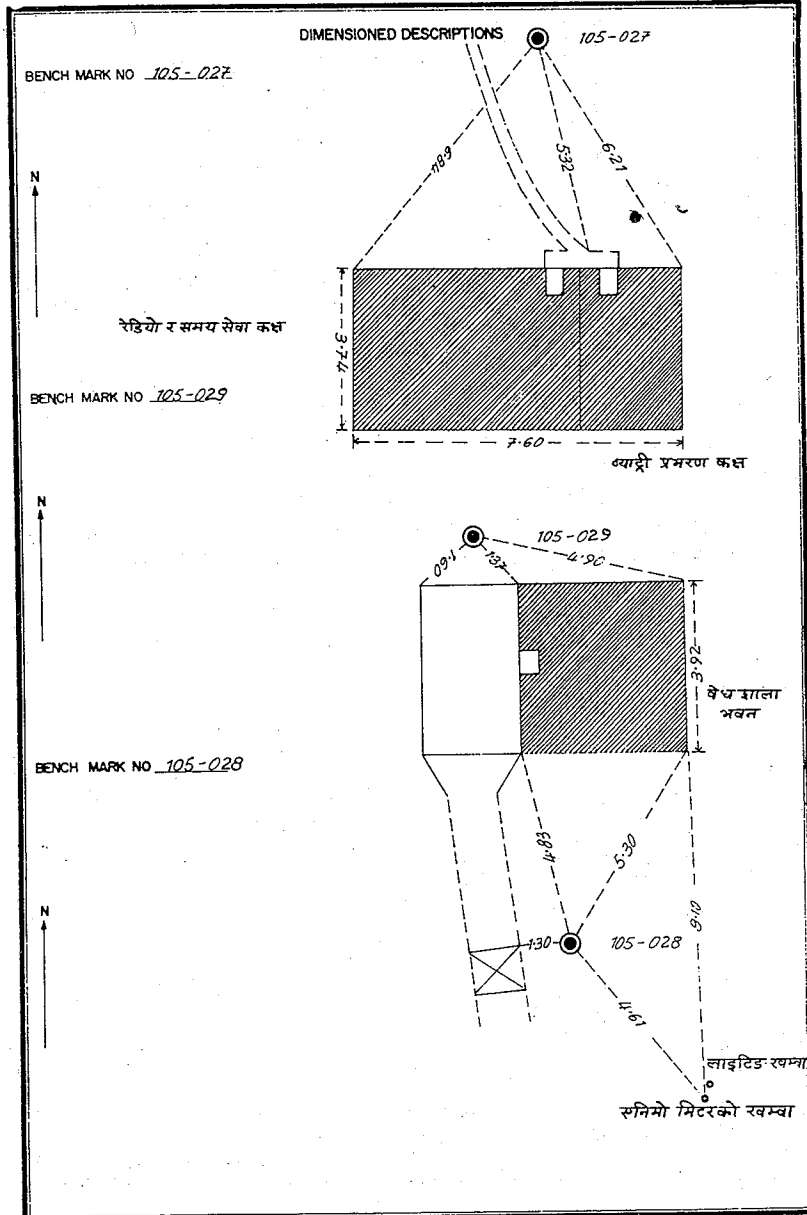
PBM NO. 105-029  
TYPE: BRASS MARK / PILLAR ✓  
KIND OF LAND UNCULTIVATED  
LANDOWNER SURVEY DEPARTMENT  
SIGNATURE \_\_\_\_\_

MONUMENTATION:  
DATE 26 / 9 / 1976  
DONE BY Chandra Bdr. Waj.  
SIGNATURE \_\_\_\_\_

## GENERAL SKETCH



# APPENDIX B/2



# APPENDIX B/3

## H.M.G. SURVEY DEPARTMENT DESCRIPTION OF BENCH MARKS

DISTRICT MORANG

PANCHAYAT BIRATNAGAR

WARD NO 21

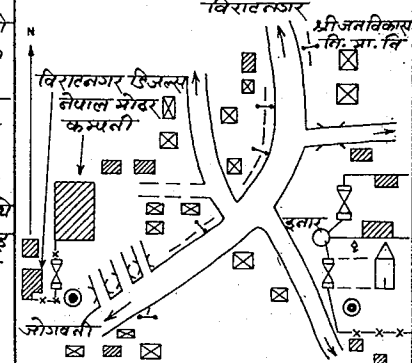
GRID SHEET 172

ONE INCH SHEET 72 N/6

| PBM NO<br>TYPE | WRITTEN DESCRIPTION                                                                                                                                                                                                             | PBM NO. | CRUDE HT. (m) | DEFINITE HT. (m) |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|------------------|
| 107-028<br>G   | यो उँचाई स्मारक चिन्ह अरुणोदय उद्योग प्रा. लि देखि दक्षिण करिब २ कि. मि डुवहरी जोगवती सडकको बायाँ पट्टि जल विकास ति. मा. वि. को दक्षिण पट्टि रामजातकी मन्दिर अगाडी पिलर स्थापित गरिएको छ। यो चिन्ह जमिन सतहबाट ०.५० m तल पर्दछ। |         |               |                  |
| 107-029<br>G   | यो उँचाई स्मारक चिन्ह राती विराटनगर मा पर्ने सडक को बायाँ पट्टि नेपाल मोटर कम्पनी भवन को पूर्व पिलर स्थापित गरिएको छ। अरु माथि को विवरण हेर्नुहोस्। यो चिन्ह जमिन सतहबाट ०.५० m तल पर्दछ।                                       |         |               |                  |

REFERRED TO

GENERAL SKETCH



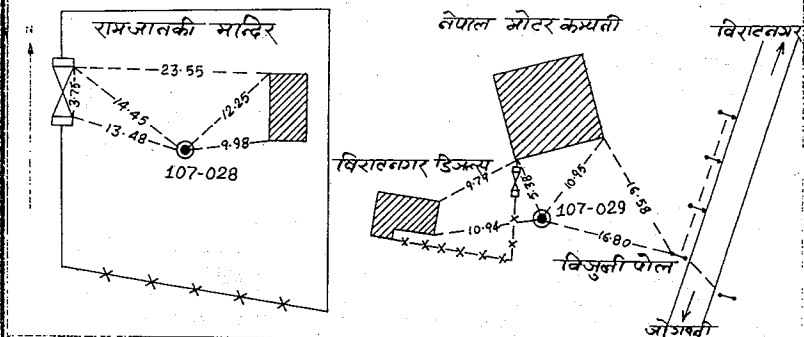
MONUMENTED ON 18.9.1980

YEAR OF OBSERVATION 1981

LANDOWNER

YEAR OF COMPUTATION

## DIMENSIONED DESCRIPTION

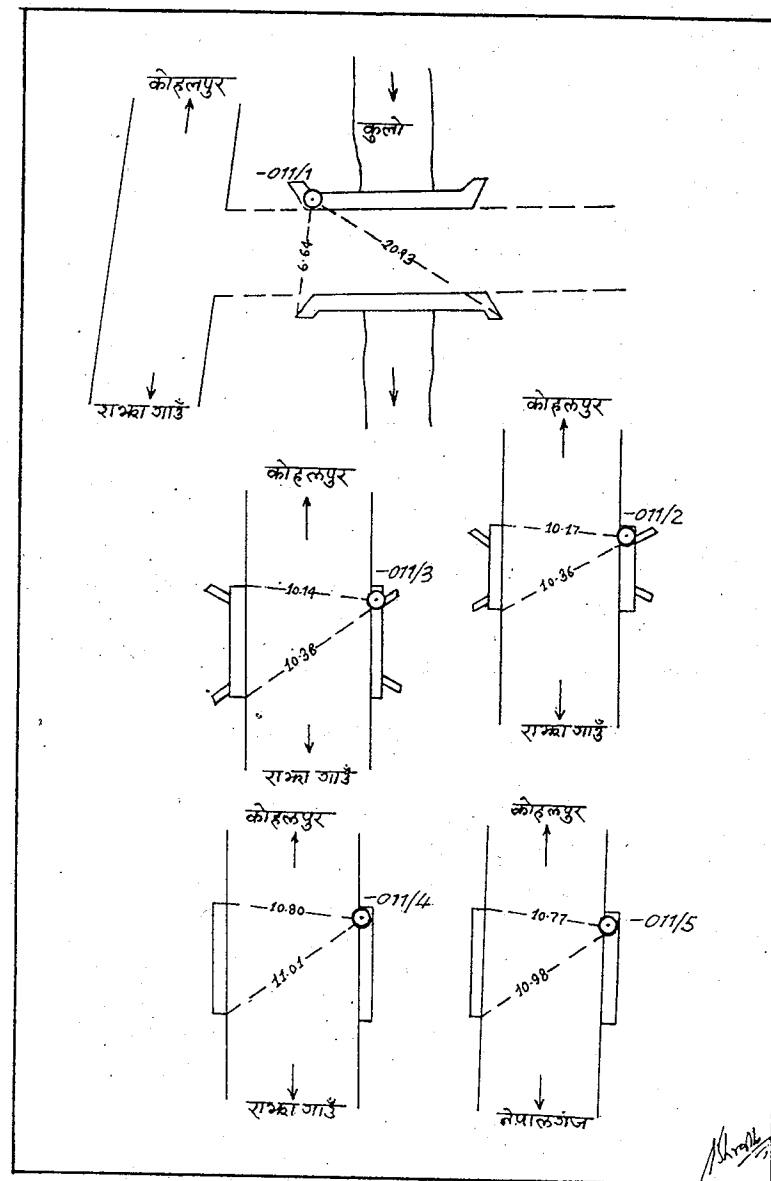


# APPENDIX B/4

| <b>H.M.G. GEODETIC SURVEY BRANCH</b><br>LEVELLING DIVISION<br>DESCRIPTION OF BENCH MARKS (TBM's)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| District... <b>BANKE</b><br>Panchayat <b>SAMSERGANJ</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Page... <b>7</b> .....<br>Grid sheet... <b>035</b> .....<br>One Inch Sheet <b>62.H/12</b> |
| <b>WRITTEN DESCRIPTION</b><br>BM No.<br>TYPE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>GENERAL SKETCH</b>                                                                     |
| <p>121-011/1 यो उचाई स्मारक चिन्ह नेपालगंज 11 Km र कोहलपुर 5 Km को दूझा देखि करिब 40 m दक्षिण मा सडक देखि पूर्व मा नहर को पुलको पश्चिम उत्तर कुनामा स्थापना गरिएको छ।</p> <p>121-011/2 यो उचाई स्मारक चिन्ह कोहलपुर 5 Km र नेपालगंज 11 Km को दूझा देखि करिब 300 m उत्तर मा पर्ने कलभर्ट को पूर्व उत्तर कुना मा स्थापना गरिएको छ।</p> <p>121-011/3 यो उचाई स्मारक चिन्ह कोहलपुर 4 Km र नेपालगंज 12 Km को दूझा देखि करिब 150 m दक्षिण मा पर्ने कलभर्ट को पूर्व उत्तर कुनामा स्थापना गरिएको छ।</p> <p>121-011/4 यो उचाई स्मारक चिन्ह हवलदारपुर गाँउ देखि पूर्व दक्षिण मा पर्ने कलभर्ट को पूर्व उत्तर कुनामा स्थापना गरिएको छ।</p> <p>121-011/5 यो उचाई स्मारक चिन्ह हवलदारपुर गाँउ देखि पूर्व पर्ने कलभर्ट को पूर्व उत्तर कुनामा स्थापना गरिएको छ।</p> |                                                                                           |
| Monumented By <b>Laxma Mishra</b><br>Date... <b>30.9.2015</b> .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Signature <b>Mishra</b><br>Checked By: <b>B.L. Karna</b>                                  |

LEV. FORM 1.22

# APPENDIX B/5



## APPENDIX C

| Conventional sign         |        |                      |        |
|---------------------------|--------|----------------------|--------|
| Description               | symbol | Description          | symbol |
| Fundamental Bench Mark    |        | Waterfall, rapids    |        |
| Datum Station             |        | Railway              |        |
| Base Point                |        | Traffic lights       |        |
| Permanent                 |        | Road metalled        |        |
| Temporary                 |        | Road unmetalled      |        |
| Trigonometrical station   |        | Foot track           |        |
| Flag pole                 |        | Telephone line       |        |
| Mail box                  |        | Power line           |        |
| Road pointer              |        | Water pipe line      |        |
| Gate                      |        | Chimney              |        |
| Milestone / Km stone      |        | Tower                |        |
| Telephone booth           |        | Hindu temple         |        |
| Post office               | P.O    | Buddhist temple      |        |
| Pump                      |        | Statue               |        |
| Water tap                 |        | Dam                  |        |
| Well                      |        | Forest               |        |
| Tube well                 |        | Dominant single tree |        |
| Spring                    |        | Bamboo clump         |        |
| Customs office            | C.O    | Banana plant         |        |
| Police station            | P.S    | Garden               |        |
| Witness mark              |        | Pond                 |        |
| Brick built building      |        | Lake                 |        |
| Clay brick built building |        | Sandy land           |        |
| Canal                     |        | Sloping land         |        |
| Road over the river       |        | Cliffs               |        |
| Bridge                    |        | Grassland            |        |
| Culvert                   |        | Stone taps           |        |
| Railway over the river    |        | Fence                |        |
| River, stream             |        |                      |        |



## APPENDIX E/2

## APPENDIX F

**GEODETTIC SURVEY BRANCH  
LEVELLING DIVISION**

ALIGNMENT NO. 315, DISTRICT Bhaktapur, FROM PBM 315-003 TO PBM 105-002, DATE 25/6/1983, WEATHER Cloudy, FOREWARD / BACKWARD, INSTRUMENT WILD2 NO. 383635, MICROMETER WITH / WITHOUT STAKES, OBSERVER Bal Mukund Samal, BOOKER Biju Chaudhary, PAGE NO. 3, LEV GROUP NO. 3.

| SIGHT | POINT NO | STAFF | BACK SIGHT |         | CENTRE HAIR |         | FORE SIGHT |        | DISTANCES |      | DIFFERENCE OF HEIGHT | REMARKS<br>(TIME OF START / END) |
|-------|----------|-------|------------|---------|-------------|---------|------------|--------|-----------|------|----------------------|----------------------------------|
|       |          |       | STADIA     | READING | MEAN        | READING | MEAN       | STADIA | BACK      | FORE |                      |                                  |
| BACK  | -003/2   | A     | 0.228      | 0.8650  |             | 1.5594  | 1.623      |        |           |      |                      |                                  |
| FORE  | 23       | B     | 0.726      | 0.8652  | 0.8651      | 1.5592  | 1.495      | 12.6   | 12.6      |      | -0.6942              |                                  |
| BACK  | 23       | B     | 1.552      | 1.5026  |             | 1.4120  | 1.534      | 11.2   | 11.2      |      |                      |                                  |
| FORE  | 24       | A     | 0.712      | 1.5206  | 1.5026      | 1.4120  | 1.421      | 23.8   | 24.1      |      | +0.0256              |                                  |
| BACK  | 26       | A     | 1.352      | 1.2015  |             | 1.4120  | 1.961      | 30.1   | 29.7      |      |                      |                                  |
| FORE  | 26       | B     | 0.301      | 1.2016  | 1.2016      | 1.4122  | 1.664      | 53.9   | 53.8      |      | -0.6106              |                                  |
| BACK  | 26       | B     | 1.604      | 1.4420  |             | 1.4120  | 1.292      | 32.1   | 31.9      |      |                      |                                  |
| FORE  | 26       | A     | 1.283      | 1.4420  | 1.4420      | 1.4122  | 1.317      | 86.0   | 85.7      |      | -0.0362              |                                  |
| BACK  | 26       | A     | 0.321      | 1.4420  | 1.4420      | 1.4122  | 1.630      | 29.9   | 29.9      |      |                      |                                  |
| FORE  | 26       | B     | 1.336      | 1.4420  | 1.4420      | 1.4122  | 1.331      | 115.9  | 115.6     |      | +0.0052              |                                  |
| BACK  | 26       | B     | 0.222      | 1.4655  | 1.4655      | 1.4122  | 2.284      | 29.8   | 30.0      |      |                      |                                  |
| FORE  | -003/1   | A     | 1.561      | 1.4421  |             | 2.1344  | 1.994      | 29.8   | 29.8      |      |                      |                                  |
| BACK  |          |       | 1.263      | 1.4122  | 1.4122      | 2.1342  | 0.300      | 145.7  | 145.6     |      | -0.7221              | $\Delta H = -2.0303$             |
| FORE  |          |       | 0.288      | 7.9100  |             | 9.9403  |            | 29.13  |           |      |                      |                                  |
| BACK  |          |       |            |         |             |         |            |        |           |      |                      |                                  |
| FORE  |          |       |            |         |             |         |            |        |           |      |                      |                                  |

LEV FORM 17

**H.M.G. SURVEY DEPARTMENT  
LEVELLING OF HIGH PRECISION  
CHECKING OF SECTIONS**

PAGE NO. 45

GROUP NO. 2 FROM PBM 112-046 TO PBM 111-029  
CAMP Banepa PAGE NO. FROM 167 TO 225  
TEAM I, FORWARD OBSERVER Rishi Ram Shrestha  
DATE FROM 30/12/1982 TO 1.1.1983  
TEAM II, BACKWARD OBSERVER Dinesh Prasad Karra  
DATE FROM 22/12/1982 TO 24.1.1983

| BENCH MARK                       | FROM    | TO | TEAM I       |     |       |     | TEAM II      |    |       |     | DIFF. | TOL. | REMARKS |      |     |                  |
|----------------------------------|---------|----|--------------|-----|-------|-----|--------------|----|-------|-----|-------|------|---------|------|-----|------------------|
|                                  |         |    | $\Delta H_1$ |     | $D_1$ |     | $\Delta H_2$ |    | $D_2$ |     |       |      |         |      |     |                  |
|                                  |         |    | m            | cm  | m     | cm  | m            | cm | m     | cm  |       |      |         |      |     |                  |
| 111-046                          | 10      | 13 | 64           | 098 | 234   | 57  | 10           | 13 | 64    | 101 | 235   | 51   | -3      | 53   |     |                  |
| 1                                | 2       | 10 | 15           | 29  | 046   | 320 | 24           | 10 | 15    | 29  | 016   | 318  | 04      | +30  | 62  |                  |
| 2                                | 3       | 10 | 13           | 57  | 087   | 240 | 51           | 10 | 13    | 57  | 114   | 240  | 137     | -27  | 54  |                  |
| 3                                | 4       | 10 | 6            | 19  | 282   | 205 | 62           | 10 | 6     | 19  | 289   | 205  | 134     | -7   | 50  |                  |
| 4                                | 5       | 10 | 15           | 55  | 272   | 276 | 06           | 10 | 15    | 55  | 135   | 275  | 142     | -23  | 58  |                  |
| 5                                | 6       | 10 | 14           | 26  | 186   | 232 | 42           | 10 | 14    | 26  | 192   | 232  | 142     | -11  | 53  |                  |
| 6                                | 7       | 10 | 17           | 20  | 016   | 265 | 02           | 10 | 17    | 20  | 024   | 267  | 55      | -9   | 52  |                  |
| 7                                | 8       | 10 | 9            | 28  | 624   | 240 | 97           | 10 | 9     | 28  | 553   | 240  | 71      | +11  | 54  | MEAN = -974.568  |
| 8                                | 9       | 10 | 11           | 51  | 327   | 238 | 78           | 10 | 11    | 51  | 441   | 238  | 53      | -116 | 54  | MEAN = -1151.364 |
| 9                                | 111-049 | 4  | 8            | 08  | 042   | 124 | 99           | 4  | 8     | 08  | 144   | 123  | 148     | +1   | 39  |                  |
| REPETITION                       |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| 2                                | 8       | 9  | 9            | 28  | 583   | 234 | 79           | 9  | 9     | 28  | 633   | 234  | 43      | -50  | 53  | MEAN = -974.608  |
| 8                                | 9       | 9  | 11           | 51  | 336   | 238 | 43           | 9  | 11    | 51  | 351   | 238  | 38      | -15  | 54  | MEAN = -1151.363 |
| SUM                              |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| 111-049                          | 111-049 | 24 | 125          | 70  | 388   | 237 | 165          | 24 | 125   | 70  | 501   | 237  | 94      | -113 | 169 |                  |
| CHECKED BY Guriy Bahadur Basnet  |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| DESIGNATION Computer             |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| SIGNATURE Basnet                 |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| RE-CHECKED BY Ganapati Chaudhary |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| DESIGNATION Survey Officer       |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |
| SIGNATURE Chaudhary              |         |    |              |     |       |     |              |    |       |     |       |      |         |      |     |                  |

LEV FORM 18

## H.M.G. SURVEY DEPARTMENT

## TOLERANCE TABLE

## HIGH PRECISION/FIRST ORDER

TOLERANCE:  $=\sqrt{12.1 D}$  1/100 mm.

D = Distance in metres

| D  | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D    | TOL. | D    | TOL. | D    | TOL. | D    | TOL. | D | TOL. |
|----|------|-----|------|-----|------|-----|------|------|------|------|------|------|------|------|------|---|------|
| 3  | 007  | 82  | 032  | 264 | 057  | 549 | 082  | 937  | 107  | 1429 | 132  | 2024 | 157  | 2723 | 182  |   |      |
| 5  | 008  | 87  | 033  | 273 | 058  | 563 | 083  | 955  | 108  | 1451 | 133  | 2050 | 158  | 2753 | 183  |   |      |
| 6  | 009  | 93  | 034  | 283 | 059  | 576 | 084  | 973  | 109  | 1473 | 134  | 2076 | 159  | 2783 | 184  |   |      |
| 7  | 010  | 98  | 035  | 293 | 060  | 590 | 085  | 991  | 110  | 1495 | 135  | 2103 | 160  | 2813 | 185  |   |      |
| 9  | 011  | 104 | 036  | 303 | 061  | 604 | 086  | 1009 | 111  | 1517 | 136  | 2129 | 161  | 2844 | 186  |   |      |
| 11 | 012  | 110 | 037  | 313 | 062  | 618 | 087  | 1027 | 112  | 1540 | 137  | 2156 | 162  | 2875 | 187  |   |      |
| 13 | 013  | 116 | 038  | 323 | 063  | 633 | 088  | 1046 | 113  | 1563 | 138  | 2182 | 163  | 2905 | 188  |   |      |
| 15 | 014  | 123 | 039  | 333 | 064  | 647 | 089  | 1065 | 114  | 1585 | 139  | 2209 | 164  | 2937 | 189  |   |      |
| 17 | 015  | 129 | 040  | 344 | 065  | 662 | 090  | 1083 | 115  | 1608 | 140  | 2236 | 165  | 2968 | 190  |   |      |
| 20 | 016  | 136 | 041  | 355 | 066  | 677 | 091  | 1103 | 116  | 1631 | 141  | 2264 | 166  | 2999 | 191  |   |      |
| 23 | 017  | 142 | 042  | 365 | 067  | 692 | 092  | 1122 | 117  | 1655 | 142  | 2291 | 167  | 3031 | 192  |   |      |
| 25 | 018  | 149 | 043  | 377 | 068  | 707 | 093  | 1141 | 118  | 1678 | 143  | 2319 | 168  | 3063 | 193  |   |      |
| 28 | 019  | 156 | 044  | 388 | 069  | 723 | 094  | 1161 | 119  | 1702 | 144  | 2346 | 169  | 3094 | 194  |   |      |
| 31 | 020  | 164 | 045  | 399 | 070  | 738 | 095  | 1180 | 120  | 1726 | 145  | 2374 | 170  | 3126 | 195  |   |      |
| 35 | 021  | 171 | 046  | 411 | 071  | 754 | 096  | 1200 | 121  | 1750 | 146  | 2403 | 171  | 3159 | 196  |   |      |
| 38 | 022  | 179 | 047  | 423 | 072  | 770 | 097  | 1220 | 122  | 1774 | 147  | 2431 | 172  | 3191 | 197  |   |      |
| 42 | 023  | 186 | 048  | 434 | 073  | 786 | 098  | 1240 | 123  | 1798 | 148  | 2459 | 173  | 3224 | 198  |   |      |
| 46 | 024  | 194 | 049  | 446 | 074  | 802 | 099  | 1261 | 124  | 1823 | 149  | 2488 | 174  | 3256 | 199  |   |      |
| 50 | 025  | 203 | 050  | 459 | 075  | 818 | 100  | 1281 | 125  | 1847 | 150  | 2517 | 175  | 3289 | 200  |   |      |
| 54 | 026  | 211 | 051  | 471 | 076  | 835 | 101  | 1302 | 126  | 1872 | 151  | 2545 | 176  | 3322 | 201  |   |      |
| 58 | 027  | 219 | 052  | 484 | 077  | 851 | 102  | 1323 | 127  | 1897 | 152  | 2575 | 177  | 3356 | 202  |   |      |
| 63 | 028  | 228 | 053  | 496 | 078  | 868 | 103  | 1343 | 128  | 1922 | 153  | 2604 | 178  | 3389 | 203  |   |      |
| 67 | 029  | 237 | 054  | 509 | 079  | 885 | 104  | 1365 | 129  | 1947 | 154  | 2633 | 179  | 3424 | 204  |   |      |
| 72 | 030  | 245 | 055  | 522 | 080  | 903 | 105  | 1386 | 130  | 1973 | 155  | 2663 | 180  | 3456 | 205  |   |      |
| 77 | 031  | 255 | 056  | 536 | 081  | 920 | 106  | 1407 | 131  | 1998 | 156  | 2693 | 181  | 3490 | 206  |   |      |
| 82 |      | 264 |      | 549 |      | 937 |      | 1429 |      | 2024 |      | 2723 |      | 3524 |      |   |      |

## APPENDIX G/2

## H.M.G. SURVEY DEPARTMENT

## TOLERANCE TABLE

## SECOND ORDER

TOLERANCE:  $\pm \sqrt{40 D}$ 

1/100 mm.

D = Distance in metres

| D  | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D   | TOL. | D   | TOL. |
|----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|
| 10 | 51   | 124 | 228  | 363 | 529  | 726 | 955  | 121 | 146  | 171 | 196  | 221 | 246  | 271 | 296  | 321 | 346  |
| 11 | 21   | 54  | 127  | 232 | 369  | 536 | 735  | 122 | 147  | 172 | 197  | 222 | 247  | 272 | 297  | 322 | 347  |
| 12 | 22   | 56  | 131  | 237 | 375  | 543 | 743  | 123 | 148  | 173 | 198  | 223 | 248  | 273 | 298  | 323 | 348  |
| 13 | 23   | 58  | 135  | 242 | 381  | 551 | 752  | 124 | 149  | 174 | 199  | 224 | 249  | 274 | 299  | 324 | 349  |
| 15 | 24   | 61  | 138  | 247 | 387  | 558 | 761  | 125 | 150  | 175 | 200  | 225 | 250  | 275 | 300  | 325 | 350  |
| 16 | 25   | 63  | 142  | 252 | 393  | 566 | 770  | 126 | 151  | 176 | 201  | 226 | 251  | 276 | 301  | 326 | 351  |
| 17 | 26   | 66  | 146  | 257 | 400  | 573 | 778  | 127 | 152  | 177 | 202  | 227 | 252  | 277 | 302  | 327 | 352  |
| 18 | 27   | 68  | 150  | 262 | 406  | 581 | 787  | 128 | 153  | 178 | 203  | 228 | 253  | 278 | 303  | 328 | 353  |
| 20 | 28   | 71  | 154  | 267 | 412  | 589 | 796  | 129 | 154  | 179 | 204  | 229 | 254  | 279 | 304  | 329 | 354  |
| 21 | 29   | 74  | 158  | 273 | 418  | 596 | 805  | 130 | 155  | 180 | 205  | 230 | 255  | 280 | 305  | 330 | 355  |
| 23 | 31   | 77  | 162  | 278 | 425  | 604 | 814  | 131 | 156  | 181 | 206  | 231 | 256  | 281 | 306  | 331 | 356  |
| 24 | 32   | 79  | 166  | 283 | 432  | 612 | 823  | 132 | 157  | 182 | 207  | 232 | 257  | 282 | 307  | 332 | 357  |
| 26 | 33   | 82  | 170  | 288 | 438  | 620 | 832  | 133 | 158  | 183 | 208  | 233 | 258  | 283 | 308  | 333 | 358  |
| 28 | 34   | 85  | 174  | 294 | 445  | 628 | 841  | 134 | 159  | 184 | 209  | 234 | 259  | 284 | 309  | 334 | 359  |
| 29 | 35   | 88  | 178  | 299 | 452  | 636 | 851  | 135 | 160  | 185 | 210  | 235 | 260  | 285 | 310  | 335 | 360  |
| 31 | 36   | 91  | 182  | 305 | 459  | 644 | 860  | 136 | 161  | 186 | 211  | 236 | 261  | 286 | 311  | 336 | 361  |
| 33 | 37   | 94  | 187  | 310 | 465  | 652 | 869  | 137 | 162  | 187 | 212  | 237 | 262  | 287 | 312  | 337 | 362  |
| 35 | 38   | 97  | 191  | 316 | 472  | 660 | 878  | 138 | 163  | 188 | 213  | 238 | 263  | 288 | 313  | 338 | 363  |
| 37 | 39   | 100 | 195  | 322 | 479  | 668 | 888  | 139 | 164  | 189 | 214  | 239 | 264  | 289 | 314  | 339 | 364  |
| 39 | 40   | 104 | 200  | 327 | 486  | 676 | 897  | 140 | 165  | 190 | 215  | 240 | 265  | 290 | 315  | 340 | 365  |
| 41 | 41   | 107 | 204  | 333 | 493  | 684 | 907  | 141 | 166  | 191 | 216  | 241 | 266  | 291 | 316  | 341 | 366  |
| 43 | 42   | 110 | 209  | 339 | 500  | 693 | 916  | 142 | 167  | 192 | 217  | 242 | 267  | 292 | 317  | 342 | 367  |
| 45 | 43   | 113 | 213  | 345 | 507  | 701 | 926  | 143 | 168  | 193 | 218  | 243 | 268  | 293 | 318  | 343 | 368  |
| 47 | 44   | 117 | 218  | 351 | 514  | 709 | 936  | 144 | 169  | 194 | 219  | 244 | 269  | 294 | 319  | 344 | 369  |
| 49 | 45   | 120 | 223  | 357 | 522  | 717 | 945  | 145 | 170  | 195 | 220  | 245 | 270  | 295 | 320  | 345 | 370  |
| 51 | 46   | 124 | 228  | 363 | 529  | 726 | 955  | 146 | 171  | 196 | 221  | 246 | 271  | 296 | 321  | 346 | 371  |

Lev. Form 1:15

## APPENDIX H/1

H.M.G. SURVEY DEPARTMENT  
STANDARDIZATION OF THE STAFF

PAGE NO. 27

DATE 4/6/1985

OBSERVER K. L. Chaudhary

STAFF NO. 4664 X B INARMETER NO. 22208

| NO.  | MICROSCOPE A |                 |                  | MICROSCOPE B |                 |                  | TEMP  | L-005<br>(1-20) + L | Δ+L    | V2       |
|------|--------------|-----------------|------------------|--------------|-----------------|------------------|-------|---------------------|--------|----------|
|      | MARK         | LEFT<br>1/10 mm | RIGHT<br>1/10 mm | MARK         | LEFT<br>1/10 mm | RIGHT<br>1/10 mm |       |                     |        |          |
| 1    | 10           | 2.8             | 18.8             | 110          | 3.0             | 18.6             | +33.2 | +0.198              | +0.198 | 0.002225 |
| 2    | 20           | 2.0             | 18.0             | 120          | 2.0             | 18.0             | +33.2 | +0.198              | +0.198 | 0.002225 |
| 3    | 30           | 1.0             | 17.0             | 130          | 1.0             | 16.8             | +33.4 | +0.201              | +0.301 | 0.002244 |
| 4    | 40           | 2.0             | 17.8             | 140          | 2.2             | 17.6             | +33.6 | +0.204              | +0.204 | 0.002244 |
| 5    | 50           | 2.0             | 18.2             | 150          | 2.4             | 18.4             | +33.6 | +0.204              | -0.096 | 0.002244 |
| 6    | 60           | 1.8             | 17.2             | 160          | 1.8             | 17.6             | +33.6 | +0.204              | +0.004 | 0.002244 |
| 7    | 70           | 2.6             | 18.6             | 170          | 2.6             | 18.4             | +33.6 | +0.204              | +0.304 | 0.002244 |
| 8    | 80           | 3.2             | 19.2             | 180          | 3.0             | 18.8             | +33.2 | +0.198              | +0.198 | 0.002225 |
| 9    | 90           | 3.6             | 19.2             | 190          | 3.6             | 19.2             | +33.2 | +0.198              | -0.102 | 0.002225 |
| 10   | 100          | 3.2             | 19.0             | 200          | 3.4             | 19.4             | +33.4 | +0.201              | +0.201 | 0.002244 |
| 11   | 110          | 1.8             | 17.4             | 210          | 1.8             | 18.0             | +33.4 | +0.201              | +0.501 | 0.002244 |
| 12   | 120          | 2.4             | 18.2             | 220          | 2.0             | 18.0             | +33.4 | +0.201              | +0.401 | 0.002244 |
| 13   | 130          | 1.4             | 17.2             | 230          | 1.4             | 17.2             | +33.4 | +0.201              | +0.304 | 0.002244 |
| 14   | 140          | 1.4             | 17.0             | 240          | 1.4             | 16.8             | +33.4 | +0.201              | +0.109 | 0.002244 |
| 15   | 150          | 3.0             | 19.0             | 250          | 3.2             | 19.0             | +33.4 | +0.201              | +0.201 | 0.002244 |
| 16   | 160          | 1.6             | 17.2             | 260          | 1.4             | 17.4             | +33.4 | +0.201              | +0.201 | 0.002244 |
| 17   | 170          | 1.4             | 17.2             | 270          | 1.4             | 17.2             | +33.4 | +0.201              | +0.101 | 0.002244 |
| 18   | 180          | 3.0             | 18.8             | 280          | 3.0             | 19.0             | +33.4 | +0.201              | +0.301 | 0.002244 |
| 19   | 190          | 1.8             | 17.8             | 290          | 1.8             | 17.6             | +33.4 | +0.201              | +0.301 | 0.002244 |
| SUM  |              |                 |                  |              |                 |                  |       |                     |        |          |
| MEAN |              |                 |                  |              |                 |                  |       |                     |        |          |

CORRECTION =  $\pm 0.0213$  mmM.S.E. =  $\frac{E(1)}{n(n-1)} = \pm 0.004$  mmBOOKED BY: B. N. Yadav  
CHECKED BY: B. P. Yadav

LEV FORM 1:0

# APPENDIX H/2

## H.M.G. SURVEY DEPARTMENT LEVELLING OF HIGH PRECISION

INSTRUMENT N3 NO 366672 DATE 1.1.8 1982 WEATHER Clear PAGE NO. 5  
OBSERVER Nil Govinda Maharjan BOOKER Ram Narayan Chaudhary  
Determined value of  $\epsilon$  &  $z$  are valid for 14 days section from Bench Mark 111-132 to B.P. No. 3

### CROSS ERROR (e)

| STADIA | m   | cm  | Reading | CENTRE HAIR | Mean      |           |
|--------|-----|-----|---------|-------------|-----------|-----------|
| U      | 135 | 135 | 130.142 | 130.142     | S1        |           |
| L      | 126 | 124 | 142     | 130.142     | S2        |           |
| D=U-L  | 10  | 11  | 142     | 000.000     | S1-S2     |           |
|        |     |     |         | 130.142     | Mean = M1 |           |
|        |     |     |         | 130.142     | S3        |           |
|        |     |     |         | 144         | S4        |           |
|        |     |     |         | 130.142     | S3-S4     |           |
|        |     |     |         | 142         | 000.001   | Mean = M2 |
|        |     |     |         | 130.142     | Mean = M2 |           |
|        |     |     |         | 000.000     | M1-M2     |           |
|        |     |     |         | 10.141      | D         |           |

$$e = \frac{(M1-M2)}{D} = \frac{0.00}{10} = 0.00 \text{ mm/100 m}$$

$$e = \text{mm/100 m}$$

### ZERO ERROR (z)

| STAFF A |             |         |                 |         | STAFF B     |         |    |  |  | V                | STAVES: A, No. <u>4652</u><br>B, No. <u>4652</u>                                           |
|---------|-------------|---------|-----------------|---------|-------------|---------|----|--|--|------------------|--------------------------------------------------------------------------------------------|
| STADIA  | CENTRE HAIR |         |                 |         | CENTRE HAIR |         |    |  |  | V                | NOTES:<br>1. Staff on PBM<br>2. Distance max. 10 m.<br>3. Sequence:<br><br>Staff A → B → A |
| m : cm  | Reading     | Mean    |                 | Reading | Mean        |         |    |  |  |                  |                                                                                            |
| 135.136 | 130.160     | 130.160 | S               | 130.160 | 130.160     | S       | -0 |  |  |                  |                                                                                            |
| 124.95  | 160         | 437.270 | B               | 160     | 437.270     | B       | -0 |  |  |                  |                                                                                            |
| 10.141  | 437.270     | 307.550 | B-S             | 437.270 | 307.550     | B-S     | -0 |  |  |                  |                                                                                            |
|         | 770         |         |                 | 770     |             |         |    |  |  |                  |                                                                                            |
|         |             | 567.820 | B+S             |         | 567.820     | B+S     |    |  |  |                  |                                                                                            |
|         |             | 301.550 | C               |         | 301.550     | C       |    |  |  |                  |                                                                                            |
|         |             | 260.320 | B+S-C           |         | 260.320     | B+S-C   |    |  |  |                  |                                                                                            |
|         | q1 =        | 130.160 | ÷ 2             | b1 =    | 130.160     | ÷ 2     |    |  |  | z = (a1 - b1)    |                                                                                            |
|         | 130.158     | 130.158 | S               |         |             | S       |    |  |  |                  |                                                                                            |
|         | 160         |         | B               |         |             | B       |    |  |  | = 0.002 mm/100 m |                                                                                            |
|         | 437.270     | 437.270 | B-S             |         |             | B-S     |    |  |  |                  |                                                                                            |
|         | 770         | 307.550 |                 |         |             |         |    |  |  |                  |                                                                                            |
|         |             | 567.827 | B+S             |         |             | B+S     |    |  |  |                  |                                                                                            |
|         |             | 301.550 | C               |         | 301.550     | C       |    |  |  |                  |                                                                                            |
|         |             | 260.317 | B+S-C           |         |             | B+S-C   |    |  |  |                  |                                                                                            |
|         | a2 =        | 130.158 | ÷ 2             | b2 =    |             | ÷ 2     |    |  |  | z = (a2 - b2)    |                                                                                            |
|         | +000.002    | q1 - a2 | - Tol = 0.003 - |         |             | b1 - b2 |    |  |  |                  |                                                                                            |
|         |             | S       |                 |         |             | S       |    |  |  |                  |                                                                                            |
|         |             | B       |                 |         |             | B       |    |  |  | = mm/100 m       |                                                                                            |
|         |             | B-S     |                 |         |             | B-S     |    |  |  |                  |                                                                                            |
|         |             | B+S     |                 |         |             | B+S     |    |  |  |                  |                                                                                            |
|         |             | 301.550 | C               |         | 301.550     | C       |    |  |  |                  |                                                                                            |
|         |             | B+S-C   |                 |         |             | B+S-C   |    |  |  |                  |                                                                                            |
|         | a3 =        |         | ÷ 2             | b3 =    |             | ÷ 2     |    |  |  | z = (a3 - b3)    |                                                                                            |
|         |             | 02-03   | - Tol = 0.003 - |         |             | b2 - b3 |    |  |  | = mm/100 m       |                                                                                            |

$$z = (a_1 - b_1)$$

$$z = 0.00 \text{ mm/100 m}$$

$$z = (a_2 - b_2)$$

$$z = \text{mm/100 m}$$

$$z = (a_3 - b_3)$$

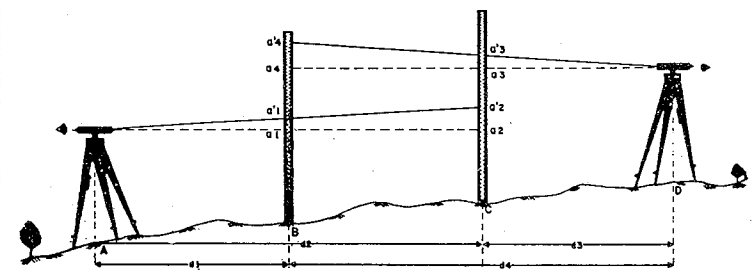
$$z = \text{mm/100 m}$$

# APPENDIX H/3

## H.M.G. SURVEY DEPARTMENT LEVELLING OF HIGH PRECISION

INSTRUMENT N3 NO 366660 DATE 10.1.4 1982 WEATHER Clear PAGE NO. 1  
OBSERVER Nil Govinda Maharjan BOOKER Gurj Bahadur Basnet  
Determined value of  $c$  is valid for 14 days section from Bench Mark 111-034 to 105-031

### COLLIMATION ERROR & ITS CORRECTION (c)



| a'1    |        |             |         |     |         | a'2         |     |        |       |        |      | a'1 - a'2 |                      |
|--------|--------|-------------|---------|-----|---------|-------------|-----|--------|-------|--------|------|-----------|----------------------|
| STADIA |        | CENTRE HAIR |         |     |         | CENTRE HAIR |     |        |       | STADIA |      | v         | DIFF OF HEIGHT<br>cm |
| m      | cm     | Reading     | Mean    |     | Reading | Mean        |     | m      | cm    |        |      |           |                      |
| U      | 106.62 | 92.568      | 92.568  | S   | 23.022  | 73.022      | S   | 23.022 | U     | +1     | S    | + 25.687  |                      |
| L      | 04.49  | 1570        | 401.118 | B   | 1982    | 375.431     | B   | 63.191 | L     | +1     | B    | + 25.687  |                      |
| D=U-L  | 101.13 | 1118        | 301.569 | B-S | 375.430 | 301.549     | B-S | 72.174 | D=U-L | -0     | Mean | + 25.687  |                      |
| U      |        |             |         | S   |         |             | S   |        | U     |        | S    |           |                      |
| L      |        |             |         | B   |         |             | B   |        | L     |        | B    |           |                      |
| D=U-L  |        |             |         | B-S |         |             | B-S |        | D=U-L |        | Mean |           |                      |

| a'3    |        |             |         |     |         | a'4         |     |         |       |        |   | CHECK :-<br>d1 = d3 ± 10 cm<br>= ± d2<br>= ± d4 |
|--------|--------|-------------|---------|-----|---------|-------------|-----|---------|-------|--------|---|-------------------------------------------------|
| STADIA |        | CENTRE HAIR |         |     |         | CENTRE HAIR |     |         |       | STADIA |   |                                                 |
| m      | cm     | Reading     | Mean    |     | Reading | Mean        |     | Reading | Mean  |        | m | cm                                              |
| U      | 122.01 | 116.808     | 116.808 | S   | 142.610 | 142.610     | S   | 152.74  | U     | -0     |   |                                                 |
| L      | 111.79 | 1612        | 018.458 | B   | 1612    | 144.160     | B   | 132.49  | L     | +1     |   |                                                 |
| D=U-L  | 101.23 | 1459        | 301.550 | B-S | 160     | 301.549     | B-S | 201.25  | D=U-L | -1     |   |                                                 |
| U      |        |             |         | S   |         |             | S   |         | U     |        |   |                                                 |
| L      |        |             |         | B   |         |             | B   |         | L     |        |   |                                                 |
| D=U-L  |        |             |         | B-S |         |             | B-S |         | D=U-L |        |   |                                                 |

| a'5    |    |             |      |     |         | a'6         |     |         |      |        |   |    |
|--------|----|-------------|------|-----|---------|-------------|-----|---------|------|--------|---|----|
| STADIA |    | CENTRE HAIR |      |     |         | CENTRE HAIR |     |         |      | STADIA |   |    |
| m      | cm | Reading     | Mean |     | Reading | Mean        |     | Reading | Mean |        | m | cm |
| U      |    |             |      | S   |         |             | S   |         |      |        |   |    |
| L      |    |             |      | B   |         |             | B   |         |      |        |   |    |
| D=U-L  |    |             |      | B-S |         |             | B-S |         |      |        |   |    |

| a'3 =  |    |             |      |     |         | a'4 =       |     |         |      |        |   |    |
|--------|----|-------------|------|-----|---------|-------------|-----|---------|------|--------|---|----|
| STADIA |    | CENTRE HAIR |      |     |         | CENTRE HAIR |     |         |      | STADIA |   |    |
| m      | cm | Reading     | Mean |     | Reading | Mean        |     | Reading | Mean |        | m | cm |
| U      |    |             |      | S   |         |             | S   |         |      |        |   |    |
| L      |    |             |      | B   |         |             | B   |         |      |        |   |    |
| D=U-L  |    |             |      | B-S |         |             | B-S |         |      |        |   |    |

| a'1 - a'2 |    |             |      |     |         | a'3 - a'4   |     |         |      |        |   |    |
|-----------|----|-------------|------|-----|---------|-------------|-----|---------|------|--------|---|----|
| STADIA    |    | CENTRE HAIR |      |     |         | CENTRE HAIR |     |         |      | STADIA |   |    |
| m         | cm | Reading     | Mean |     | Reading | Mean        |     | Reading | Mean |        | m | cm |
| U         |    |             |      | S   |         |             | S   |         |      |        |   |    |
| L         |    |             |      | B   |         |             | B   |         |      |        |   |    |
| D=U-L     |    |             |      | B-S |         |             | B-S |         |      |        |   |    |

| a'5 - a'6 |    |             |      |     |         |
|-----------|----|-------------|------|-----|---------|
| STADIA    |    | CENTRE HAIR |      |     |         |
| m         | cm | Reading     | Mean |     | Reading |
| U         |    |             |      | S   |         |
| L         |    |             |      | B   |         |
| D=U-L     |    |             |      | B-S |         |

NOTE: If the value of "c" is too large, repeat and conform the value again.

$$\text{CORRECTION } c = \frac{c}{4} = -0.74 \text{ mm/100 m}$$

CHECKED BY Chaudhary

## H.M.G. GEODETIC SURVEY BRANCH

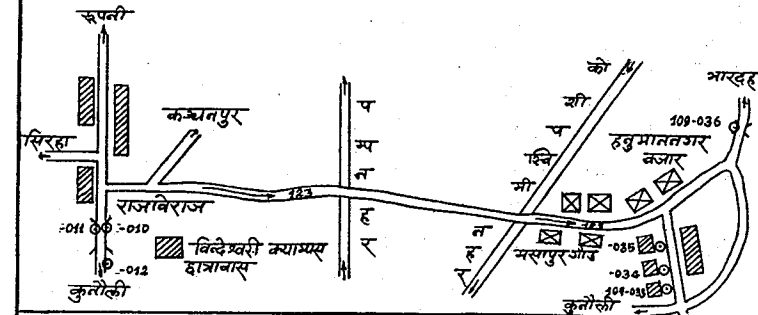
LEVELLING DIVISION

TECHNICAL REPORT

Page. 1

1. ALIGNMENT NO. 123.. FROM 109-012 TO 109-036  
 TOTAL LENGTH FORWARD. 75.41... KM DATE FROM 7.13/1988 TO 20.3/1988  
 TOTAL LENGTH BACKWARD. 75.41... KM DATE FROM 6.13/1988 TO 20.3/1988

## 2. SKETCH OF THE ALIGNMENT



## 3. STARTING JUNCTION POINT

(i) Few words about its location, condition

109-010, 109-011 र 109-012 :- यी उँचाई स्मारक चिन्हहरू राजविराज-कुनौली सबक हुदै जाँदा महेन्द्र विन्देश्वरी क्याम्पस को दात्रावास नजिक को पुलको बायाँ पट्टि, सोही पुलको दायाँ पट्टि माथको जंगमा र बायाँ पट्टि निजुली को पोलको जंगमा कायम छन। सबै उँचाई स्मारक चिन्हहरू राम्रो अवस्था मा छन्।

(ii) Comparison of (New-old) results

| Bench Mark |         | New Result<br>$\Delta H$ | Old Result<br>$\Delta H$ | Difference<br>(N-O) | To lorence | Remarks                                                     |
|------------|---------|--------------------------|--------------------------|---------------------|------------|-------------------------------------------------------------|
| From       | To      |                          |                          |                     |            |                                                             |
| 109-012    | 109-011 | + 50.592                 | + 50.604                 | -12                 | 45         | 109-012<br>+ 109-011<br>fallies with<br>previous<br>results |
| "-011      | "-010   | + 9.460                  | + 9.549                  | -89                 | 26         |                                                             |
|            |         |                          |                          |                     |            |                                                             |
|            |         |                          |                          |                     |            |                                                             |
|            |         |                          |                          |                     |            |                                                             |
|            |         |                          |                          |                     |            |                                                             |
|            |         |                          |                          |                     |            |                                                             |

## 4. LEVELLING LINE

(Few words about the nature of the terrain, weather condition, overall condition of BM, List of destroyed BM and about observation)

यो लेभल लाइनको प्राकृतिक स्वरूप समतल छ।  
 तुरै लाइन मोटर चल्ने पक्की सडक छ। सर्वेक्षण समय मा मौसम जाडो र  
 सुफा छ। सबै स्मारक चिन्हहरू राम्रो अवस्था मा छन्। यस लेभल लाइन मा  
 सर्वेक्षण कार्य बिहान ६:३० देखि ६:३० सम्म र बेलुका २:३० देखि २:३० सम्म गरिएको छ।

## APPENDIX 1/2

# TECHNICAL REPORT

Page 2

### 5. END JUNCTION POINT

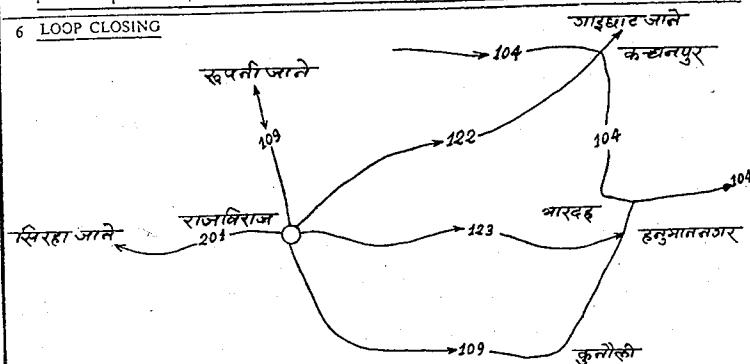
(i) Few words about its location and condition.

109-033, 109-034, 109-035 र 109-036 :— श्री उन्नाई स्मारक चिन्हहरू कुमौली-भादवह सडकमा कुमौली बाट भारदह जोडा क्रमशः हनुमाननगर बजारमा भन्सार कोणस्थलको अगाडी, मोहनलाले उनको घर अगाडी, सिक्नु गहारको घर अगाडी र हनुमाननगर बजार बाट भारदह जोडा जोगमोघा गाउँ तलजि कलभर्वमा कायम छन्। सबै उन्नाई स्मारक चिन्हहरू राप्ती अक्सरमा छन्।

(ii) Comparison of (New-Old) result

| bench Mark |         | New Result | Old Result | Difference | Tolerance | Remarks                                                     |
|------------|---------|------------|------------|------------|-----------|-------------------------------------------------------------|
| From       | To      | $\Delta H$ | $\Delta H$ | (N-O)      |           |                                                             |
| 109-033    | 109-034 | + 38.354   | + 38.480   | -126       | 31        |                                                             |
| 11-034     | 11-035  | + 12.238   | + 12.730   | -492       | 29        |                                                             |
| 11-035     | 11-036  | + 75.684   | + 74.992   | + 692      | 143       |                                                             |
| 11-033     | 11-036  | + 126.276  | + 126.202  | + 74       | 149 ✓     | 109-033<br>to 109-036<br>allies with<br>previous<br>result. |

## 6 LOOP CLOSING



7. BRIEF REPORT ABOUT THE TECHNICAL MANPOWER INCLUDING LIST OF STAFFS

|                                   |                                |
|-----------------------------------|--------------------------------|
| समतलत नापी समुह नं १              |                                |
| नापी अधिकृत जगदिश प्रसाद यादव     |                                |
| टोली प्रमुख नारायण बहादुर बिस्    | कम्प्यूटर सुख बहादुर वस्तेत    |
| झांझर अशोक कृष्ण मानन्धर          | अदली इच्छा राम बिस्            |
| सर्वेक्षण टोली नं १               | सर्वेक्षण टोली नं-२            |
| सर्भेक्षक देवतारायण चौधरी         | सर्भेक्षक कृषिराम श्रेष्ठ      |
| बुकर लक्ष्मण मिश्र                | बुकर विरेन्द्र लाल कर्ण        |
| स्टाफ सुपर भाइजर निरोधर प्र. यादव | स्टाफ सुपर भाइजर काशीकाल चौधरी |
| रेकी " " बिजु चौधरी               | रेकी " " बन्नी नारायण यादव     |

8. STATISTICAL DATA Please Fill in the Lev. Form No. 130 and attach herewith.

STATISTICAL DATA Please Fill in the Lev. Form No. 130 and attach herewith  
Prepared By. G. B. Basnet Group / Team No. 1 ..... Camp. Mansapur  
Checked By. N. B. Bista Date. 28.1.7.1988 

Lev. Form 1.29

## APPENDIX J

H.M.G SURVEY DEPARTMENT  
LEVELLING DIVISION  
STATISTICAL DATA

DATE FROM 7/3/1988

TO 20/3/1988

Page No 1

ALIGNMENT NO. 123

[illegible]

Checked By G.B. Bassett

LEV. FORM 1.30



## APPENDIX L

List of technical datas to be send to headquarters from the field

1. Description cards of PBMs with a duplicate copy  
(Lev. Form 1.20)
2. Description cards of TBMs with a duplicate copy  
(Lev. Form 1.22)
3. Observation books of Forward and Backward teams  
(Lev. Form 1.11)
4. Checking of section with a duplicate copy  
(Lev. Form 1.16)
5. Record of Bench Marks. (Lev. Form 1.24)
6. Error checking forms:
  - (a) Collimation error. (Lev. Form 1.13)
  - (b) Cross and zero error. (Lev. Form 1.12)
  - (c) Staff standardisation. (Lev. Form 1.10)
7. Work programme.
8. Technical Report. (Lev. Form 1.29)
9. Statistical data. (Lev. Form 1.30)
10. Result of loop closing with diagram.
11. Map indicating the location of Bench Marks.